Not meeting the necessary conditions

Necessary conditions

In order to prove an argument, certain supporting reasons or evidence will be essential to it. These are called *necessary conditions*. A necessary condition is just as it sounds: it is an essential requirement. If it is not present, there is a gap in the argument, and the outcome could be different. If the outcome could be different, then the argument isn't proved. It is important to bear in mind that there may be many necessary conditions, or requirements, for proving a case.

'Without this, then not that ...'

You can check whether a reason forms a necessary condition by rephrasing the argument and seeing whether it still holds true. Necessary conditions are expressed in statements such as:

- *If this doesn't happen, then that won't occur.*
- *If this isn't true, then that can't be true either.*
- *If this isn't present, then that won't be present.*
- *If A isn't present, then B can't be true.*
- *If it doesn't have A, then it can't be B.*
- *If it doesn't do A, then B won't result.*

This is easier to grasp through concrete examples.

Example 1

If you don't make advance arrangements for a taxi to come to the house to take you to the station, then a taxi won't arrive in time for you to catch your train.

A necessary condition, or requirement, for the taxi arriving in time, in this case, is that arrangements are made in advance. This is a sound argument.

Example 2

One way of making a lot of money is by winning the lottery. In order to win the lottery, you have to have a lottery ticket for the draw. John has a lottery ticket for the draw so he will make a lot of money.

One necessary condition, or requirement, for making money through the lottery is to have a relevant lottery ticket. John has met this necessary condition by having such a ticket.

Checking for necessary conditions

When you are checking for necessary conditions, it can help to rephrase some or all of the reasons, and see whether the argument still holds.

Example 3

Proposition: Birds have wings. The item has wings. The item is a bird.

To check whether wings are a necessary condition of the item being a bird, apply a statement such as: *If it doesn't have A, then it can't be B*, and check whether this is true or false. In this case:

> *If it doesn't have wings, then it can't be a bird. True or false?*

This is true: if an item did not have wings, it would be hard to argue that it was a bird.

However, it is important to take the context into consideration: if a bird had lost its wings in an accident, or had been born without wings, it would be flawed to argue that this prevented it from being a bird. For example, the underlying DNA that leads birds to have wings would be able to determine that this was a bird.

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'Necessary' is not enough proof

Necessary and sufficient conditions are different. Even if a necessary condition is met, this might not be sufficient to prove a case: there may be other conditions that must be met. You need to consider whether the 'conditions' are sufficient to support the conclusion. If not, then the argument is not yet proved.

For example, simply having a ticket for the lottery draw is not a sufficient condition for making money: the ticket might not win. This illustrates the difference between necessary and sufficient conditions.

'If this, then that ...'

Sufficient conditions form the totality of all those conditions that must be met in order to secure a particular argument. If sufficient conditions are met, then a particular set of consequences *must* follow. Sufficient conditions are expressed in statements such as:

- *If this is true, then that must always be true.*
- *If A is present, then that proves B.*
- *If this is true, then that must always follow.*
- *If A is present, then B must be true.*

Example 1

The lottery prize money was £10 million. John held the only winning ticket. He met the rules of the competition. Therefore, John made a lot of money.

In Example 1, some necessary conditions for John to make a lot of money are met: the prize was for a large sum, and John was the sole winner. However, if he lost his ticket, didn't claim his prize, or the lottery company went bankrupt, sufficient conditions would not have been met for John to make a lot of money.

Example 2

Bacteria usually have very short life spans. However, in 1989, the skeleton of a well-preserved, 11,000-year-old mastodon was found in Ohio. Scientists found an intestinal bacterium in its rib cage that they believe was its last meal. The bacterium was not found in the surrounding peat. Therefore, the bacterium must be over 11,000 years old.

The author is here arguing that bacteria may live much longer than was assumed. A necessary condition is that bacteria found in the skeleton are not also found in the surrounding peat. If they are,then the bacteria might have travelled from the peat to the skeleton only very recently, and might not have been present in the rib cage 11,000 years ago. However, this is not a sufficient condition to prove the age of the bacteria. We don't know, for example, whether the bacteria were blown by the wind into the skeleton at any intervening point during the last 11,000 years, without making contact with the surrounding peat.

Sufficient and/or necessary

When you are checking for sufficient conditions, it can help to rephrase some or all of the reasons, and see whether the argument still holds true. To check whether wings are a sufficient condition to prove that something is a bird, apply a statement such as: *If A is present, then that proves B,* and check whether this is true or false.

Example 3

Proposition: Birds have wings. The item has wings. Therefore, it is a bird.

If wings are present, then that proves this is a bird. True or false?

The answer is 'false'. Its having wings is not sufficient proof that this is a bird. Other necessary conditions would be that it was, or had been, a living creature, with feathers, and that it had the DNA of a bird. A winged item could simply be an aeroplane.

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Activity: Necessary and sufficient conditions

For each proposition, decide:

- Whether the reasons given meet the necessary conditions to support the conclusion. Write *Yes* or *No* in the box headed *Necessary?* Give reasons for your answer.
- Whether the reasons given to support the conclusion are sufficient. Write *Yes* or *No* in the box headed *Sufficient?* Give reasons for your answer.
- An example is given in the first box.

Answers: see pp. 295–6.

Ex.	Proposition	Necessary?	Sufficient?
	Example: Birds have wings. The item has wings. Therefore, it is a bird.	*Yes. Wings are a necessary condition for the item being a bird.*	*No. The reasons given to support the argument that the item is a bird are not sufficient to satisfy the definition of a bird. This would include: usually flies, is animate, lays eggs, has two legs, has feathers. The information given is not sufficient to rule out an aeroplane or a toy.*
1	The report makes reference to branches. It must be about a tree.		
2	The boxer doesn't eat meat or fish. He does eat dairy products and vegetables. The boxer is a vegetarian.		
3	Amir is under the age of 20. Teenagers are less than 20 years old. Amir must be a teenager.		
4	Claire does not play any musical instrument. Therefore, she is not a musician.		
5	The bishop arrived on a vehicle with two wheels, one in front of the other. The bishop must have been on a bicycle.		
6	A television usually costs more than a radio. This one costs less than a radio, so it must be a bargain.		
7	Li Yeung had the benefit of an exceptionally happy childhood. She must be a very happy adult.		

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False analogies

An analogy is a comparison made to draw out similarities between two things.

Creative comparisons

Authors can attempt to persuade their audience through using comparisons. In creative writing such as poetry and fiction, it is legitimate to compare two items that seem at first to be dissimilar in order to produce a literary effect such as surprise, humour or an unexpected perspective. In creative writing, it may be permissible to say 'it was raining wellington boots', or 'the moon is a goddess riding her chariot of clouds'. Literary critics have to decide whether such comparisons work to create the desired effect on the audience.

Valid comparisons

For most types of critical thinking, comparisons must be valid, and add to our understanding of the situation. In scientific terms, for example, it doesn't help to think of the moon as a goddess or clouds as chariots. Comparisons draw attention to those aspects which are similar. As two things are never identical, it takes critical evaluation and judgement to decide whether a comparison is valid for the context. If the comparison helps to give a more accurate understanding, then it is likely to be valid.

Example

The heart works as a pump, moving blood through the body by opening and constricting.

For most purposes, the comparison with a pump helps us to understand the action of the heart, so this is valid.

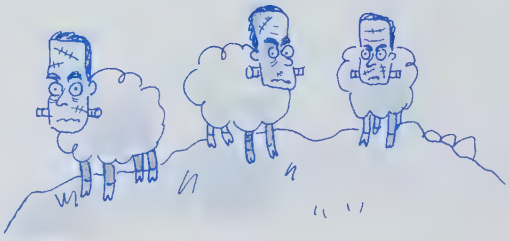
An analogy is not valid if:

- the two items being compared are not sufficiently similar; or ...
- the comparison is misleading; or ...
- the item used for comparison is described inaccurately.

Before reading on, check whether you can identify the weaknesses in the analogy in the example below.

Example

Cloning of human cells should never be allowed: it will create another Frankenstein's monster. We do not want such monsters.



The author's position on cloning is clear: that it is wrong and should be stopped. It may be that the idea of cloning is 'monstrous' to many people and the author is playing on that sentiment.

However, the analogy used is not valid as it doesn't compare like with like. A clone is an exact copy of an original. Frankenstein's monster wasn't an exact reproduction or copy of anything, but was, rather, an assembly of pieces. Moreover, by using the term 'another Frankenstein's monster', the author is implying we should have learnt our lesson from the past. However, Frankenstein's monster was only a character in a book. The author wants us to think that a clone will be a 'monster', but if the original used for the clone was not a monster, an exact copy should not be a monster either.

If an author uses a false analogy well, the argument may seem convincing. This is especially true if one half of an analogy seems easy to prove (that Frankenstein's creation was a monster) and the other isn't (the outcomes of cloning). It is easy to assume that because one half of the analogy is true, the other half must be too.

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Activity: False analogies

Activity

For each of the passages, identify:

- What the analogy is: which two things are being compared?
- Whether the comparisons are valid.

Passage 7.10

The earth's atmosphere is like a blanket of gases around the earth. It is only a thin layer but it helps to maintain the temperature of the earth, keeping us warm. It also offers a layer of protection from the intensity of the sun.



Passage 7.11

It may not seem likely that the new political party will be successful in the upcoming elections but we remain optimistic. It is true that the formal membership is small and the party does not have much money with which to campaign. However, what it lacks in these areas, it makes up for in other areas, such as the skill of its politicians and their commitment to success. The party is like a new David, taking on Goliath. It may be small, but it can take on those much bigger than itself.

Passage 7.12

As the basis of an argument, the premises are like the foundations of a building. If the premises are not well founded, the argument is likely to collapse.

Passage 7.13

There was no way the defendant was able to help himself. He had been under excessive strain for some time and his emotions had been building up like steam under pressure. The witness had been goading the defendant, knowing he was likely to get angry. The defendant was like a pressure cooker, just waiting to explode. Eventually, he just reached boiling point and an explosion became inevitable.



Passage 7.14

Investors in certain businesses lost a great deal of money in recent years as their stocks and shares wavered in the financial markets. Investors may not have a right to compensation for the knocks and bruises they have suffered on the stock market but they should be reimbursed for major accidents and serious lapses in the health of the financial markets.

Answers: see pp. 294–5.

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Deflection, complicity and exclusion

Language can be used skilfully to construct a powerful argument. However, it can also be used in ways that are unfair and which produce flaws in the line of reasoning. Language can be used to lull the audience into a false sense of security about whether an argument is valid, or can divert the audience from the line of reasoning. Some of these tricks of language are examined below.

Deflective language

An author can use language to suggest there is no need to prove the argument, deflecting the audience from critically evaluating the reasoning.

Suggesting the argument is proved

Use of words such as: *obviously, of course, clearly, naturally* suggests that the argument is so obvious there is no need to evaluate it.

Appeals to modern thinking

Another way of deflecting the audience from the reasoning is by referring to the date, as if that, in itself, added weight to the argument.

Examples

- *We're not in the nineteenth century now!*
- *It's no longer 1940!*
- *It's like being back in the ark!*

As the date is factually accurate, the audience is already drawn into part agreement with the argument. This approach attempts to discredit anyone who disagrees with the argument as being old-fashioned and out-of-date.

Encouraging complicity

Everybody knows

This is a particular form of deflective language where the author acts as if the reader were already part of a group of like-minded thinkers. This can be a powerful way of enticing the audience into agreement.

Examples

- *As we all know … we all know that …*
- *Surely, we all share the view that …*
- *Everybody knows that … Everyone believes …*
- *It is well established that …*

If 'everyone' believes something, then the audience would seem unreasonable not to agree.

'People like us': In-groups and out-groups

Another version is to suggest that people with certain attributes, such as 'decent people' or 'anyone with any intelligence', are more likely to agree with the argument. This can be especially convincing if coupled with an appeal to commonly held assumptions and prejudices.

Example

Anyone with any sense knows that women are naturally better at housework than men.

Tajfel (1981) wrote about the way people divide into 'in-groups' and 'out-groups'. The in-group tends to make the out-group appear inferior and undesirable so that others want to avoid being associated with them. Authors can present opponents of their argument as an 'out-group'. The audience is more likely to be persuaded by the arguments of an in-group and less likely to consider the views of the out-group. Appeals to decency, morals, shared values and shared identity can be examples of this.

Examples

- *All decent people would agree that X is immoral.*
- *As British people (or black people/Muslims/Catholics/deaf people, etc.), we all want …*

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Other types of flawed argument

There are many ways in which an argument may be flawed. As you become more used to critical analysis, you will become more attuned to spotting the weak parts of an argument. You could use a checklist such as that for Practice 2 on page 244 (or on the companion website at www.palgravehighered.com/cts3e) to help you identify the main flaws. However, you would need a very long checklist to cover *all* potential weaknesses. It is more useful to develop an increased sensitivity to potential flaws, so that you can recognise them in different kinds of circumstance.

The following sections look at some further potential distortions and weaknesses to watch out for. To help develop the sensitivity mentioned above, this section presents texts first and encourages you to find the flaws yourself, if possible, before reading the commentaries that follow.

Activity

Before reading about further types of flawed argument, see if you can identify these for yourself in the passages that follow.

You don't need to worry about whether there are technical names for the flawed arguments: just see if you can recognise when and why the argument isn't watertight. There may be more than one flaw in each passage.

Passage 7.15

Community centre

Closing the community centre will leave our poor little children with nowhere to play after school. Parents are rightly furious. After the death of five children from the area on a school canoeing trip, feelings are running very high. The neighbourhood just cannot take any more. If the community centre closes, parents will worry that their children are being left to suffer all over again.

Passage 7.16

Internet copying

Although it is possible to devise software to catch people who copy on the internet, it is unlikely that everyone who does this could be charged. If you can't enforce a law, then there isn't any point in passing it. If there isn't a law, then there isn't a crime. If there isn't a crime, then nobody has done anything wrong.

Passage 7.17

Tolls

More people should travel by public transport, as this would improve traffic flows in the city. If there were tolls for using roads, people would use public transport. Polls indicate that most people want the traffic flow to be improved. This shows that people would be willing to support the introduction of tolls. Therefore, the council should introduce heavy tolls.

Passage 7.18

Identity cards

Personal identity cards don't present any real dangers to human rights. They add to our security, by making it easier for the police to track and catch criminals. Opponents of identity cards are wishy-washy liberals who live in leafy areas and haven't a clue what it is like to live in run-down areas where crime is rife.

Passage 7.19

The managing director

The rugby team has had a chequered season. The team started badly and although it has picked up now, it seems unlikely that it can still win the championship. The managing director says that two new acquisitions will make a great difference to the team's performance for the end of the season. However, the board should give little credence to anything he has to say on the matter. After seeing his seedy affair with the TV quiz hostess broadcast all over the media, despite his constant denials, fans shouldn't give him any further credibility as a manager.

Answers: read pp. 100–1.

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Unwarranted leaps and castle of cards

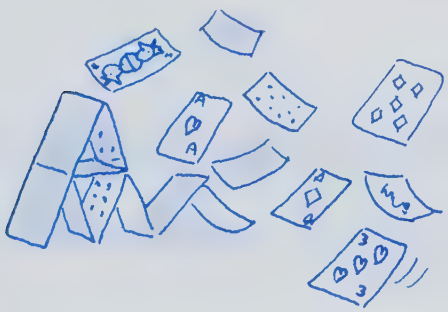
Unwarranted leaps

Where there are unwarranted leaps, the author appears to add two and two to make five. The argument races ahead, leaving gaps in the reasoning, and relying on unsubstantiated assumptions.

Castle of cards

In *castle of cards* types of argument,

- the author uses a set of interconnected reasons;
- the argument becomes precariously balanced, and depends on the previous reasons being accepted;
- if one reason or assumption is proved incorrect, the argument collapses easily.



Passage 7.17 Tolls (p. 99)

Passage 7.17 contains examples of both unwarranted leaps and *castle of cards* reasoning. The argument relies on a set of interconnected reasons and assumptions and is very delicately balanced. There are unsubstantiated assumptions which could be challenged such as that:

- the traffic problem is caused by the number of cars on the road, rather than, for example, road works or a one-way system;
- if a toll was introduced, people would respond by using public transport.

There is an unwarranted leap to the conclusion that, because a poll shows people want the traffic flow to be improved, they would also support tolls. We are not told whether the poll asked questions about tolls, so we do not know that a toll would be welcomed. The public might have preferred a different solution, such as bus shuttles or car-sharing.

Sleight of hand

A sleight of hand is a 'cunning trick' that can go unnoticed. In Passage 7.17, *Tolls*, the author jumps from a line of reasoning that appears to be discussing tolls, to a conclusion that argues for *heavy* tolls. This slight change of wording is an example of a 'sleight of hand'.

Passage 7.16 Internet copying (p. 99)

The *castle of cards* approach is also evident in Passage 7.16. This makes unsubstantiated claims such as that everybody who is caught copying on the internet could not be charged. This is not proved. On the contrary, large-scale fining is possible, and is used for minor traffic offences and for not having a television licence.

The author then argues that if a law can't be enforced, it shouldn't be passed. This is a matter of opinion and the author hasn't proved the law can't be enforced. Using this argument as the next stepping stone, the author argues that without a law there isn't a crime. There is a sleight of hand here, as the author hasn't mentioned whether a law against such copying is already in place at the time of writing.

The author makes a final leap to argue that if there isn't a crime, nobody has done anything wrong. This is not the case. Right and wrong are questions of ethics, not law. Some acts which are wrong might not yet be enshrined in law. For example, when there is a new invention or an advance in medical technology, it can take time for these to result in changes in the law.

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Emotive language; attacking the person

Emotive language

Emotive language uses words, phrases and examples that intend to provoke an emotional response. Some subjects such as children, parents, national pride, religion, crime and security are emotive. Using these unnecessarily as arguments can manipulate the audience's emotions.

People tend to trust their own emotional responses. Strong emotions are usually a signal to the body to act quickly rather than to slow down and use reasoning. If an author can elicit an emotional response, then the audience is likely to be less critical of the reasoning. Where subjects are emotive, it is particularly important to check the underlying reasoning carefully.

Attacking the person

We saw in Chapter 3 that an argument should take counter arguments into consideration. This means making a critical analysis of the line of reasoning, not using personal attacks on those with opposing views. Attacks on the person rather than the argument are often used to undermine the credibility of an opposing point of view – but it is not a valid method of critical reasoning.

The exception is where there is a valid reason for showing that the opponents either have a history of being dishonest or have not revealed their vested interests in the debate.

Passage 7.15 Community centre (p. 99)

In Passage 7.15, *Community centre*, the author appeals to the emotions using words such as 'poor little children' and references to 'feelings running high' and 'suffering'. The passage reminds the audience of a disaster that had happened to other children in the area. The incident was very sad but its relevance to the current argument is not clear. That accident happened away from the area, and when there was already a community centre where children could play. There may be a good case for keeping the community centre open, but the author does not present a reasoned argument to support it.

Passage 7.18 Identity cards (p. 99)

This passage attacks everyone who opposes the introduction of identity cards on personal terms. It also makes unsubstantiated assumptions about the backgrounds and economic circumstances of opponents, in order to undermine their credibility. As the passage relies on these unacceptable methods rather than reasons and evidence, it demonstrates flawed reasoning.

The passage also encourages complicity in the audience (see page 98). By abusing opponents, the author encourages a division between in-groups and out-groups, or 'people like them' and 'people like us'. Furthermore, the passage draws on emotive subjects, referring to crime and security to win over the audience.

Passage 7.19 The managing director (p. 99)

This passage attacks the person of the manager rather than evaluating his judgements about the likely impact of the new players. It attacks the manager on the grounds of his personal life, not his expertise in managing a rugby team. We may not agree with decisions the manager takes in his personal life, but the passage does not show the relevance of this to managing the club. As the manager denies what is in the media, it may not even be true. The use of the term 'seedy' is emotive, suggesting there is an illicit side to the relationship, but this is not substantiated.

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More flaws

Just as you did for page 99, check whether you can identify the flaws in the following passages. There may be more than one flaw in each passage, including flaws covered in previous sections. The answers will be on the following pages.

Passage 7.20

Nature or nurture

Those who argue that intelligence is not in-born do a disservice to the truly bright individual and hinder attempts to discover excellence. Many of us had intensive training on an instrument such as the piano when we were children, but we obviously did not all turn out to be a Beethoven or Mozart. We are all able to recognise brilliance when we see it. Proponents of the view that intelligence can be nurtured are too ready to blame society or the education system for not turning out more geniuses. They want us to believe that any of our children could be a genius, which is unfair on parents and teachers alike.

Passage 7.21

Curfews

Juvenile crime has risen sharply in cities. Young people are out of control. There are only two options in a situation like this. Either we agree to put up with savage assaults on our persons and property, or we place a curfew on all young people after 10 o'clock.

Passage 7.22

Einstein

Einstein was not very good at maths when he was at school. Many school-children today could solve maths problems that he used to struggle with. The accolade of 'great scientist' shouldn't be ascribed to someone who struggled with basic numerical problems.

Passage 7.23

Health training

The public's knowledge of health is poor and more money is needed for education in this area. Increased sums of money should be spent on courses to make people aware of personal health issues. People don't always know what they can do to take care of their health so further investment is needed in training on health matters.

Passage 7.24

Advantages of maths

More people should be informed of the value of studying maths to a higher level at school or university. A mathematical education can be very advantageous. Therefore, the guidance given to young people should emphasise the benefits of choosing maths.

Passage 7.25

Selling assets



The opposition party is wrong to condemn the leader of the council for selling off public assets at a low price to its own supporters. When the opposition had a majority in the council, they sold off cemeteries and houses below the commercial price, benefiting their own supporters. If they can do it, then the current council can do it too.

Passage 7.26

Stealing at work

Mr Malcolm's employers pay their stylists much lower wages and expect them to work much longer hours than owners of other salons. Mr Malcolm supplemented his income by taking equipment and styling products from the workplace and selling these in his own area. He was justified in stealing from his employer because his employer was exploiting him.

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Misrepresentation and trivialisation

One way of distorting an argument is by presenting the options or opposing arguments in an unfair or unbalanced way. Misrepresentation can be engineered in several ways. Three are given below. A consequence of misrepresentation is that important matters can be made to appear trivial.

Ignoring the main opposing reasons

An author can misrepresent an opposing argument by focusing on its minor points and ignoring its chief supporting reasons. If the minor points are not sufficient to support the conclusion, the opposing argument will appear very weak. Sometimes, authors may simply attribute beliefs and arguments to their opponents without any evidence.

Presenting restricted options

Another form of misrepresentation is to present an argument in such a way that it looks as if there are only two possible conclusions or options for action. This approach relies on selecting one conclusion or option that appears very weak and one that seems preferable. The weakness of the alternative conclusion or option makes the author's case appear stronger than it really is.

Misrepresenting a person

A poor form of argument consists of focusing on certain characteristics of a person, especially those irrelevant to the main argument, and ignoring more relevant information about that person.

Passage 7.20 Nature or nurture (p. 102)

Passage 7.20 misrepresents the opponent's arguments. The author's position is clearly one that supports the view that levels of intelligence are innate (i.e. there from birth). The passage attributes arguments to the opponent: 'They want us to believe ...', 'Proponents ... are too ready to blame society ...' No evidence is given to show that this is what is believed by people who argue that intelligence can be nurtured. Other reasons that people might have for believing that intelligence is not simply a question of birth are not considered. For example, there is no consideration of research evidence.

The argument is trivialised by focusing on relatively rare cases of 'genius' rather than on how intelligence operates for most people. Rather than presenting a well-reasoned case, the author uses emotional devices, using an emotive subject such as unfair treatment of teachers and parents. There is an appeal for complicity through assertions aimed at drawing in the audience ('We are all able to recognise brilliance') and by references to potentially common experiences such as childhood piano lessons. These further trivialise the subject.

Passage 7.21 Curfews (p. 102)

The argument in Passage 7.21 is flawed in several ways. The main flaw is that it offers only two options, curfew or assaults. Other options, such as improved policing or changes in lighting, are not considered. 'Out of control' and 'savage' are strong statements using emotive language, but no definitions or explanations are given to substantiate these. It also assumes the crime occurs mostly after 10 o'clock.

Passage 7.22 Einstein (p. 102)

Passage 7.22 misrepresents Einstein by focusing on his early difficulties with maths and ignoring all the discoveries for which he is considered a great scientist. It overlooks that all the people who were better at maths when Einstein was young did not go on to develop such advanced scientific theories.

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Tautology; two wrongs don't make a right

Tautology

A line of reasoning should take an argument forward. Tautological arguments, on the other hand, merely repeat the same points in different words, without advancing the argument. Tautology means using different words to repeat the same concept, as in 'the car was reversing backwards'.

Two wrongs don't make a right

Another form of flawed argument is to argue that an action is acceptable simply because someone else acted in a similar way. Similarly, it is usually considered to be flawed reasoning to argue for consistent treatment when this would mean that an injustice or an illogical outcome was perpetuated by doing so. For example, if one person cheats in an exam, then it is not reasonable to argue that other people should be able to cheat too. If one person lies, it doesn't make it right for others to tell lies.

Passage 7.23 Health training (p. 102)

Passage 7.23 is tautological. Each sentence merely repeats what is said in the other sentences, using different words. 'Spending more money on courses' equates to 'investment in training'; 'make people aware' implies that 'people don't know what they can do'. The argument does not progress, as no further reasons, details or evidence are provided.

Passage 7.24 Advantages of maths (p. 102)

Passage 7.24 is another example of tautology. The empty repetition makes the argument appear to go round in circles. The author doesn't present reasons to substantiate the case for learning higher maths. No details of the potential advantages are given. For example, it could have been argued that a higher qualification in maths can lead to a greater choice of careers or a better income. The author might have included information such as that surveys suggest employees in careers that require higher levels of maths have greater job satisfaction than employees in most other occupations.

Passage 7.25 Selling assets (p. 102)

Passage 7.25 is an example of 'two wrongs not making a right'. It is wrong for any party to sell public assets cheaply in order to secure political advantages for their party. Just because a previous party did so, this does not make it right for other parties to follow suit. It may appear hypocritical to cast blame on another party for behaviour that one's own party has engaged in. However, it would still be in the public interest for an apparently hypocritical politician to expose current wrong-doing. Otherwise, even more public assets would be wasted.

Passage 7.26 Stealing at work (p. 102)

Passage 7.26 is another example of 'two wrongs do not make a right'. The employers may have been in the wrong in the way they treated their employees. However, stealing was not the appropriate response. It isn't either ethical or legal. The argument would not stand up in court.

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Summary

This chapter introduces many of the most common types of flawed reasoning. Such flawed reasoning may be a deliberate ploy on the part of an author who intends to deceive the audience. However, flawed reasoning is often the result of insufficiently rigorous critical thinking: many people are not aware of errors in their reasoning.

One group of flaws covered by the chapter relates to the concept of causality. It is a common mistake to assume that if two things appear to be connected in some way, the nature of that connection is one of cause and effect. However, the items may be linked by a third item, a distant relationship, through correlated trends, or simply by coincidence.

A second set of flaws relates to statements or arguments that do not meet the necessary and sufficient conditions to establish proof. If necessary and sufficient conditions are not met, an alternative conclusion could be drawn so the argument is not yet proved.

The third set of flaws is concerned with accuracy and validity in the way language is used to establish an argument. There are many ways that the language used to communicate the argument can distort or conceal. Some examples of this covered within the chapter are: making false analogies, attempting to draw the reader into collusion with the author, using language to conceal gaps in the reasoning, using emotive language with the aim of distorting the audience's response, and misrepresenting opponents' views.

Being able to recognise flaws in an argument is a useful skill. It helps you to identify weak points in other people's arguments and to pinpoint areas for you to investigate more closely so you can make more informed decisions. If you are evaluating an argument within your writing, or as part of a debate, knowing the flaws in the opponent's arguments helps you to formulate better counter arguments. If you are able to recognise such flaws in your own arguments, you are in a better position to put forward more convincing arguments in their place.

Information about the sources

For more about mastodons: Postgate, J. (1994) *The Outer Reaches of Life* (Cambridge: Cambridge University Press).

For more about 'out-groups': Tajfel, H. (1981) *Human Groups and Social Categories* (Cambridge: Cambridge University Press).

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Chapter 8

Where's the proof?

Finding and evaluating sources of evidence

Learning outcomes

This chapter gives you opportunities to:

- recognise the difference between primary and secondary sources
- understand what is meant by a literature search
- understand concepts such as authenticity, validity, currency, reliability, relevance, probability, and controlling for variables, as applied to research evidence
- identify ways of evaluating samples used in research projects
- recognise potential weaknesses in oral testimony

Introduction

We do not always need to be an expert in a subject to evaluate an argument. In many instances, we will still be able to evaluate whether the reasons support the conclusion and whether the line of reasoning is ordered in a logical way.

However, in order to evaluate many arguments, we have to know whether the evidence used to support the reasoning is true. This means that we need to go to other sources, either people or material resources, to check the facts that underlie the reasons given.

Evidence may be convincing in one context, such as in everyday conversation or a magazine, but not in others, such as in a court of law or for academic or professional writing. In the latter cases, it is expected that greater efforts are made to check that evidence is all that it appears to be.



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Primary and secondary source materials

Most types of evidence can be divided into one of two categories:

- primary sources: the 'raw material' for the subject, such as data and documents;
- secondary sources: materials such as books and articles based on, or written about, primary sources.

Primary source materials

Primary source materials are those that originate from the time and place of the events being investigated. Primary sources can include:

- contemporary letters, documents, prints, painting and photographs;
- newspapers, books and materials published at that time;
- TV, film and video footage from the time;
- recordings of radio broadcasts;
- remaining body parts, sources of DNA, finger prints and footprints;
- artefacts such as tools, pottery, furniture;
- testimonies of witnesses;
- the raw data from experiments;
- autobiographies;
- material on the internet if the internet or materials on it are the focus of the study;
- individual responses to surveys and questionnaires.

Secondary sources

Secondary sources are any materials written or produced about the event, usually some time later. These include:

- books, articles, web pages, documentaries about an event, person or item;
- interviews with people reporting what they heard from witnesses;
- biographies;
- articles in magazines;
- papers and reports using the results of surveys, questionnaires and experiments.

Crossing between categories

Whether something is a primary source depends on how far it was part of the events at the time. Secondary sources in one circumstance may be primary sources in another. For example, a biography is normally a secondary source, but may reproduce copies of original letters that are primary sources. The biography of a prime minister is a secondary source of information about the political leader but could be a primary source about the life of the author. Magazine articles written in the 1950s were secondary sources when published, but are primary sources for present-day research into life in the 1950s, for example, paraphrasing interviews held with celebrities. In the 1950s they were writing about people, events and objects of the time, whereas for present-day researchers they are artefacts in their own right, telling us what things were considered important at that time.

Activity  **Primary sources**

What are the main primary sources for your subject?

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Searching for evidence

Critical thinking generally requires an active approach to seeking out the most relevant evidence to support your own arguments, and to checking the evidence used by other people.

Checking other people's evidence

When you are reading, or watching a programme, or listening to a lecture, you may encounter a line of argument that is so interesting or relevant that you want to discover more. Alternatively, you may consider that the evidence cited does not sound very credible and you may want to check it for yourself. The higher the level of study or research, the more important it is to check the key evidence, especially if there is any doubt about its being reported accurately.

Use the references

When reading articles and books, you will see a short-hand reference in the text such as '(Gilligan, 1977)' and a more detailed list of references at the end of the text. These references provide the details you need in order to find that source for yourself.

Good references enable any reader who wishes to do so, to check whether:

- the source material really does exist;
- the author represented the source material in an accurate way, and the source really says or contains what the author claimed;
- the source contains any additional information that readers can use for their own projects.

When critically evaluating an argument, don't be afraid to go back to some of the sources and check whether these stand up to scrutiny. Often, it is not possible to form a judgement about an argument until you have more information about the subject.

Evidence for your own arguments

When looking for evidence to support your own arguments, the first questions you are likely to ask are:

- Has anything been written about this already?
- Where can I find that information?
- Which are the most relevant and authoritative sources for this subject?

For everyday purposes

If you need information for casual purposes, such as for a personal project or for contributing to a debate, you may need only to do one or two of the following:

- browse an introductory chapter of a book;
- use a search engine such as *Google* for information about the subject;
- read recent newspapers, or read papers on the internet, using a source such as *www.theguardian.com*;
- ask an expert in the area, such as a librarian;
- visit the websites of relevant bodies, such as campaign groups, charitable bodies, or government sites.

For academic and professional purposes

If you are looking for material as background for a professional report or for academic work, you will need to conduct a 'literature search'. The rest of this chapter focuses on finding and critically evaluating potential sources of evidence.

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Literature searches

A literature search gives you an overview of previous research on the subject. Usually, the larger the project, the more extensive the search should be. For smaller projects, or where there are word restrictions for the report or essay, careful selection is especially important.

Doing a literature search means:

- finding out what has been written on the subject (secondary sources);
- collating a list of the sources that are potentially relevant for your subject;
- paring down the list, selecting sources for initial investigation to check for relevance;
- browsing selected items to help you select the most useful sources;
- selecting the most relevant sources for more detailed investigation.

Online literature searches

Many reputable sources are now available on line. If you know the names of journals, government papers or other relevant authoritative sources, enter these as part of your search. Otherwise, enter several key words to help pinpoint exactly what you want. Your search will be more effective if you use a relevant search engine. If you are at university, your tutors are likely to recommend the most useful websites and search engines. Some useful starting places are given in the Appendix (p. 274).

Using abstracts

Browsing the abstracts of journal articles is a particularly useful way of gaining a sense of all the recent research in the field. The abstract summarises the main argument, research methods, findings and conclusions, which helps you decide whether the article is worth reading in depth. Note, especially, the section which summarises the background literature for that report. This can indicate important leads for your own project.

Deciding whether to use a secondary source

Examine secondary sources critically to decide whether, for your purposes, they are likely to be sufficiently:

- well researched;
- trustworthy;
- recent;
- relevant.

This is especially important if you are considering purchasing books or borrowing them from a library, as it helps you to avoid unnecessary costs and time delays.

Basic questioning of the evidence

Critical thinking is a questioning process. When evaluating evidence, ask such questions as:

- How do we know this is true?
- How reliable is this source?
- Are the examples given truly representative of the whole area?
- Does this match what I already know?
- Does this contradict other evidence?
- What motive might this person have for saying this?
- What are we not being told?
- Are any other explanations possible?
- Do the reasons support the conclusion?
- Is the author's line of reasoning well substantiated by the evidence?

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Reputable sources

For academic study and for professional life, evidence is roughly divided into 'reputable sources' (or 'authorities') and then everything else. A reputable source is basically one that:

- has credibility: it can be believed with a high degree of certainty;
- is likely to give accurate information;
- is based on research, first-hand knowledge or expertise;
- is recognised in the field or academic discipline as an authority.

Journal articles

Articles in journals are usually regarded as the most reputable sources as, in order to be published, they have to be reviewed and selected by other leading academics. This is known as 'review by peers'. There is a great deal of competition to get published in leading journals, so articles that succeed in passing such a peer review are generally well regarded.

Subject differences

A reputable source for one subject may not be a reputable source in another field of study. Each academic discipline has its own conventions. For some subjects, such as in science, law, medicine, and accountancy, 'hard' data such as facts and figures are generally regarded as superior forms of evidence. On the other hand, in subjects such as art, music and psychotherapy, qualitative evidence can be regarded as more important: 'feeling the subject' may be more valuable than 'number-crunching'. However, this is not a hard and fast rule, and it can depend on the nature of the subject being studied and the evidence that is available.

Questions to consider

When deciding whether a text is worth reading, consider:

- Has it been recommended by a source you trust, such as your tutor or a reputable journal or a review in a quality newspaper?
- Is there a clear line of reasoning, with supporting evidence?
- Does it include a detailed list of references, or a bibliography, indicating thorough research?
- Does it provide clear references to its sources of information, so that other people could check these? If not, this may not be a suitable text for use in academic contexts.
- Does it use source materials that look reputable, such as journals and relevant books, rather than the popular press?

Using recognised 'authorities'

Older sources, especially those regarded as authorities, may have made a significant contribution to the area of study. It is important then to check:

- exactly how the source contributed to knowledge in the field – don't dismiss something just because it sounds old;
- which parts of the original arguments and evidence are still applicable, and which are not;
- how later research used the source as a stepping stone to further findings – and in what ways the original ideas or data have been refined or superseded;
- more recent authorities, to see whether the source is still exerting an influence on research.

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Authenticity and validity

Authentic evidence

Authentic evidence is of undisputed origin. This means that it can be proved that it is what it is claimed to be, or that it really was written or produced by the persons claimed. It isn't always possible to check for authenticity when hearing or reading an argument, but it is possible to maintain an open mind about whether the evidence is likely to be authentic.

Activity  **Authenticity**

Consider whether each of the following references is likely to be authentic or inauthentic.

1. A medieval illuminated manuscript found in the stacks of a cathedral library.
2. A medieval illuminated manuscript that turns up in a local second-hand bookshop.
3. A collection of 1000 autographs of Elvis Presley being sold over the internet.
4. An unpublished diary written by Shakespeare, in the possession of a second year student.
5. Letters written by Napoleon Buonaparte, dated 1809, contained in a large collection of French Revolution memorabilia.
6. A set of five previously unknown Van Gogh paintings discovered in a garage on a housing estate.
7. Decaying remnants of a Viking ship found in recently drained marshland.
8. Letters and artwork written by prisoners in the nineteenth century, in the care of a prison governor.

Answers: see p. 297.

Validity

Valid evidence meets the requirements agreed, or the conventions that are usually followed, for the circumstances. What is valid will vary depending on the circumstances. Evidence may not be valid if, for example, it is not authentic, if it is incomplete, is unreliable, or if it isn't based on sound reasoning.

Examples

1. A defendant confessed to a crime but the confession wasn't considered valid because it became evident that the defendant had been forced to make it. Legal requirements would not regard a confession exacted under duress as valid evidence of committing a crime.
2. To gain a particular qualification, students were required to write eight essays as their own work. Although one student handed in eight essays on relevant subjects, the examiners found that three were too similar to essays available on the internet. These were not accepted as valid evidence of the student's own work, so the requirements of the qualification were not met.
3. An athlete argued that she was the fastest runner in the world. Although she had reliable evidence of her running times, these were not considered valid evidence that she was the fastest runner, as they were gained in unusually favourable wind conditions.
4. A report claimed that people who smoke are more likely to drink alcohol. The evidence wasn't considered valid as all the participants who smoked were selected in places that sold alcoholic drinks, whereas non-smokers were selected in the street. This meant that the selection of participants was already weighted in favour of the smokers being more likely to drink alcohol. This doesn't meet agreed research conventions, which aim to avoid weighting the evidence.

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Currency and reliability

Currency

If a source is described as 'having currency', this means it is still relevant *in the present*. This may be because:

- it was published recently;
- it was updated recently;
- it has been produced in a new edition that takes account of the latest research;
- the material covered is relatively stable and unchanging over time, so that it remains relevant for a long time. Examples of this would be anatomy, biographies, or descriptions of how machinery used to work in the past.

It is always worth checking whether a source is still up to date: new research can appear on any topic at any time.

'Currency' is a term that is applied to secondary sources. Primary sources are contemporary to an event, so may be relevant or not relevant to a topic, but questions of currency are not usually appropriate.

Seminal works

Seminal works are those that are so original or far-reaching in their findings that they continue to exert an influence for a long time. A seminal work could be a text, a film, music, art, architecture or commercial design, or any other item that had a strong impact on the thinking and research in a discipline over time. It helps our understanding of our subject discipline if we have first-hand experience of the seminal works that influenced its research base and theoretical perspectives. We are in a better position to recognise the theoretical perspective informing other research, and to recognise the influence of those works in later works.

Activity

Which works are considered seminal for your area of research or the subjects you are studying this year?

Reliability

Evidence is reliable if it can be trusted. This may be because the source of the evidence is:

- someone you know to be trustworthy;
- a recognised expert;
- a person with no vested interest in the outcome;
- a reputable source (see p. 111).

Reliability also refers to whether the evidence is stable over time, so that it can be used to make reasonably secure predictions. In other words, if you have evidence that something worked once, is this sufficient to show that it will work next time?

Example

Climatic conditions are relatively stable for large areas and time-periods and can be used to predict general trends in temperature or rainfall. On the basis of evidence of climatic change, we can predict that the Sahara region is likely to remain hot and dry for many years. Weather, on the other hand, changes quickly, and is less reliable for making predictions. It will rain in the Sahara, but it is hard to predict when or how much rain will fall.

Replication

In scientific writing, you may see references to the results being 'replicated' or 'not replicated'. This means that the results of a survey or experiment were re-tested to see whether they held true. If they didn't, the original outcome might simply have been the result of chance.

It is useful to know whether research was repeated and the findings replicated. If the outcomes were similar, this increases the probability that the findings are reliable.

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Selecting the best evidence

A summary of your background reading, or reasoning based on secondary sources, is normally required as an early section in a report and for dissertations and doctoral theses.

Which sources should I refer to?

It is usually the case that there is a great deal to say about the source materials, but there are word restrictions that limit what can be said. This means you need to consider very carefully the sources to which you will refer.

Be selective

- Include sources regarded as the leading authorities on the issue.
- Refer in brief to any other sources. Select evidence that demonstrates the main pathway, or set of stepping stones, leading up to your own project.

Sources contributing to your argument

The main source materials to which you refer should be those that contribute most to supporting your own line of reasoning. There may be one or two seminal works that you refer to in some detail, a small selection of key works that you cover at some length, and several others that you refer to in passing. It is important, when writing academic reports, to show you can discriminate appropriately between the most significant sources and those of peripheral importance.

Passing references

References to other research add weight to your own reasoning. A passing reference may be a major study in its own right, but contribute only background detail to your own argument. Usually, you would use a passing reference to support a step in your line of reasoning or to substantiate a minor point in your argument. You do this by either:

- writing a sentence summarising the research findings and naming the source and date; or
- writing your point and then adding a reference in brackets.

Examples

Miles (1988) argues that British Sign Language is a language in its own right.

Sign languages are also languages with their own traditions (Lane, 1984; Miles, 1988).

What should I say about sources?

Most writing tasks have word restrictions. You will usually need to allocate most of your word allowance to critical evaluation of the argument and your sources of evidence, and very few words, if any, to describing them. If you are uncertain of the difference between descriptive and analytical writing, see pages 48–9.

When selecting sources, ask:

- Did this contribute a major theoretical perspective to the discipline?
- Has this changed thinking in the subject, or made a significant contribution to the questions debated in the discipline?
- Does this provide a contribution to the path of research evidence that leads up to my own project? If so, how? Is this a direct or an indirect link? Is it a key contribution that needs to be discussed or a lesser contribution requiring a passing reference?
- Does this source challenge what was said before or provide an alternative way of thinking about the issue?
- Does it use research methods that are novel or that I could use for my project?

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Relevance and irrelevance

Relevant evidence is that which is necessary to give a good understanding of the issues. An author can provide evidence that:

(1) supports the conclusion;
(2) is relevant to the subject, but which may not be relevant to the conclusion: in this case, the evidence might even contradict the conclusion;
(3) is relevant neither to the conclusion nor to the subject.

Example 1

People need to improve their understanding of how language works so that they can use it more effectively. Research studies (Bloggs, 2003; Bloggs, 2006) show that the study of a foreign language improves our understanding of the structure of language, providing a way of comparing different language structures. Therefore, people who only speak one language should be encouraged to study a second language.

Here, the research evidence about the benefits of studying a foreign language is relevant to the conclusion that people who speak only one language should be encouraged to study a second language.

Example 2

People need to improve their understanding of how language works so that they can use it more effectively. Research studies (Bloggs, 2003; Bloggs, 2006) show that many people cannot describe the different components of their own language. A surprising number of people have difficulties remembering the rules even of their mother tongue. Therefore, people who only speak one language should be encouraged to study a second language.

Here the evidence that people have difficulties in their own language could be interpreted to suggest that people who have difficulties with one language should not be encouraged to learn a second. The evidence is relevant to the debate, but does not support the argument. Further information would be needed to support the conclusion.

Example 3

People need to improve their understanding of how language works so that they can use it more effectively. Research studies (Bloggs, 2003; Bloggs, 2006) show that people can recognise concepts in a foreign language even when there is no word for that concept in their mother tongue. Therefore, people who only speak one language should be encouraged to study a second language.

Here, the evidence about recognising concepts in a foreign language is loosely related to the topic about languages. However, it has a completely different focus. It has no apparent relevance to the debate about using language effectively or the conclusion that people should learn a second language in order to use language more effectively.

Relevance to the conclusion

In considering whether evidence is relevant, your main focus should be on whether the conclusion would be different if that evidence (or reason) was different or not available?

Check

When evaluating an argument, check:

- Is the evidence relevant to the topic?
- Is it needed to substantiate the reasoning?
- Does it make a difference to the conclusion?
- If so, does it support it or contradict it?
- Is the evidence needed to substantiate interim conclusions?

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Activity: Relevant and irrelevant evidence

Activity

For each of the following passages, identify whether the evidence and reasons are relevant to the conclusion. Then read the *Commentary* opposite.

Passage 8.1

Ice Age

Winters are getting colder. Opinion polls show that most people think there is a new Ice Age on the way. Therefore, we need to take measures to ensure that fuel resources are managed so that nobody is left to suffer from extreme cold during forthcoming winters.

Passage 8.2

Mr Charlton was given information, in confidence, that the price of shares in MKP2 Oils would rise suddenly if news of the new promotion reached the press before the share price was adjusted. Mr Charlton bought 50,000 shares in MKP2 Oils and leaked news of the promotion to the press. As a result, he made 10 million pounds personal profit. We can conclude that Mr Charlton abused the trust of the company and cheated it financially.

Passage 8.3

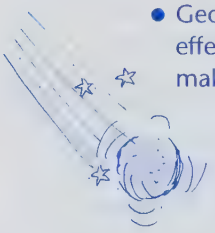
Major catastrophes, rather than gradual evolution, may be the main cause of change. Such a view did not seem plausible in the past as it was assumed that the process of geological change took place in a gradual way, just as it appears to today. However, evidence now suggests that change can be rapid and extreme. Geological evidence indicates that an enormous meteor collided with the earth several hundred million years ago, making most life-forms extinct. Geological science now attracts more funding than it did in the past. Archaeological evidence suggests that sudden changes in the environment brought about the rapid collapse of ancient civilisations.

Commentary

For Passage 8.1, the first reason, that winters are getting colder, is relevant to the conclusion about managing fuel resources. However, no evidence is given to substantiate this reason. The evidence from polls shows opinions, not facts, and this does not support the conclusion. An opinion is still only an opinion, even if held by a lot of people. The validity of an argument or of evidence does not normally rest on a majority decision.

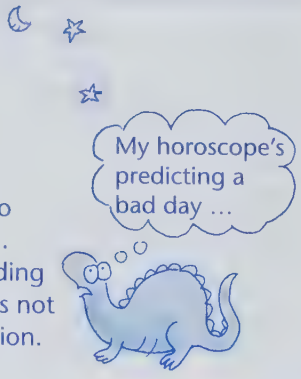
For Passage 8.2, all of the evidence given is relevant to the subject and to the conclusion that Mr Charlton abused the trust of the company and cheated it financially. He betrayed a secret to the press so that he could make money at the company's expense.

In Passage 8.3, the conclusion is that major catastrophes, rather than gradual evolution, may be the main cause of change. The relevant pieces of evidence given to support this are:



- Geological evidence about the effects of a meteor collision in making life-forms extinct.
- Archaeological evidence about the effects of sudden environmental change leading to the fall of ancient civilisations.

The section about the plausibility of this view in the past is useful background information, but does not provide evidence to support the conclusion. Information about funding for geological science is not relevant to the conclusion.



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Representative samples

Most research topics cannot be tested using very large numbers of people or circumstances. This would usually be too expensive, time-consuming, complicated to organise and unnecessary. Instead, surveys and research projects rely on selected samples. A representative sample is one which gives due consideration to the potential variety of relevant groups and circumstances.

Example

Four animal charities wished to know the views of the public on whether pets taken overseas should be held in quarantine before being allowed to re-enter the country. Each one selected the sample in a different way.

Sample 1
Charity 1 chose 1000 dog-owners from across the nation. The survey was balanced to ensure that roughly equal numbers were interviewed in every part of the country.

Sample 2
Charity 2 chose 1000 dog-owners from across the nation. The survey was balanced to ensure that more people were included in the survey in parts of the country which had large populations, and fewer representatives were questioned if the population was low.

Sample 3
Charity 3 chose 1000 pet-owners from across the nation. The sample was chosen to ensure that a broad range of pet-owners were included, including owners of snakes, budgies and tropical spiders.

Sample 4
Charity 4 chose 1000 people, representing a variety of pet-owners and people who do not own pets. The sample was selected from every county, weighted to include more people from heavily populated areas.

Differing principles of sample selection

Each of these samples selected participants according to a different principle. Sample 1 ensures that all geographical areas are represented equally, whereas sample 2 is more concerned that the sample is representative of population size. Sample 3 aims to ensure that different kinds of pet-owners are represented, whereas sample 4 is representative of both pet-owners and non-pet-owners.

Depending on the aim of the research, any of these methods of selection may be appropriate. For example, if it were known that 99 per cent of pets affected by quarantine were dogs, and that people from poorly populated rural areas were particularly affected, then the approach in sample 1 would be the most appropriate choice. Otherwise, a weighting according to population size is preferable.

If a wide variety of pets were subject to quarantine, then the approaches taken in samples 3 and 4 would be more representative of those affected. Samples 1–3 assume that people without pets do not need to be consulted, whereas sample 4 is more representative of the population in general. Sample 4 is more typical of the kinds of sample you will see in research projects and in articles. Usually, samples need to be representative of several different perspectives.

Check

When reading the 'Methods' section of research papers, articles and reports, check whether the most appropriate sampling method was used. If a group was not represented in the sample, then the findings may not be applicable to it.

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Activity: Representative samples

Activity 

Consider the following passages and decide in what ways the sample used in each is representative, and the ways it is not. Then read the *Commentary* opposite.

Passage 8.4

The experiment aimed to prove that eating carrots improves night vision in people under the age of 45, excluding children below school age. The sample consisted of 1000 people; 789 were women and the rest were men. For each sex, 25 per cent of participants were from the different age groups, 6–15 years, 16–25 years, 26–35 years and 36–45. Participants ate three capsules of carrot extract every day for 10 weeks.

Passage 8.5

The survey set out to discover whether consumers preferred soap perfumed with almond essence or soap perfumed with aloe vera. The sample consisted of 1000 people. Of these, 503 were women and 497 were men; 50 per cent of the sample were aged between 25 and 40, and the rest were aged between 41 and 55.

Passage 8.6

The research project tested the hypothesis that people who receive six sessions of counselling following a bereavement are less likely to take time away from work in the following 12 months than people who do not receive counselling. The sample consisted of 226 participants, in two groups that were matched for age, sex and ethnicity. Group 1 consisted of the 37 participants who opted to receive six sessions of counselling. Group 2 consisted of those who opted not to have counselling.

Commentary

The sample in Passage 8.4 is representative of the age group it set out to test, as it has taken care to ensure a good age distribution. It is not representative in terms of gender, as it includes far more women participants than men. It does not appear to be representative of people with different kinds of eye-sight, which would be important for this experiment.

In Passage 8.5, the sample is representative in terms of gender. Although the numbers of men and women are not exactly the same, the difference is small and not likely to be significant. The sample is not representative in terms of age. The survey does not state that the intention is to discover the preferences of people of a particular age range. It is not representative of people aged under 25 years or over 55 years. It is not clear whether the sample represented people from different economic, social, racial or geographical backgrounds.

In Passage 8.6, the two groups were 'matched' for age, sex and ethnicity. This means the sample was chosen so that a similar proportion of each of the two groups were men and women, from similar age groups and backgrounds. That is useful for ensuring the findings are not the result of differences in the composition of the groups. However, we do not know whether the samples were representative in terms of age, sex or ethnicity. For example, each group might consist entirely of white women aged 25–30. No details are given about whether the sample is representative in any other way, such as by type of job, geographical area or relationship with the deceased person. Most importantly, as only a small number of people received counselling, this is not a balanced sample.

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Certainty and probability

Certainty

Arguments cannot always be proved with 100 per cent certainty. Chapter 7 looked at how necessary and sufficient conditions may need to be met in order to prove a conclusion. In many circumstances, it is difficult to prove that sufficient conditions have been met, as there are so many exceptions to the rule.

Reducing uncertainty

Uncertainty is not very satisfying and does not help in decision-making. Academics aim to reduce uncertainty in a number of ways, including:

- selecting reputable sources, which are more likely to be credible;
- critically analysing the evidence, looking for the kinds of flaws outlined in previous chapters;
- calculating the level of probability;
- increasing the level of probability as far as they can.

Probability

When evaluating an argument, the audience needs to decide on a general level of probability. This means deciding whether the evidence is likely to be credible and authentic and, if so, whether the conclusions are likely to follow from the line of reasoning and its supporting evidence. Any conclusion may lie on a spectrum from impossible, to possible, to probable, through to certain. As Chapter 10 shows, academic writing is reluctant to express certainty, even when it has taken significant steps to ensure a highly probable finding.

Probability spectrum

Impossible — possible — probable — certain

Calculating the level of probability

The level of probability is related to the likelihood that something occurred because of the reasons given, compared with how far the outcome could have occurred by chance. If you throw a coin a hundred times so that it lands flat, there are only two options for the way it can fall, heads or tails. The probability is that the coin will land on heads about 50 times and tails about 50 times. This outcome is not certain, but it shouldn't surprise us if it occurs.

To win the lottery, the chances are much less probable. If there are 14 million options for the winning set of numbers, and you have only one set of numbers, the chances of your set being selected are one in 14 million.

Statistical formulae or specialist software can be used to calculate how likely it is that a particular outcome occurred by chance or coincidence. This can be expressed as 'The probability of this happening by chance is ...'

- less than one in 10;
- less than one in 100;
- less than one in 1000.

Expressing levels of probability

You are likely to see probability expressed as:

- p = <0.1 (less than a 1 in 10 chance that the outcome could have occurred by chance);
- p = <0.01 (less than a 1 in 100 chance);
- p = <0.001 (less than a 1 in 1000 chance);
- p = <0.0001 (less than a 1 in 10,000 chance).

The words 'The probability of this happening by chance' are abbreviated to 'p ='.
The words 'less than' are abbreviated to '<'.
The numbers are usually expressed as decimals smaller than the number 1.

The more zeros that follow the decimal point, the less likely it is that the outcome is a random one, that could have occurred just by chance.

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Sample sizes and statistical significance

Sample size

The larger the sample size, the greater the degree of probability. The smaller the sample size, the more likely it is that the outcome could have occurred by chance. The appropriate size of sample varies.

An appropriate sample size depends on:

- how essential it is to reduce the element of coincidence;
- whether it is a question of health and safety: a very small sample may suffice to prompt action;
- how necessary it is to be representative of many ages, backgrounds and circumstances;
- the funding available;
- how likely it is that a smaller sample will give reliable results.

Example

Clinical trials on a thousand volunteers indicate a success rate of over 95 per cent. Most patients made a complete recovery and, so far, few side effects have been identified. These trials offer hope of pain relief to a significant proportion of current patients.

Here, a thousand may seem like a significant number of people. However, that sample is unlikely to be representative of all those who may take the drug in future and of the circumstances which would ensure the drug was safe for them. If you needed to take the drug, you would be more reassured if you knew it had been tested on people who share similar circumstances to yourself, such as your blood group, age group, ethnic group, and people with similar allergies or medical conditions.

A study of heart attacks reported in *The Times* (31 August 2004) involved 29,000 participants in 52 countries over 10 years. Other medical surveys may be much smaller. Opinion polls are usually based on surveys of about 1000 people.

Statistical significance

When there are very small samples, such as surveys which include fewer than 16 people in each category, it is hard to say that the outcome wasn't just a coincidence. When the sample is small, or the differences between groups are small, we say that these are 'not statistically significant'.

Look for

When evaluating evidence, look out for expressions such as: 'the results are significant at $p = <0.0001$ (see p. 119). This shows the level of statistical significance: a one in 10,000 chance. The more zeros after the decimal point, the more reliable the finding and the less likely it is that the result occurred as a coincidence.

If, on the other hand, you see an expression such as 'the results were not statistically significant', this means that the results, or the differences between two things, may just be a coincidence.

Small samples

A small sample may be necessary:

- when surveying people who are unusual in some way, such as people who are exceptionally successful or with rare medical or neurological conditions;
- if it is dangerous to gain larger samples, such as when working at depth under the ocean, travelling into space, exposed to chemicals, or living with extreme sleep deprivation;
- in unusual circumstances, such as large numbers of multiple births.

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Over-generalisation

Generalisations are useful as they help us to see patterns and to make judgements more quickly when this is needed. However, a generalisation should be well founded, based on a reasonable sample.

An over-generalisation is one based on too small a sample to justify the generalisation.

Example

My first child slept through the night but the second one was a very poor sleeper. First-born children are better at getting to sleep than their younger brothers and sisters.

Here, the generalisation about first-born children is made on the basis of only two children. This is a database of two, which is a very small sample. If thousands of other first-born and second-born children showed the same sleeping pattern, then the generalisation might be valid. However, when only two children are involved, there is a large element of chance. The family next door might find that both their children sleep well.

Generalising from a single case

Generalising from a single case means forming a general conclusion on the basis of one instance. This is rarely acceptable.

Example

Some people say that calling people names because of the way they look is offensive. My friend is very overweight and people call him names for being fat. He says he doesn't mind as he finds horrible things to call back. This shows there is no harm in calling people names as they can just retaliate if they want to.

Just because one person appears not to mind offensive language, this does not mean that all other people will react in the same way.

An exception can disprove a rule

However, some generalisations can be made on the basis of a single instance, and be accurate. This is true, for example, when a general rule is already in existence, such as that objects, when dropped, will fall towards the ground. A single case that contradicts that rule would show that the generalisation wasn't universally true: for example, a helium balloon would rise. In such cases, the rule then has to be reconsidered and refined to account for the exception. Much of science and law has progressed by refinements to rules so that they are more accurate about the exact circumstances in which they apply.

Example

Clinical trials showed the drug to be very successful. However, this patient had a severe allergic reaction to the new drug. This means that doctors need to be aware that some people may react negatively to the drug.

Here, a single example is sufficient to necessitate a carefully worded generalisation. Over time, as more exceptions emerge, the generalisation will change to become more precise and accurate.

Example

This drug can create a severe allergic reaction in asthma sufferers and people taking the drug BXR2.

These examples illustrate that a small sample, even a single example, can disprove a theory based on a much larger sample. A single example can disprove a theory or rule. When this happens, the rule or theory has to be re-examined and reformulated to take account of the exception. However, it is also important to bear in mind that a generalisation means 'most of the time' and may be useful in helping to understand a situation despite the exceptions.

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Controlling for variables

What are 'variables'?

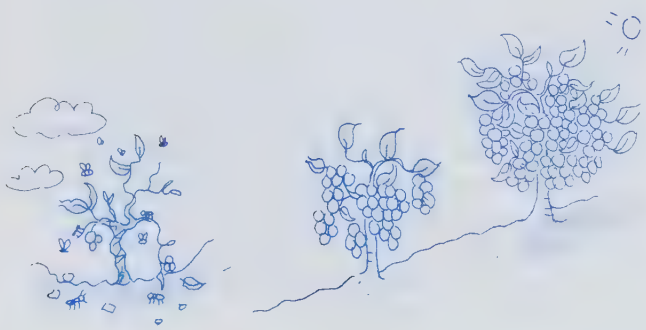
'Variables' are all those circumstances that might affect the outcome in intended or unintended ways. When evaluating evidence, it is useful to consider whether the author has taken steps to identify potential unintended variables and to prevent them affecting the outcome of the research.

Example

During trials in South Africa, the yield of grapes on a new vine was twice the usual level for red grapes. The yield produced twice the volume of wine. Cuttings of the vine were transported to California to an area with similar soil and rainfall. However, the vine didn't produce the same yields in California.

In this case, the producers controlled for some variables such as soil and rainfall, but these were not enough. In order to find out why the vine yielded more in one area than the other, the producers would need to grow it under controlled conditions, changing just one aspect of the conditions each time, until they isolated the special conditions that doubled the yield. Such variables might include:

- the total hours of daylight available;
- minerals and trace elements in the soil that had been overlooked;
- when the rainfall occurs during the growing process;
- the slope of the land;
- other plants growing nearby and their effect on insects and pests.



When you read research reports or journal articles, check what steps were taken to control for variables. In an article, this will be found in the section on methods. If the research doesn't take steps to control for variables, then the results may have been attributed to the wrong cause.

Control groups

One way of checking that the results support the conclusion is by using a control group. The control group is treated differently from the experimental group and provides a point of reference or comparison. If an experiment was testing for sleep deprivation, the experimental group might be denied sleep for 60 hours, whereas the control group might be allowed to sleep as usual.

Example

A company claims that its SuperVeg juice reduces the incidence of colds and flu. 100 people drink a bottle of SuperVeg every day for a year, and a control group, also of 100 people, is given flavoured water in a SuperVeg bottle.

The flavoured water is known as a 'placebo'. Participants should not know which group they are in, as that can influence their response: participants might wish either to help the experiment along or to sabotage it.

Activity **Controlling for variables**

Look again at passages 8.4–8.6 (p. 118). For each example, identify what kinds of control groups or controlled conditions are needed.

Answers: see p. 297.

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Facts and opinions

Opinion

An opinion is a belief that is believed to be true, but which is not based on proof or substantial evidence. An opinion may be a personal point of view or held by a large number of people, even if it runs contrary to the evidence.

Example

Opinions



Facts

Facts are basically items of information that can be checked and proved through experience, direct observation, testing or comparison against evidence. However, as knowledge of an area increases, facts can later be disproved. A fact checked against reputable evidence generally carries more weight than personal opinion, but that doesn't mean it is true.

Example

Facts

The coroner stated that the time of death was between 2 a.m. and 4 a.m. in the morning. The body was found at 6.30 a.m. by the cook. The footman reports that there were six people in the house overnight. The butler reports that four other people have keys and could have entered the house and left again before 6.30 a.m.

The facts in the example above are:

- The time of death, as given by the coroner. That is likely to be reliable.
- The time the body was found by the cook; however, somebody else could have found the body earlier and remained silent.
- The footman reported certain information.
- The butler reported certain information.

The details of the reports by the footman and the butler may not be facts: these could be personal opinions, or they may have been lying.

False appeals to the 'facts'

People's opinions can vary about what is a fact and what is an opinion.

Example

The butler was in the house all night. His employer was murdered during the night. The butler says he was a loyal servant but maybe he wasn't. I think he was lying and that he had some sort of vendetta against his employer. The facts say he is the murderer.

In this case, the facts appear to be:

- The butler was in the house all night.
- His employer was murdered during the night.
- The butler says he was a loyal servant.

These do not prove that the butler was either a loyal servant or a murderer: either or even both could be true. However, note that the author states his opinion, that the butler is the murderer, as if it were a fact.

Expert opinion

'Expert opinion' is based on specialist knowledge, usually acquired over time or based on research or direct experience. It is often used in court to help a judge or jury to understand the issues. Experts are often asked for their own judgements. This, in itself, is not taken as 'proof', as even experts can be wrong.

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Eye-witness testimony

Eye-witness testimony may be useful in a number of circumstances, such as:

- people who saw or experienced accidents, crime and disasters first-hand;
- people who lived through historic events including the more distant past;
- clients' accounts of experiences and/or services received;
- patients' accounts of their experiences.

Levels of accuracy

Untruth

Personal testimonies can provide invaluable evidence, but they are not always accurate.

Interviewees may not reveal the true case because they:

- may want to be helpful, so say what they think the interviewer wants to hear;
- may not like the interviewer;
- may be trying to protect somebody;
- may not remember anything, but like the attention of being interviewed;
- may have a vested interest in the outcome, so benefit from concealing the truth;
- may be being bullied or intimidated and be scared of speaking out;
- may have promised to keep a secret.

If using interviews to gather evidence, remember that the interviewee may have complex motivations for presenting the picture that they give.

Lack of expertise and insider knowledge

The witness may lack information such as expert knowledge or details of why something was taking place which would enable them to make sense of what they saw. They may have seen a camera crew filming a fight in the street as they passed by one afternoon. However, they would not necessarily know whether they were watching a real fight at which a camera crew happened to attend, or whether the fight was staged deliberately for a TV drama. It may also be the case that the interviewee misunderstood what was asked of them.

The limits of memory

Loftus, in *Eyewitness Testimony* (1979), demonstrated, for legal use, how unreliable the memory can be. In one experiment, participants were shown a film of an accident and some were then asked how fast a white car was travelling when it passed a barn. A week later, 17 per cent of those who had been asked this question reported that they had seen a barn in the film, even though there had been no barn. This compared with only 3 per cent of the other viewers. Common memory mistakes include the following.

- Errors in perception: making mistakes about what you have seen and heard.
- Errors in interpretation: misinterpreting what you have seen.
- Errors of retention: simply forgetting.
- Errors of recall: remembering the event inaccurately. Our memory may be altered by going over the event in our mind, discussing it, hearing other people's accounts, or hearing about similar events.
- Composite memories: our brain can blend aspects from several events into one, without us being aware this is happening.

Corroborating sources

It is usually necessary to find other sources of information that corroborate a witness testimony, for example:

- official records from the time;
- other witness testimony;
- TV footage of the events;
- newspaper, police, social work or court records;
- photographs taken at the time;
- information about similar events that happened elsewhere but which might throw light on the event being considered.

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Triangulation

What is triangulation?

Triangulation means checking and comparing different sets of evidence against each other, to see whether they support and complement each other, or whether they contradict each other. This is especially important when relying on first-hand accounts.

Triangulation is something that most of us tend to do in everyday contexts to check whether something is true.

Example

John told his mother that his sister Mary hit him. John was crying and called Mary a bully.

John may or may not be telling the truth. Before his mother took action, she is likely to have triangulated the evidence by:

- listening to Mary's side of the story;
- looking for evidence that John was hit;
- considering John's and Mary's usual ways of recounting events;
- checking for alternative explanations.

Example

A head teacher says that a school's record of achievement is better than ever, that most pupils succeed, and that this is because of improvements in teaching at the school.

This statement could be triangulated with:

- published government records over several years to check for general improvement over time at all schools;
- comparing the school's achievement rates with the average for all schools;
- comparing the school's achievement rates with those of schools of a similar type. For example, if the school was situated in an area of high economic deprivation, it is likely to be more appropriate to compare it with schools in similar areas.

You might also wish to investigate whether there are any other reasons for changes to the school's rates of achievement. For example, if the school had started to set difficult entry tests, this might have attracted a very different type of pupil to the school and excluded those less likely to achieve. The improved achievement rates might be because the pupils were different and not because of improvements in teaching.

Comparing like with like

When triangulating information, it is important to check that the different sources used are also referring to the same subject and interpreting words in the same way. If not, you may not be comparing like with like. For example, the head teacher in the example may be talking about sports achievement, not academic, so this would require triangulation with a different set of sources, such as sports records not government records.

Activity  **Triangulation**

What kinds of evidence would be needed to triangulate the following sources?

(1) A person at the bus stop mentioning that cheap tickets will be available at the door, on the night, to see a band that you really like.

(2) A report by a car manufacturer that new brakes fitted in their latest model of car were safer than other brakes available.

(3) A chapter in a book that argued that, in the past, there were very severe legal penalties for begging.

Answers: see p. 297–8.

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Evaluating a body of evidence

When you are researching a subject, or producing an academic assignment, you are likely to refer to many sources of evidence. However, you are not likely to evaluate all of these in the same way.

You can evaluate some sources ...

- *by browsing*, to evaluate whether they are sufficiently relevant to your research topic and sufficiently reputable for the level of research;
- *by focusing* on the most relevant items, evaluating how these support specific aspects of your line of reasoning;
- *by selecting and carefully evaluating* a relatively small number of key sources, weighing the arguments, and looking for flaws and gaps in the evidence;
- *by comparing and contrasting* different sources, checking for inconsistencies.

The following activity gives you the opportunity to work with a set of short texts to practise discriminating appropriately between them. These texts are also used as the basis for further activities in Chapters 9 and 11.

Activity **Identifying reputable sources**

Read through the texts on pages 229–33.

(a) Identify which are the most reputable sources of evidence. Categorise these as:
- very reputable;
- fairly trustworthy;
- little authority

(b) For which texts might the authors have a vested interest in the outcome?

(c) Which are the most reliable sources for indicating what internet users believe about file sharing?

Answers: see p. 298.

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Summary

This chapter has looked at some key concepts in evaluating evidence from the point of view of both conducting your own projects, and examining the evidence used by other people.

If you are conducting your own research, whether for a project, report or essay, you will need to ensure that you collect and select the most appropriate evidence, and subject it to critical scrutiny. This chapter introduced the principles of making a literature search. It looked at ways of whittling down a large number of potential sources of evidence to a manageable number for deeper scrutiny. It also showed how to recognise the difference between primary and secondary sources.

When using secondary sources as evidence to support your own arguments, you need to be able to understand the evidence base used by those sources and have criteria you can use to evaluate it. For example, you need to be alert to whether the evidence is what it is claimed to be, checking that it is authentic, accurate, reliable and up-to-date. You also need to understand its significance in terms of probability and the methods taken to ensure reliable findings. When first starting to analyse materials critically, it can seem as though there are a great many aspects to check. However, many of these, such as selecting reputable sources, become automatic. Others are useful to hold lightly in mind whenever you hear or read an argument. It is often useful, and sometimes necessary, to go back to the original sources or published versions of these, to check for accuracy. If sources are well referenced, this makes the task of checking for details much easier.

The earlier section of the chapter looked at ways of analysing individual sources to check for aspects such as their reliability and validity. Later sections of the chapter looked at using one source to check another. Cross-comparison, or triangulation, is something that many of us do naturally in our everyday lives. However, many people take at face value what they read or hear in one source, without checking how this compares with what other sources say. Comparing materials doesn't necessarily lead to the truth, but it often shows where there are different points of view and therefore room for error and further investigation.

You will find that some of the concepts introduced in this chapter will be more relevant for your subject than others. Each academic subject has well-established research methods that develop specialist skills for analysing source materials. Some will use:

- carbon-dating to check the age of materials;
- knowledge of medieval Latin and allegory in order to read and interpret original documents;
- advanced skills in semiotics in order to interpret the meaning of texts;
- specialist equipment to make precise measurements or detect micro-organisms;
- statistical approaches and formulae to analyse the kinds of data relevant to your subject.

Such advanced skills are likely to be taught within the subject. However, for most subjects, the basic skills in critical thinking will also apply.

Information about the sources

Lane, H. (1984) *When the Mind Hears: A History of Deaf People and their Language* (Cambridge: Cambridge University Press).

Loftus, E.F. (1979) *Eyewitness Testimony* (Cambridge, MA: Harvard University Press).

Miles, S. (1988) *British Sign Language: A Beginner's Guide* (London: BBC Books).

Palmer, T. (2004) *Perilous Planet Earth: Catastrophes and Catastrophism through the Ages* (Cambridge: Cambridge University Press).

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Chapter 9

Critical reading and note-making

Critical selection, interpretation and noting of source material

Learning outcomes

This chapter offers you opportunities to:

- develop strategies for reading selectively
- understand the relation of theory to argument
- categorise arguments and theories
- check whether interpretations of texts are accurate
- develop strategies for selective and critical note-making

Introduction

Although critical thinking can be used in any context, it is likely that you will apply it most when using written materials. The material presented in previous chapters is relevant to critical reading. This chapter focuses on applying critical thinking skills when reading for a specific purpose, such as writing a report or assignment. It looks at issues such as:

- identifying theoretical perspectives;
- categorising information to assist with its selective use;
- using a critical approach to note-making when reading.

Critical reading is different from other kinds of reading such as skimming or scanning text. The latter are useful strategies for locating information in a text and to develop a general feel for a subject. These strategies offer a useful starting place, although used on their own, they would usually result in a more superficial reading of the material.

Critical reading requires you to focus your attention much more closely on certain parts of a written text, holding other information in mind. As it requires analysis, reflection, evaluation and making judgements, it usually involves slower reading than that used for recreational reading or for gaining general background information. As you develop your critical reading abilities, these reading skills will become faster and more accurate.

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Preparing for critical reading

It is not usually easy to make sense of any information taken out of context. When reading new material, some basic preparation can help you to:

- see how the main argument fits together;
- better remember the overall argument;
- better comprehend specific pieces of information;
- recognise how reasons and evidence contribute to the main argument.

The following sections offer suggestions on actions you can take to orientate yourself to a text, in order to facilitate critical reading.

Books

Preliminary skim



First, skim through the book to get a feel for what it contains. Glancing through as you flick the pages a few times, or scanning each page quickly in turn, can give you an initial impression of what the book is about and where relevant information may be located.

Scan the introduction

Check whether the introduction indicates the author's position or refers to the overall argument. Such information can direct you to the most relevant chapters and help you to make sense of detailed information presented in these.

Scan the final chapter

Look at any conclusions drawn at the end of the book. Check whether the final chapter sums up the argument, reasoning and evidence. If so, this is invaluable for keeping track of the line of reasoning when reading about the more detailed evidence in other chapters.

Scan beginnings and ends of chapters

Scan the introductions and final sections of relevant chapters: these are likely to orientate your thinking to the material in the chapter.

Articles

- Browse the abstract to see if the article looks relevant.
- If it does, read the abstract slowly, to identify the main argument.
- If the article is about a research project, the research hypotheses sum up what the author is trying to prove. The results will tell you what they found. The discussion indicates what the author considers to be significant about the research and its findings.
- Use the abstract to locate the most relevant information for you. Decide whether you need to know more about the methods used, the results, the discussion of the results, or the recommendations, depending on your purpose.

Find the argument

Once you have worked quickly to locate where the information is in general terms, apply the critical thinking methods covered in earlier chapters in order to identify the arguments:

- Identify the author's position: what does the text want you to do, think, accept or believe?
- Look for sets of reasons that are used to support conclusions.

Once you have located the argument, you are likely to need to read more slowly and carefully, applying further critical thinking strategies outlined in this chapter and earlier in the book. In particular, you would need to look out for the quality of the evidence used to substantiate the arguments presented.

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Identifying the theoretical perspective

What is a theory?

A theory is a set of ideas that helps to explain why something happens or happened in a particular way, and to predict likely outcomes in the future. Theories are based on evidence and reasoning, but have not yet been proved conclusively.

Everyday and academic use of 'theory'

We use the term 'theory' in everyday language to suggest we don't know yet, for certain, either the reasons or the outcomes.

Examples

The flight still hasn't been announced. My theory is that a storm is brewing so they think they can't take off.

Everyday use of the word tends to be an expression of opinion, but it shares the characteristics of academic theory in being:

- an attempt to provide an explanation, or a prediction of likely outcomes;
- an idea, or set of abstract ideas, that haven't been fully proved;
- based on the facts as far as they are known at the time, and acknowledging there is still more to find out.

Knowing the theory helps fill the gaps

Most things that we do are based on some kind of theory, but we are not always aware that our opinions fit a theoretical perspective. In Chapter 6, we saw that what we say or write often contains unstated assumptions – which may be unrecognised theories. If we can identify an author's theoretical perspective, we are in a better position to recognise gaps in the reasoning as well as unstated assumptions.



Theory in research and academic life

In professional research and academic thinking, a theory is usually an elaborated system, or 'school', of ideas, based on critical analyses of previous theories and research. Much research sets out to test or further refine existing theories so that they are more useful in providing explanations, and for creating models for future action.

Finding the theoretical position

In the best research and texts, the theoretical position will be stated by the author in an explicit way to assist the reader. In books, this is usually outlined in an early section, or at the beginning of chapters. In articles, reports, dissertations and theses, the theoretical position will be indicated by the following:

- The research hypothesis: this should be stated near the beginning of the research and provides the key theoretical position that the research sets out to prove.
- The literature that has been selected for the literature search: authors' analysis of this should draw out the theories which have influenced the research.

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Arguments can be based on theories

A theory may be used as the basis of an argument.

Example

Marx's theory of economics argues that wealth will become concentrated into a few hands. This research project is based on an interpretation of Marx's theory, and argues that although the denationalisation of public services in Britain led to more companies being set up in the short term, over a few decades, mergers and buy-outs have resulted in many smaller companies closing. As a result, the wealth of those industries is now in the possession of a small number of 'super-companies'. The research hypothesis is that after three decades, 75 per cent of the wealth of former British nationalised industries will, in each case, be in the hands of three or fewer super-companies.

In the example above, the main argument is that after a few decades, industries that were once nationalised, but were later sold to private companies, will become part of a few 'super-companies'. The author is explicit that the argument is based on an interpretation of a particular economic theory. Here, the theory is used to develop the research hypothesis.

The inclusion of numbers and proportions helps to make a general theory more specific and measurable. However, the general argument and theory could be valid, even though the specifics differ, providing that the trend was clearly in the direction predicted.

Theories as arguments

Theories can also be arguments in their own right if they offer reasons and conclusions and attempt to persuade. However, you may find that when theories are used as the basis of an argument, as in the example above, the author refers only to the conclusions or key aspects of the theory. To examine the line of reasoning behind the theory, it may be necessary to return to the original text rather than using second-hand accounts.

An argument is not necessarily a theory

Note that arguments are not always theories. In the example below, the argument for going into town is supported by two reasons, but does not represent a theory.

Example

I know you are keen to return home quickly, but it would be a good idea to go to the shops first. We need to buy a present for Serina's birthday. We also need to get some food for tonight.

Activity Identifying theory

Identify which of the texts on pages 229–33 have an explicit (openly stated) theoretical position.

State what the theoretical position is in each case.

Answers: see p. 299.

Subject-specific schools of thought

There will be specific theories, usually organised into schools of thought based around a few key researchers or approaches, for your own subjects. These might be clustered around broad theoretical approaches such as: nativism, humanism, chaos, catastrophism, functionalism, psychodynamics, systems, constructivism, Marxism, feminism, postmodernism and so on.

Activity Schools of thought

What are the main schools of thought for your own subject discipline and areas of interest?

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Critical choices

Research tasks, including reading for reports and assignments, can require us to cover a great deal of information. We can only make active use of a proportion of what we read, but it may seem that everything is useful and interesting.

Critical thinking requires us to make decisions about:

- where to allocate available reading time;
- where to focus our critical thinking;
- what to note for future reference;
- what material to use in our own report or assignment, and what to leave out.

Critical choices involve selection, and selection is made easier if we are skilled at categorising information. Practice in categorising information was provided in Chapter 2.

The importance of categorising information

It is easier to make critical choices when we have organised information not simply in files, but within our thinking. Categorising information is an essential process that helps us to recognise links between different kinds of information. This enables us to:

- compare information more easily;
- contrast information more easily;
- refer to sets of information as a group, so that our account is more succinct.

Categorising theory

We saw, above, that identifying the theoretical position helps us to fill in gaps in a line of reasoning. If we can categorise texts according to their theoretical position, we will be better able to:

- sort the information required for our analysis of the literature;
- track how one piece of research builds on previous research;
- better understand why further research into a subject has been undertaken, seeing how it fits into a bigger picture. Often, a piece of research can only examine part of the picture.
- Group information under headings that help to clarify our understanding. This also helps us to remember the information.

Generic types of theory

There are some generic headings that are useful as points of reference when starting to group information. It is worth checking whether theories or arguments are, primarily, one or more of the following:

- *aesthetic*: related to an appreciation of art;
- *cultural*: related to the ideas, customs and artefacts of a particular society;
- *economic*: related to an economy;
- *ethical*: a question of right and wrong;
- *financial*: considerations of money;
- *legal*: related to the law; what the law says;
- *historical*: resulting from past circumstances;
- *humanitarian*: with the interests of mankind at heart;
- *philanthropic*: acts of kindness to others;
- *philosophical*: related to the study of knowledge;
- *political*: related to government or state;
- *scientific*: resulting from a systematic and/or experimental approach that can be repeated;
- *sociological*: related to the development or organisation of human society;
- *sophistical*: arguments that seem clever but are misleading.

Activity **Categorising arguments**

Read through the texts on pages 229–33. Each text contains one or more types of argument. Categorise these using the generic themes listed above. More than one may apply to each text, or 'none of these' might apply.

Answers: see p. 299.

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Reading style and accuracy

Accurate interpretation is particularly important to critical thinking. Donaldson (1978) found that people often get questions wrong because they do not adhere closely enough to the detail of what is asked or stated.

Incorrect interpretations can arise because reading is either over-focused on small details or it pays insufficient attention to details. Some common mistakes are:

- *Over-focused reading*: the reading is too slow, focusing excessively on individual words and sections of the text. Although close reading is a necessary part of critical reading, it is also important to interpret specific details in the wider context of the argument and the theoretical perspective.
- *Insufficient focus*: the reading is too superficial, taking in the big picture but lacking a sense of how the main theories and arguments are supported by specific details and evidence.
- *Insufficient attention to the exact wording*: missing out essential words such as 'not', or not following the exact sequence closely.
- *Failing to draw out correctly the implications* of what is stated.

It follows that, in order to interpret texts accurately, it helps to vary the focus of attention when reading, alternating between:

- the big picture and the fine detail;
- a consideration of the exact words and unstated implications and assumptions.

Activity

Read the interpretations given in the passages of specific texts on pages 229–33. In each case, decide whether the passage:

A makes an accurate interpretation of the writer's overall argument;

B misinterprets the writer's position.

Give reasons for your response, identifying the overall argument.

Passage 9.1

(Interpretation of Text 1, p. 229)
The author is a true artist who is offering a service to smaller artists who cannot find distributors.

Passage 9.2

(Interpretation of Text 2, p. 229)
The author argues that as giving garden cuttings is regarded as acceptable and little concern is shown for royalty issues, then downloading music without paying should also be regarded as acceptable.

Passage 9.3

(Interpretation of Text 3, p. 229)
Piracy is not usually acceptable and most customers should be prepared to go without an item if they are not willing to pay for it.

Passage 9.4

(Interpretation of Text 6, p. 230)
When people make free copies of music they put the future of distributors of independent artists at risk.

Passage 9.5

(Interpretation of Text 7, p. 230)
This argues that Plants Breeders are only likely to take action against large companies, so the important issue for gardeners is that they are safe from prosecution.

Passage 9.6

(Interpretation of Text 10, p. 231)
Individuals should stand up for what they believe is right and stop obeying the law, as it is undemocratic.

Answers: see p. 299.

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Making notes to support critical reading

Why make notes?

Note-making is a good idea. It has several benefits over simply reading without making notes:

- If done properly, it breaks up a continuous reading task into many shorter reading sessions alternated with note-making. This rests the eyes and the parts of the brain involved in reading. This is especially useful given the intense reading activity used for critical reading.
- Writing involves the motor memory, making it easier to remember information.
- Many people find it easier to recall information that is written in their own handwriting.
- Selecting what to write, rather than writing everything, means greater interaction with the material, which helps us to recall it in the future.
- Making notes draws together the information that is relevant on the subject, so you have less to read over than all the material contained in the various source materials.
- You can make notes on a copy of the text if it is your own copy, but this doesn't help draw the key ideas into one place.
- The thought that goes into note making, and your particular style and choices, help to make the subject material your own.

How do you make notes to support critical reading?

The notes you make should support your main purpose. Avoid making notes on related topics just because they are interesting or might be useful one day. It is possible to write notes to fulfil several different purposes, such as to support a current project and to contribute towards a future project or assignment. If you do this, either use separate sets of notes for each project, or use clear headings in your notes to help you find what you need easily for each.

Reflect as you note

It is worth making a conscious effort to reflect on what you have read.

- What does this really mean?
- Do the reasons support the argument?
- Is there any supporting evidence?
- Does this match what I know about the subject already?
- Does it fit what other people say about the subject?
- Is this relevant and useful to my current purpose?
- How does this add to previous research on the subject?
- Are there any flaws in this?



TIP

Read without a pen in your hand. This helps to avoid writing lots of unnecessary notes that you haven't thought through, and to avoid accidental plagiarism.

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Making notes for analysing an argument

If your main purpose is to keep notes to analyse an argument, use headings or a pro-forma such as that on page 137, to note the following:

- Details for finding the source again easily.
- The author's position/theoretical stance.
- The main argument, or hypothesis.
- The conclusion(s).
- A list of the reasons used to support the conclusion. Number these. If the author repeats a reason in different words, make sure you include it only once on your list.
- Your evaluations of the strengths and weaknesses of the line of reasoning and supporting evidence.

Notes for assignments and reports

When making notes from a book, there is a danger of losing critical focus by taking down information indiscriminately, rather than selecting the most relevant points.

TIP

If you like to make lots of notes about facts and supporting details, keep these on separate sheets from your notes for critical analysis, or write them on the reverse side. If your critical analysis pages remain empty and your background information pages begin to mount up, this will alert you that you are neglecting to evaluate the information for relevance and to select the most salient points. It may also indicate that you have slipped into copying from the text.

The pro-forma on page 138 provides a model for critical note-making when reading books. It may not suit all your purposes. However, it is structured so that there is, deliberately, very little space for general background. It is rare that you can use more than very minimal background information on any one source material for either academic or professional purposes.

Note-making when reading journal articles

The main difference in note-making when reading from research articles is that you are more likely to make a close analysis of the particular contribution that the research findings or methodology make towards advancing knowledge within the subject area. Such articles tend to be based on a single piece of research and you may be especially interested in the methodology and the discussion of findings. The pro-forma offered on page 139 puts the emphasis on your analysis rather than on background information.

Choose quotations carefully

Use few quotations and keep them short

Avoid long quotations as they eat into the word limit without providing any additional marks. Select a few short quotations that:

- in a secondary source, sum up a point well in a few words;
- in a primary source, provide direct evidence for your argument;
- are relevant and the best. Use sparingly.

Make quotations stand out in your notes

Develop the habit of using a particular coloured pen, such as red, blue or green, for any copied text such as quotations. This will make it immediately obvious to you, when you read your notes at a later date, what you have copied and what are your own words and ideas.

Note the source of quotations

Note down exactly where the quotation comes from. See pp. 144–5.

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Concise critical notes: Analysing an argument

Names of author(s)/source			
Title of book/programme			
Website address		Date downloaded	
Date and/or time		Edition	
Publisher/channel		Place published	
Volume of journal		Issue	
Author's position/ theoretical position?			
Essential background information			
Overall argument or hypothesis			
Conclusion			
Supporting reasons	1	5	
	2	6	
	3	7	
	4	8	
Strengths of the line of reasoning and supporting evidence			
Flaws in the argument and gaps or other weaknesses in the argument and supporting evidence			

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Concise critical notes: Books

Names of author(s)			
Full title of book			
Author of chapter			
Chapter title			
Year published		Edition	
Publisher		Place published	
Theoretical position or type of theory?			
Essential background information			
Key arguments			
Reasons and evidence to support the arguments			
Strengths of the arguments			
Weaknesses in the arguments			
Comparison or contrast with other sources			

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Concise critical notes: Articles and papers

Names of author(s)			
Full title of article			
Full title of journal			
Year published		Month	
Volume number		Issue number	
Hypotheses: What is the paper setting out to prove? Are research hypotheses supported?			
What is the theoretical position underlying the research? Type of theory?			
What is the key literature used as background to the article or paper?			
Which research methods are used?			
What kind of sample is used?			
Key results			
Key conclusions or recommendations			
Strengths of the research: • How does it advance our understanding of the subject or how to research it? • Are there appropriate hypotheses, methods to test the hypotheses, sample sizes or types, controls for variables, recommendations?			
Weaknesses of the research: • In what ways is it limited? When and where would it not apply? • What, if any, are the flaws in the research, in the hypotheses, research design and methods, sample size and type, conclusions drawn on the basis of the results?			

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Critical selection when note-making

Activity

Below, are notes made on Texts 1–11 (pp. 229–231). The purpose of the notes is for use in a report entitled:

'Unfair treatment: The law only seems to apply to business these days.' Discuss.

Look through the sample notes that were made and underline any sections that are relevant to the report. Give reasons why the notes are relevant, and comment on whether the notes are made in the note-maker's own words.

Then read the commentary and compare your answers.
NB This activity is focusing on the content of the notes, not their layout.

Sample notes for 'Unfair treatment: The law only seems to apply to business these days.' Discuss.

Evidence that supports the statement

- Legal proceedings are usually only instigated against businesses, not individuals. This is true for file sharing via the internet, when businesses sell well below market prices (Spratt, 2014, Text 4) and for plant breeding (Johl, 2016, Text 7). BUT: this doesn't mean that the law only applies to big business, just that it is likely to be applied unevenly. This may appear to be unfair to big business. NB Johl (Text 7): the combined effect of plant sharing is a large financial loss to plant breeders – so business is targeted just because it is easy.

Evidence that contradicts the statement

- Big publishers are only interested in music that has broad appeal because they hope to make large profits (Text 1).
- Cuttle (2015): Publishers, including big businesses, can choose the price at which they sell. NB: this can be much higher than it costs to make the item, so in this respect the law supports business (Text 3).
- The law is scrambled together over time and is often contradictory. There is very little debate on what we want as our concept of justice (Piaskin, 1986, Text 10).
- Isn't it mainly big businesses that can afford to use the law against copying – not small artists?

Commentary on the notes

The notes on Text 1 are about big business but it is not clear why the note-maker considers these relevant to the question. Furthermore, this is not a reputable source for a report.

The notes on Text 10 are about the law in general, but not about business in particular. It is not clear why the note-maker considers these notes relevant to the discussion question.

The notes on Texts 1 and 10 are copied almost word for word from the texts, showing no critical selection. If these were reproduced in a report or assignment, and it was discovered, it would be regarded as plagiarism (unacceptable copying).

The notes on Texts 3, 4 and 7 are better as they are relevant, written in the note-maker's own words, along with reflection that could be used to make a relevant point in the report.

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Activity: Critical selection – Notes A

Activity

Look at the two sets of notes, A and B, which are related to the texts on pages 229–33. In each case, decide:

- Has the note-maker selected information that is relevant for the purpose given?
- Have they selected the most relevant information?

Notes A. Purpose: 'The internet is corroding moral values.' Discuss.

Points For: Internet corrodes moral values

(1) Text 3 Cuttle (2015) argues that people who make illegal copies from the internet try to rationalise this rather than see it as wrong, using arguments such as 'everybody else does it'.

(2) Text 1 (Carla, 2016): Comments made by internet users support Cuttle as they use rationalisations:

e.g. 'it isn't really stealing to file share on the internet'.

Also, sending and accepting copies without paying for them is: 'performing a useful service' to the arts and individual artists.

(3) Text 1. NB: as music can be downloaded off the internet for free, it provides a temptation for people like Carla and encourages them to look for excuses to justify taking without asking.

(4) Text 9 (KAZ, 2015): Internet user's flawed reasoning to defend non-payment, such as that if you are unlikely to be charged, 'there isn't a crime'.

(5) Texts 4 and 7: The law is mainly used to prosecute other businesses, and not people who only make a few copies of files such as for friends.

(6) Text 8: Moral decay isn't just a few people on the internet: even professors now offer flawed reasoning in favour of taking without payment.

e.g. Lee (2015) argues in favour of taking material for free from the internet just because people were not caught and punished for copying off the radio ('nobody bothers about this ...').

Points Against: Internet does not corrode values

(1) Text 2 (Potter, 2016) draws a comparison between downloading from the internet for free and giving away plant cuttings.

(2) If we can compare making cuttings of plants with downloading from the internet, and both are wrong, then the internet is only offering a different way of expressing similar values, not 'corroding' them.

e.g. Text 7 (Johl, 2016) shows plant breeders also suffer from the practice of giving free cuttings. Plant cutting preceded the internet – so these values can't be blamed on the internet.

See commentary on p. 143.

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Activity: Critical selection – Notes B

Notes B. Purpose: essay 'Stealing is always wrong.' Discuss this view.

Stealing is wrong

(1) Text 6: Yes, because we don't always know who all the victims are. Kahliney (2013): small music distributors can be severely affected by even a few people losing them royalties, when they copy music to their friends. People's livelihood depends on fair-trading.

(2) And Text 7 (Johl, 2016): Plant breeding is very expensive – so even small royalties which build up over time help producers to invest in new varieties.

(3) Text 3 (Cuttle, 2015): Producers of all media are entitled by law to recoup the costs of their labour or outlay.

Arguments used to support stealing

(1) Text 1: It's other people's fault, such as large publishers that are only interested in music that will give them large profit margins (but see Text 3 above).

(2) Text 1: It can provide a useful service, e.g. sharing music from the internet for free helps bring innovative and radical music to more people, which is better for true artists who want their music to reach as many people as possible. (BUT: Text 6: Small distributors are not necessarily helped.)

(3) Text 2: It's acceptable if you can get away with it – e.g. Ivan Potter (2016): Plants and file sharing.

(4) Text 8 also implies that if nobody gets caught or prosecuted, such as for copying from the radio, then stealing doesn't matter. BUT: stealing isn't defined by whether you get caught.

Where stealing might be acceptable?

(1) Text 11 (Soyinka, 2013): No, there can be instances when people don't realise they are stealing, such as students plagiarising – e.g. because the rules are complicated. BUT: ignorance of the 'law' is not accepted as an excuse.

(2) Text 12 (Ebo et al., 2014): Research shows people's behaviour is affected by how easy it is to act in an ethical way. The research hypotheses were that most young people who downloaded music for free pay to download music if this is made easy, and people are less willing to pay for music if they are high earners. The participants were 1206 people aged 15–25, matched for age, sex, and ethnic background across groups and conditions. An advertisement for an alternative website where the music could be downloaded for free appeared when the participant was on line. Damblin and Toshima (1986) used a sample of 200 senior citizens and found significant differences in ethical behaviour depending on medical conditions. Several research studies show external conditions can have more impact on behaviour than has ethical understanding (Singh et al., 1991; Colby, 1994; Miah and Brauer, 1997).

Issues of right and wrong are not clear cut

(1) Text 10 (Piaskin, 1986): Right and wrong are 'more properly regarded as dilemmas'. There may be occasions when stealing is wrong in itself, but less wrong than not stealing?

- A person could be stealing and yet not acting in an immoral or unethical way, e.g. to save a life; OR
- 'stealing within the law', which would then be an ethical issue, a matter of conscience, not law.

See commentary on p. 143.

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Commentary on critical selection activity

Notes A. (p. 141) Collated for 'The internet is corroding moral values.' Discuss.

In this case, most of the notes taken are relevant to the subject.

The notes on Text 3 are relevant as they point out that people look for reasons to justify taking without paying. Searching for reasons to justify an act that may not be right is known as 'rationalisation'.

The note-maker makes a good selection of brief quotations from primary sources, Texts 1 and 9, to illustrate the point. As the quotations are underlined, these stand out properly from the other notes, showing immediately that the words are copied directly from elsewhere. Although personal websites such as these are not normally reputable authorities as secondary sources, for this question they are relevant sources of primary evidence, illustrating what some ordinary internet users are saying. It is evident that the note-maker is using Text 8 to provide a contrasting source that also argues for non-payment, which is also relevant.

The notes benefit from being divided into points 'for' and 'against' the argument, but this means that more complex points are not included. For example, when considering values, it would have been useful to consider Text 5 and the issues it raises about the differences between what people say and do. The author argues as if downloading for free is acceptable, although he does pay for the music he downloads himself.

One strong feature of the notes is that they show that the note-maker is thinking whilst reading, and jotting down relevant reflections.

The main weakness of the notes is that they do not select the most relevant material:

- They do not refer to all the relevant sources, such as Texts 10 and 12.
- The note-maker gives no indication why notes on Texts 4 and 7 are relevant to the topic (see note 5, under 'Points For').

Notes B. (p. 142) Collated for 'Stealing is always wrong.' Discuss.

Most of the notes made on the early texts show good critical selection. The note-maker has chosen the most relevant material for the purpose.

The main exception is the set of notes made for Text 12. These are too detailed and are too close to the original text. It is not clear why the note-maker considered that all the details about the research were relevant: these notes do not show evidence of critical selection.

Strong points about these notes are:

- Information is grouped to support different points.
- The notes include evidence of the critical thinking process, which can then be used in reports or essays.

Weak points of these notes are that:

- Quotations and notes copied from the texts do not stand out, and could easily be copied accidentally into a report or essay later without proper acknowledgement. This is especially so for the notes on Texts 1 and 12.
- The words noted down are too close to those used in the original texts, suggesting that the reader has slipped into 'automatic' note-making, or copying, rather than focusing on selecting the most relevant information.

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Note your source of information (1)

All of your notes should make it very clear where information comes from.

Long hand and short hand

The first time you use a source, it is useful to write its title and other key details in full, preferably in an electronic store, so you can cut and paste it as needed. In your notes, write the details in full the first time you use a source and then use a recognisable abbreviation as a short-hand. Note exact page references or website addresses so you can find information again easily when needed.

Details for references

If you are writing an assignment for college or a report for a company, you will be required to make references to the material you use so that your readers know what influenced your thinking and where you found your evidence. Universities and companies usually recommend a particular style of referencing, such as the Harvard or Vancouver system or a house style. These vary in the fine detail, such as whether you write the authors' initials or their full names. Make sure you note the information needed for the referencing system you are required to use. The Harvard system is outlined below. See also pages 167–8.

For books

Note:

- Who wrote it? (Full surnames, followed by the initials, of all writers that appear on the front cover.)
- When did they write it? (See pages inside the cover. Give the date it was first published or the date of the current edition, but not the date of the reprint.)
- What exactly is it called, including the subtitle?
- Which edition is it (if not the first edition)?
- The city where it was published. (See the pages at the front of the book that give the address of the publisher.)
- The name of the publisher.

Example

Details to note about books

Crane, T. (2001) *Elements of Mind: An Introduction to the Philosophy of Mind* (Oxford: Oxford University Press).

Example

Multiple authors

Pears, R. and Shields, G. (2016) *Cite Them Right* Tenth edition (London, Palgrave).

For a chapter from a book

If you are noting information from a book where each of the chapters is written by different authors, note:

- The name of the chapter's author, the date and then the name of the chapter.
- The name of the editor, and the title of the book. Note that the initials of the editor are written before the surname when citing this in references.
- The page numbers, following the title of the book.
- Where it was published and the name of the publisher, in brackets. You may need to note where a chapter was first published.

Example

Chapters from a book

Willis, S. (1994) 'Eruptions of Funk: Historicizing Toni Morrison'. In L. Gates Jr. (ed.), *Black Literature and Literary Theory* (pp. 263–83) (New York: Methuen).

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Note your source of information (2)

For articles

Note:

- Who wrote it? (Full surname, followed by the initials of all writers that appear at the top of the article, in the order they appear.)
- When did they write it?
- What exactly is it called?
- What is the exact name of the journal the article comes from?
- In which volume and/or issue of the journal did the article appear?
- Page numbers of the entire article.

Example

Articles
Shulman, L. (1986) 'Those who Understand: Knowledge Growth in Teaching'. *Educational Researcher*, **15** (2), 4–14.

For electronic sources

Note:

- The authors' names.
- For online journals, give the full surnames of the authors, followed by their initials, in the order they appear.
- The date it was written. If no date is given, consider whether this is a good source to use.
- The name of the item (if there is one).
- The name of the journal, and its volume and issue details, for articles.
- If the material is available only on the internet, give exact details of the web page so somebody else could open that site and page.
- The date it was downloaded from the internet.

Example

Electronic sources
Collins, P. (1998) 'Negotiating Selves: Reflections on "Unstructured" Interviewing'. *Sociological Research Online*, **3** (3). www.socresonline.org.uk/socresonline/3/3/2.html; January 2001.

Newspapers

Note the name of the author, article, the name of the newspaper, the date and the pages.

Example

Farrar, S. (2004) 'Old Sea Chart is so Current', *Times Higher Educational Supplement*, 16 July, p. 5.

If the author's name is not given in the newspaper, give the name of the newspaper first, then the year, then the name of the article, followed by the date and page numbers.

Example

Times Higher Educational Supplement (2004) 'Old Sea Chart is So Current', 16 July, p. 5.

Other sources

There are many other sources of information that you may need to use. Make notes of any details that will help you and others locate that particular source. This might include the name of the library and/or collection, volume numbers and folio numbers. Give exact details of what the source is.

Example

Letter in a collection
Papers in the Bodleian Library. Curzon Collection, vol. 22, ff. 89–90. Letter from Henry Peter Lord Brougham to C. H. Parry, 3 September 1803.

Example

Government and official sources
National Committee of Inquiry into Higher Education (1997) *Higher Education in the Learning Society* (London: HMSO).

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Summary

Critical thinking needs to be incorporated at many different stages in the process of producing a critical piece of work. This chapter has focused on applying a critical approach to reading and related note-making.

It is not unusual for people to suspend critical thinking when reading and making notes. For example, they often assume it is acceptable to read and make notes in a non-selective or non-critical way, amass a pile of notes, and then apply critical thinking to the notes that have been made. Whilst this is not an unacceptable strategy, it is not efficient in terms of time management. Using such methods, you are more likely to read and take notes on material you will not use, and then repeat your reading of such unnecessary material in order to select what is needed.

Making notes in an uncritical way is also a risky strategy. It is much easier to become confused about which notes have been taken down verbatim from the text and to include these, by accident, in your own work. This would leave you open to charges of cheating and/or plagiarism.

This chapter recommends strategies which, if followed, are more likely to save you time, and to help you develop critical thinking skills as part of an ongoing process when reading and writing. Guidance on referencing your source materials is included: critical readers will want to ensure that they can find the source of information again in the future if they need it. If the material is to be used within a piece of writing, these details will be needed to refer the reader to the source materials.

Critical reading is assisted by identifying certain key pieces of information that can direct and focus your attention. Earlier chapters identified certain components of an argument, such as identifying the conclusion, as useful ways of finding the argument within a passage. This chapter draws attention to the importance of identifying the underlying theoretical perspective, where possible, in order to better evaluate the significance of the material to the author's point of view.

This chapter also emphasises the importance of developing skills in categorising and selecting information as component skills within critical thinking. Such skills contribute to more effective reasoning abilities, as they require you to find comparisons and exceptions, to look for factors that link and connect information, to develop an understanding of the relative significance of different pieces of information, and to make evaluative judgements.

Information about the sources

Donaldson, M. (1978) *Children's Minds* (London: Fontana).

For background on plant cuttings and PBRs: Hogan, C. (2004) 'Giving Lawyers the Slip'. *The Times*, 24 August, p. 26.

On moral issues: Kohlberg, L. (1981) *Essays on Moral Development*, vol. 1 (New York: Harper & Row).

Peters, R.S. (1974) 'Moral Development: a Plea for Pluralism'. In R.S. Peters (ed.), *Psychology and Ethical Development* (London: Allen & Unwin).

Gilligan, C. (1977) 'In a Different Voice: Women's Conceptions of Self and Morality'. *Harvard Educational Review*, **47**, 418–517.

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Chapter 10
Critical, analytical writing
Critical thinking when writing

Learning outcomes

This chapter offers you opportunities to:

- consider the characteristics of critical, analytical writing
- identify the appropriate language structures for indicating, or signposting, the direction of your argument
- compare pieces of writing to identify the characteristics of critical writing
- understand how critical thinking skills are applied to essay-writing

Introduction

Critical writing draws together other aspects of critical thinking in order to present a forceful case to readers. This means that it must continue the process of selecting and forming judgements about the evidence. However, the writing must be produced with its eventual readers in mind.

This chapter considers the characteristics of critical, analytical writing from the perspective of writing text, as opposed to considering written arguments from the reader's point of view. As well as looking at general characteristics, it focuses on the language used to present written arguments.

Previous chapters emphasised the importance of developing a clear line of reasoning. When speaking, it is possible to use the tone of voice, pacing and pauses, as well as body language, to help the audience to follow the argument. It is also common to repeat phrases or to raise the voice for emphasis.

These devices are not available to orientate the reader when arguments are written down, especially in formal writings. Therefore, it is all the more important to set the scene well, to summarise key points as you go through and, in particular, to use recognisable words and phrases to signpost the different aspects of the argument.

The process of re-drafting and editing writing is particularly important to critical writing. The writer needs to ensure that the final draft has the characteristics associated with critical writing. The final piece of critical writing should be clearly written and well structured. It should include devices, such as signal words, that lead readers through the evidence in such a way that they are clear about the conclusion even before they read it.

Finally, the chapter looks at how a range of critical thinking skills are applied to the process of producing essays – a particular kind of writing. Essay-writing is a requirement of most subjects, and the main vehicle used for demonstrating, and assessing, good critical thinking skills.

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Characteristics of critical, analytical writing (1)

Content

In critical writing, most of the text is dedicated to presenting a case through providing reasons, using relevant evidence, comparing and evaluating alternative arguments, weighing up conflicting evidence, and forming judgements on the basis of the evidence. Background information of a general nature is used very sparingly, and only essential details are usually included. Description is kept to a minimum.

A sense of audience

Good critical writing always keeps its future audience, or readers, in mind. The aim of an argument is to persuade others. When producing critical writing, it is important to consider how the message might be read by other people, especially people who might disagree with the evidence or the conclusions. A good critical writer knows which aspects of the argument are likely to be the most contentious, and the kind of evidence required in order to counter potential opposition within the reader.

Clarity

Critical writing should aim to be as clear as is possible. The aim is to convince the reader, so it is important that the style of writing makes it easy for the reader to see the point. Long, complicated or poorly punctuated sentences can make it difficult for the reader to follow the argument.

The language used for critical writing is generally sparse. It usually sticks to the facts and avoids emotional content, adjectives, and flowery language or jargon. The aim is to present, as far as is possible, the points in a way that an intelligent general reader can understand. Technical language can be used but should not be used simply to sound clever.

Often, an argument can sound clear in our own mind but does not come across clearly in our writing. It is not always easy to see which text may be interpreted differently when read by someone else, or what might be confusing or ambiguous. Skilful writers check through their writing several times, often by reading aloud, looking for any phrases that may be awkward to read or which could be open to a different interpretation by others.

Analysis

Analytical writing is writing that looks at the evidence in a detailed and critical way. In particular, it weighs up the relative strengths and weaknesses of the evidence, pointing these out to the reader, so that it is clear how the writer has arrived at judgements and conclusions.

Selection

Presenting too much detail can mean the main argument becomes obscured. The reader may lose interest in tracking the line of reasoning and simply conclude that the argument is weak. Usually, writers cannot include detailed critical analyses of every point that supports their arguments. On the other hand, presenting too little detail can make it sound as if there is not enough concrete evidence to support the case.

Skilful writers select the most important points, often the most controversial points, to examine in detail. They may only allude briefly to other points, sometimes grouping several together, in order to indicate that they are aware of these points. Strong critical writing uses a good balance of detailed analysis and sections that summarise arguments and evidence.

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Characteristics of critical, analytical writing (2)

Sequence

The more complicated the argument, the more important it is that the information is sequenced in a way that helps the reader. Good critical writing is planned out well so that the most important points stand out clearly. Readers can follow an argument more easily if they can see how each point is connected with the preceding point, and how each point links to the main argument. Good signposting, as described below, helps the reader to understand the sequence used by the writer.

Best order

It is generally more logical to present the points that support your own argument first, so that you establish your case early in the mind of the reader. This helps to align the audience to your position. Audiences are more likely to interpret subsequent reasoning from the perspective of the first argument presented, so it is better to present your own argument first.

However, if your argument aims to show why a well-established argument is wrong, it can make more sense to make a critique of the established argument first, in order to refute this before presenting an alternative case.

Good critical writing shows an awareness of what are the most important or controversial aspects and dedicates the most space to these. If readers are persuaded on these points, they need less convincing on other points.

Skilful critical writers consider which information their audience needs to read first so as to make best sense of the argument. They ask, repeatedly, questions such as:

- Is this the best order or could it be better?
- Where does this best fit into the argument?
- Is the argument coming across clearly?
- If I moved this information somewhere else, would it be easier to follow the line of reasoning?

Group similar points

Similar points should be located near each other in the writing. For example, the points that support one aspect of the reasoning could be grouped together, followed by the points against. Usually, you should complete your analysis of one piece of evidence before moving on to an analysis of the next. Alternatively, all the aspects of the evidence that support a contributing argument could be grouped together, followed by an analysis of those aspects of the evidence that do not support it. In each case, it is important to consider whether similar points are grouped together in a way that makes the text easy to read. The readers should not feel they are 'hopping' back and forward between points.

Signposting

Good critical writing leads the readers effortlessly through the argument so that they do not need to pause to consider where they are in the argument or whether the writer intends them to agree or disagree with a particular point. A skilful writer will use certain words and phrases as 'signposts' to indicate to the readers where they are in the argument, and how each point links to previous or subsequent points.

In critical writing, it is not usually acceptable to use graphical means to highlight important points. Critical writing avoids methods such as using italics, emboldening text, capital letters, larger font, colour or arrows to make important points stand out. Instead, it relies on good sequencing and use of language to signpost the reader through the line of reasoning.

Activity **Characteristics of critical writing**

Read through a recent essay that you have written. Use this to evaluate the following:

- Which of the above characteristics (on pages 148–9) are already strengths in your own writing?
- Which of these characteristics could you improve upon in your writing?

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Setting the scene for the reader

When presenting an argument, the author usually has to include more than simply the reasons and conclusions. The circumstances and reasons for producing the argument will usually determine what else is considered to be relevant. When evaluating the likely effectiveness of an argument, it is important to consider:

- what background information the audience needs and expects;
- what they will already know;
- what kind of reasons and evidence are likely to convince that particular sort of audience.

Conventions

For academic subjects, there are conventions which govern the presentation of a line of reasoning. Journal articles, for example, have different conventions from newspaper articles or everyday speech. Usually, the background information in articles is of two types:

1 Key details of previous research relevant to the current article.
2 Details of the methods used to gather and analyse the evidence, especially data, for the current article.

Activity



Browse through journal articles and identify the way background information is treated in your subject area. Note how much or how little detail is used in each section of the article. Consider what kind of background information is included, as well as what is not included.

Background and history

In critical writing, general background details are usually kept to a minimum, as in the Feng Shui example on page 153. The history and general background are only usually included where they form part of the argument.

Take for example the question: *How did the fish come to take over the estuary?* The history is relevant, and provides a reason that supports the conclusion, as in the example below.

Example

'Background' as a reason

Historically, the fish were subject to many large prey and laid many eggs to increase their chances of survival. When they migrated to the estuary, there were no natural predators to restrain their numbers. They continued to lay as many eggs, and so took over the estuary.

If the question was *Account for changes in banking practices over the last 10 years*, the historical background given in the example below would be unnecessary.

Example

Unnecessary detail

Banking is a very old profession. Early examples include the development of the letter of exchange by the Hansa League in the fifteenth century.

Definitions

It is typical in critical thinking to define any terms used in the line of reasoning that might be open to more than one interpretation. This enables the audience to know which interpretation the author is using and reduces misunderstandings.

Example

There has been much debate about whether only humans have consciousness but there is a growing body of research which suggests that animals and even inanimate objects share this capacity. In considering whether animals and objects have consciousness, the first point to consider is what is meant by the term *consciousness*.

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Activity: Setting the scene for the reader

Activity

How well do the authors of the following passages set the scene for an essay about a theory of food production?

Passage 10.1

'Is productionism dead?'

Productionism was a theory developed following the recession and famines of the 1930s. Theorists such as Orr, Stapleton and Seebohm Rowntree argued that if farming methods were adapted to include technology, more food could be produced and famines would become a thing of the past. This essay will argue that productionism has been successful to some extent, in that some areas that were formerly subject to famine are no longer prone to famine, and the proportion of starving people worldwide reduced year on year. However, it will also argue that despite the successes of technology in producing more food, other aspects of productionism have undermined its strength as a model for social reform. The essay examines some negative by-products of the productionist approach, such as the threat to bio-diversity, pollution, depopulation of agricultural areas, and the power that lies in the hands of retailers at the expense of small farmers. It will argue that productionism is not dead, but that a new model of food production would now better serve consumers, food producers and the global ecology.

Passage 10.2

'Is productionism dead?'

Productionism is dead. Its main proponents, such as Orr, Stapleton, Orwin and Seebohm Rowntree, were inspired by social altruism. Not for them the traditional farming methods of the past nor the harrowing scenes of famine and collapse presented worldwide in the 1930s. For them, there was a saviour and the saviour was technology. Today, technology has developed in ways that even a visionary could not have imagined in the 1930s. Nonetheless, it has not been the saviour that was predicted. A new model is needed, and social and ecological forces will ensure that productionism, as a theory, passes into the realms of history.

Passage 10.3

'Is productionism dead?'

The main problem with productionism is that it places too much hope in science when science cannot always deliver. One result of productionism, with its emphasis on producing more and more food, is that people in the developed world think that food supplies can be endless. Child obesity is one result of such an approach. Whilst some people have too much to eat, others do not have enough. A lot of food isn't even a good thing: much of the food we eat is 'junk' and contains little nourishment.

Passage 10.4

'Is productionism dead?'

Food production has always been an important aspect of human activity. Since time began, humans have looked for ways of increasing the amount of food available to them. Without food, we would not be able to survive so this is a critical consideration for any society. Unfortunately, for most of history, the spectres of hunger and famine have hung over people's heads. One period when this was particularly acute was the 1930s, when even rich economies were affected. It was in the face of such crises that productionism was born.

Answers: see p. 300.

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Writing up the literature search

Chapter 8 described methods for conducting a literature search and for identifying reputable sources. You are likely to read many more sources than you can include within your own writing. This requires careful selection of what to include as background information.

For essays

In essays, the focus is on the development of your own argument. It is not typical to include a summary of the literature at the beginning of an essay. Instead, you introduce sources at the relevant point in your argument. In essays, you need to refer to materials used as background reading in order to:

- illustrate a point you are making or add weight to a specific reason you are using to support your argument;
- argue against a point of view, if you wish to challenge what has been previously written;
- provide weight to your own argument by showing that it is supported by the research or arguments of other writers who are well known in the subject area.

For reports, dissertations and projects

It is usual when writing reports, dissertations and projects to start with a relatively brief overview of the background research. This is generally about 10 per cent of the overall piece of writing. You need to identify:

- which two or three pieces, theories, perspectives or previous research articles provide the most significant background information for your own research;
- how, if at all, these pieces of research are linked to each other. Usually, this will be by chronological order.

Focus on two to five pieces of significant published research, drawing out the key points. Provide only enough information to ensure the reader understands the significance of the research and its relevance to the rest of your report or dissertation. Allude to most of the other pieces of research in passing, or very briefly.

Accuracy

Always check the original source and/or your notes carefully before writing about the work of other people. Check:

- that you have ascribed the right theory and discoveries to the right people;
- that you have given the right dates;
- that you have spelt their names correctly;
- that you have interpreted their meaning and significance correctly.

Interpretation

Critical reading is an act of interpretation as well as selection. The recommendations made on pages 135–6 about how to combine reading with note-making make it more likely that you will produce a personal interpretation for your own assignment or report rather than simply reproducing the work of someone else. For essays, this does not mean that you must find an approach that nobody else has ever considered. Simply through the choices you make and through writing in your own words, you will be making a personal interpretation. The same applies when you are writing up the 'literature search' section for reports, projects and dissertations.

Reminder about referencing

Remember that copying from the internet or a written source is not acceptable, unless it is for a brief quotation and you reference the source correctly. The basics of citing references are given in Chapter 9, pp. 144–5 and pp. 167–8.

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Words used to introduce the line of reasoning

Words that signal the direction of an argument

At the end of Chapter 3, there was an introduction to words that indicate conclusions within an argument. Authors may use other words to point out different stages of the argument to the reader. These words signal the direction of the line of reasoning.

It can help to use these words when scanning a text to find the line of reasoning quickly. The table on page 158 summarises the words and can be used when constructing your own arguments.

Different words have different functions within an argument. Some, for example, are used at the beginning of an argument, others reinforce a point, some signal a change of perspective, others are used for conclusions. These words are sometimes known as *connectives* – as they connect the different parts of the argument.

Introducing the line of reasoning

Certain words are used to signal the opening of the argument. These include words such as *first; first of all; to begin; first and foremost; at the outset; initially; I will start by …*

Examples

- I will start by arguing that Feng Shui is important to every aspect of our lives and is not simply a question of decorative art.
- First of all, studying the size of the neo-cortex in the brains of different types of animals such as monkeys or rats can tell us a great deal about their social worlds.
- In considering the role of chemistry in the commercial world, it is important, at the outset, to recognise that chemistry is a commercially viable subject.
- Initially, we will consider whether porous rocks can ever provide solid foundations for new buildings.

Note that the introduction to the argument might not be the first sentence. It may be later in the paragraph. For example, the argument in the set of examples above might follow an introductory sentence or passage, used to set the scene, such as that in the example below.

Example

Feng Shui has formed part of Chinese life for over three thousand years and is increasingly gaining popularity in the West. The reasons for this new popularity are sometimes attributed to a growth in favour of simplicity and minimalism in house decoration. This is a mistake. I will start by arguing that Feng Shui is important to every aspect of our lives and is not simply a question of decorative art.



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Words used to reinforce the line of reasoning

Certain words can be used to indicate that new information is being introduced that further reinforces the direction of the line of reasoning. These include words such as *also; in addition; besides; too; furthermore; moreover.*

Adding similar reasons

When reinforcing a line of reasoning, the author may wish to add reasons similar to those already presented. This can be signalled by words such as: *similarly; equally; likewise; in the same way.*

Examples

- Similarly, the Chinese martial arts are not merely about fighting, but offer tools for understanding mind and motivation.
- In the same way, when we look at the neo-cortex of humans, we learn about the evolution of our own social habits.
- Likewise, applying chemical knowledge to biological problems has opened up new avenues of business and many spin-off industries.

Adding different reasons

At other times, the author may choose to reinforce the overall argument by adding new and different reasons. Authors often indicate that they are adding new reasons by using words such as *in addition; besides; as well as …; not only … but also …*

Examples

- Not only can Feng Shui help to guard your health, it is believed to protect and enhance your wealth and prosperity.
- The amount of time that animals such as chimpanzees spend on grooming each other is not only linked to the composition of the social group, but also to the size of that group.
- In addition to developments within chemistry, developments within information technology have opened up new possibilities for biochemical research at the molecular level.

Strengthening the argument

At other times, authors can use words such as *furthermore; moreover; indeed; what is more; such as* in order to indicate that they believe a reason is particularly good, or that its addition to the line of reasoning makes a more convincing case.

Examples

- Furthermore, Feng Shui is used in business in order to help keep customers and employees happy.
- Moreover, the development of language in humans may be directly related to the size of human communities, which makes grooming impossible as a key form of communication.
- Indeed, the reorganisation of scientific departments to encourage work across disciplines such as physics and material science has led to much excitement about research on the boundaries of each discipline as well as opening up new areas of entrepreneurship.

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Signposting alternative points of view

Introducing alternative arguments

A strong argument will usually critically evaluate alternative perspectives or points of view. By doing so, authors show readers that they have considered other possibilities and not simply presented the first argument that entered their heads. This approach usually strengthens an argument as it suggests that the author has researched the subject or has considered all angles.

Words used to signal that an alternative point of view is being considered include: *alternatively; others argue that …; it might be argued that …*

Examples

- It might be argued that Feng Shui has not been proved through rigorous scientific research.
- On the other hand, not everyone believes that animal behaviours have anything to tell us about human behaviours.



- Alternatively, there are those who believe that the prime role of biochemical research should be the advancement of knowledge and that this goal should not be distorted or lost through the demands of the market place.

Rebutting alternative arguments

As we saw above, it is typical, within a line of reasoning, to introduce alternative points of view in order to disprove them or indicate their weaknesses. Normally you would expect the author to show why their own point of view is the more convincing.

Words used to rebut alternative arguments are: *however; on the other hand; nonetheless; notwithstanding this.*

Examples

- However, many practitioners of Feng Shui are also scientists.
- Nonetheless, humans are closely related to other primates such as chimpanzees and apes.
- These arguments notwithstanding, there is still much to be gained from a closer alignment between science and business.
- Notwithstanding the argument that chalk is porous and porous rocks provide riskier surfaces for building, under certain circumstances, chalk can provide a solid foundation for building.

Contrasting and contradicting

When other arguments are being considered, authors may move back and forth between their own point of view and opposing arguments. They will normally either weigh up the evidence for one side and then the other for each reason in turn, or they will contrast all the evidence for one point of view against the evidence for their own line of reasoning.

Words that indicate this process of contrasting include: *although …; conversely; by contrast; on the one hand … on the other hand …; in fact.*

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Examples

- On the one hand there are those who argue that Feng Shui is based on mysterious principles such as yin and yang that people in the West cannot understand. On the other hand are those who argue that Feng Shui is based on common sense and therefore suitable for everyone.
- Although humans' verbal language can be used in sophisticated ways to express abstract ideas and reasoning, it can also be very restricted in its capacity to communicate our deepest feelings and creative thoughts.



- Some researchers argue that scientists are being forced to patent their work even when they do not want to enter commercial contracts. By contrast, others complain that they do not receive enough support in patenting their discoveries.
- Houses benefit from being built on bedrock. By contrast, houses built on beaches tend to sink over time.

Expressing results and consequences

After several reasons have been considered, the author should draw out how these should be interpreted as a whole. This would normally be found towards the end of the sequence, but the author may do this several times during the line of reasoning, to help the reader keep track of the reasoning and to reinforce the message. This was covered in Chapter 5 (on p. 65), under 'Intermediate conclusions'.

Words used to express the consequences of the evidence the author has presented include: *as a result; as a consequence; hence; thus; consequently; because of this.*

Examples

- As a result, we can see that the rules governing Feng Shui in the workplace are similar to those that apply in the home.
- Thus, the introduction of verbal communication allowed us to communicate with more of our species and more quickly.
- As a consequence of commercial backing, the infrastructure for scientific research has been improved in a number of institutions.
- Hence, as sand shifts and moves over time, a house built on sand is likely to sink.

Activity

Browse through three or four articles for your subject. What words are used to:

- Introduce the main argument?
- Move an argument along?
- Sum up the argument?

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Conclusions

All the reasons and evidence presented should lead towards the conclusion. Even when alternative arguments are put forward, these should be presented in a way that supports the main line of reasoning. Authors usually signal conclusions using words such as *therefore; in conclusion; thus; thus, we can see ...*

For longer texts, the conclusion may consist of one or more paragraphs rather than just a single sentence. These would normally be placed at the end of the piece of writing. For longer texts, a good piece of writing will usually refer clearly to the overall conclusions as it unfolds, so as to help the reader to make sense of what they read.

In shorter passages, as we have seen, the conclusion may be stated near the beginning rather than the end.

Examples

- In conclusion, Feng Shui is not a decorative art but is, rather, a sophisticated system for arranging our surroundings so that we live in greater balance and harmony with the outer world.
- Thus, we have shown that the human brain evolved as a result of our need for more effective and efficient social communication.
- Therefore, academic research can be greatly advanced by commercial partnership.
- We can conclude that it is important to ensure that sufficient tests have been carried out to check the underlying rock structures, and to consider carefully the consequences of building on surfaces other than bedrock.

Activity

Add signal words to signpost the development of the argument in the following passages.

Passage 10.5

Deaf people have their own languages, based on signs, body position and facial expressions. As few hearing people understand these languages, communication between deaf and hearing people is not usually very effective. Deaf people often form strong social and cultural groups, they are often excluded from mainstream culture and their talents are not used effectively within the economy. Hearing people can feel excluded from deaf conversations and uncertain of how to behave around deaf people. It would be in everyone's interests if sign languages were taught in school so that deaf and hearing children grew up able to communicate effectively with each other.

Passage 10.6

Globalisation appears to be inevitable but there is disagreement about whether this is a positive development. There are those who argue that increased contact between countries leads to better understanding and reduces the likelihood of future wars. They see benefits to democracy and human rights from information being widely available electronically, so that different nations can compare conditions in their country with those elsewhere. Some see globalisation as a destructive force. They argue that it leads to less powerful peoples losing their indigenous languages as the languages of more powerful countries are used internationally for business and politics. They argue that globalisation often means big business buying up resources and land in poorer countries, distorting local economies and draining their resources. Although there are some potential benefits to globalisation, some controls are needed to protect poorer economies from exploitation.

Answers: see p. 300.

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Words and phrases used to structure the line of reasoning

The signal words introduced in this chapter are summarised in the table below.

Function	Words used
Introducing the line of reasoning	
Opening phrases	*Words indicating 'first …'* first; first of all; to begin; first and foremost; at the outset; Initially, I will start by …
Developing the line of reasoning	
Reinforcing with similar reasons	*Words indicating 'similarly …'* similarly; equally; likewise; in the same way; indeed; correspondingly; in the same line; also; too; again; besides
Reinforcing with different reasons or evidence	*Words indicating 'also …'* also; in addition; besides; again; as well as; either; too; not only … but also …; neither … nor …; neither
Stronger reinforcement	*Words indicating 'furthermore'* furthermore; moreover; indeed; what is more
Introducing alternative arguments	*Words indicating 'alternatively …'* alternatively; a different perspective on this …; others argue that …; it might be argued that … (the words used in 'rebutting alternative arguments' can also be used)
Rebutting alternative arguments	*Words indicating 'however'* however; on the other hand; nonetheless; nevertheless; notwithstanding this; in any case; in spite of this; despite this; at the same time; even though …
Contrasting	*Words indicating 'by contrast …'* by contrast; although …; conversely; on the one hand … on the other hand …; in fact
Concluding	
Expressing results and consequences	*Words indicating 'therefore'* therefore; this suggests that …; this indicates …; as a result; as a consequence; hence; thus; consequently; because of this; from this we can infer that …; from this we can deduce that …
Conclusions	*Words indicating 'in conclusion'* therefore; in conclusion; thus we can see; thus

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Drawing tentative conclusions

Academic writing, such as that used for research projects, articles and books, tends to avoid words that suggest absolutes and, instead, uses words that express some tentativeness. The kind of alternatives used are indicated below.

Avoids	Uses qualifiers such as:
all, every	most, many, some
always	usually, generally, often, in most cases, so far, haven't yet
never	rarely, in few cases, it is unlikely that
proves	the evidence suggests, indicates, points to, it would appear

Example 1

During the Protestant Reformation in Britain in the sixteenth century, the kings' ministers ordered that religious ornaments such as chalices and carved rood screens found in churches be destroyed. These disappeared from churches at that time. However, during the short reign of the Catholic queen, Mary Tudor, these articles reappeared. As chalices and elaborate carved rood screens appeared again so quickly during Mary's reign, this suggests that the items had not been destroyed previously. It would appear that people had simply hidden them away. This further suggests that the Reformation had less popular support than had been previously believed, and that many people had been hoping for a return to the old Catholic ways.

Here, the author considers that the sudden reappearance of religious items suggests the items had been hidden rather than destroyed. The author then proposes that this is evidence that the old religious customs were more popular than had been previously believed.

These sound like sensible conclusions. However, the author uses tentative language in drawing these conclusions as there may be other interpretations. For example, it may be that there was a much higher level of skill in reproducing those items than was formerly believed. It is possible the items were destroyed and new items were made quickly.

Alternatively, people would have been aware that there was a possibility that the new religious ways might be overturned in the future and that they might be punished for having destroyed sacred items. They may have preferred the new religion but hidden the forbidden items away in order to protect themselves in the future.

Academic writers are always aware that there may be alternative explanations or unexpected findings that overturn even the most widely held views. In Example 1, the writer used phrases such as *this suggests, it would appear, this further suggests.*

Example 2

A small amount of hydrochloric acid was poured on each rock. The first rock then gave off the smell of hydrogen sulphide, a smell like rotten eggs, suggesting the rock was galena. The second rock fizzed, suggesting that it was giving off carbon dioxide and that the rock may be an oolitic limestone.

Example 2 is science writing. The writer is basing judgements on well-tried tests. The tests used are fairly conclusive, but the writer uses tentative language as, if the rocks did not share other known characteristics of those rocks, such as mineral content or grain size, a different judgement might be needed. It is possible, for example, that the fizzing rock was a different type of calcite rock, such as chalk or marble.

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Activity: Writing conclusions

Activity

Consider how well do these passages express their conclusions in a suitably tentative manner? Then check the commentary below.

Passage 10.7

Interpreting new discoveries

We have seen that when explorers found new lands, they tended to interpret what they saw as evidence of what they had intended to find. Travellers to the 'Americas' in the fourteenth and fifteenth centuries sent home reports of finding giants and green men. Earlier, Marco Polo, who had hoped to find unicorns on his travels to China, believed the one-horned creature he found in Java was indeed a unicorn, despite the animal, a rhinoceros, bearing no other resemblance to the fabled beast. However, unlike those who claimed to see giants, or later explorers who really believed they had heard orangutans talking, Marco Polo appears to have described rhinoceroses exactly as he found them. This suggests that not everyone responded to new discoveries by using the same approach. Moreover, it is possible that with the number of discoveries made in recent decades, people are now more likely to take new discoveries in their stride.

Passage 10.8

RNA does the hard work

Although we hear more in the press about DNA, especially after work on mapping human genes, we hear much less about the role of RNA in cell reproduction. RNA, or ribonucleic acid, is essential to the functioning of our genes. One type of RNA reads the messages encoded in the DNA. Various types of RNA are involved in making proteins and carrying these to where they are needed in the body's cells, so that the cell can function as it should, including growing and reproducing. Although the DNA holds encoded messages which help define the nature of the next generation, these would not mean much without RNA. Therefore, it is RNA that appears to do the really hard work in reproduction.

Commentary

Passage 10.7 examines the way people, historically, tried to make sense when they discovered things that were new to them and their cultures. It is difficult to write with absolute certainty about approaches, attitudes and beliefs, and even more so when these took place in the distant past. The writer uses the phrases 'this suggests' and 'it is possible' to indicate the tentative nature of the conclusions being drawn. It is possible, for example, that people today think that there is little more to find out, so are even more surprised by discoveries. The writer uses tentative language appropriately.

Passage 10.8 makes a judgement about the relative importance of RNA in reproduction. Scientific judgements can usually be stated with more certainty, as they can be tested, replicated and measured more exactly than matters such as attitudes and responses. However, even science mainly sets out to support hypotheses and test what appear to be laws. Science recognises that further research can overturn scientific laws, at least under specific conditions. Most of this text is written in more certain language than Passage 10.7, as befits a scientific subject, but the overall conclusion is suitably tentative as it is possible that future research will reveal hitherto unknown roles for DNA or RNA.

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Critical analysis for essays: Essay titles

Essays are exercises in critical thinking. In academic contexts, they are set primarily for you to demonstrate your understanding of an issue, drawing on your critical analysis of, and engagement with, source materials and a range of theoretical perspectives.

As a rule, essays are set for students on issues where there are multiple perspectives. You are expected to:

- become aware of what those different perspectives are, through attendance at taught sessions and especially by reading around the subject;
- understand how and why the major differences in perspective have arisen;
- understand the theoretical underpinnings for each major area of difference in perspective, and how these compare and contrast with each other;
- critically evaluate the evidence base for the different perspectives and theories, and their applicability to different contexts;
- draw together your critical judgements to form a conclusion or set of conclusions that indicate your own considered position, based on the relative quality of the evidence base.

Critical analysis of the essay title

Essays set for students are usually carefully worded so as to encourage a focus on a particular complex or controversial issue. Before launching into an essay, look carefully at the wording of the title so as to do the following:

- Tease out how many parts or subsections there are to the essay and the relative weighting of each. If there is a word limit, consider how many words you can allocate to each section. This will, in turn, give you a sense of how much time to spend reading and writing for each section.
- Identify the main focus of the essay: what is it that the tutors want you to address? If this isn't clear, it may become so once you have read around the subject.
- Consider which theoretical perspective or schools of thought to call upon to explore the issues – you will be expected to engage in a 'critical dialogue' with these. Again, these may not be evident until you start reading around the subject.

Setting your own essay title

If you are setting your own essay title, you need to ensure that you set one that encourages you to work in the same way as for an essay title set by a tutor:

- decide on the key issue that you want the essay to address;
- ensure that there are multiple perspectives on the issue, each of which presents good arguments and evidence;
- look for a topic that contains an element of controversy or other complexity;
- ensure that there is a good range of quality reading material on these different perspectives;
- beware of creating an essay title where there is good reading material or evidence for only one perspective – as this will not help you to demonstrate your own critical judgements well.

Resource

An explanatory list of terms commonly used for essay titles is provided on page 162.

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Academic keywords used in titles

These words indicate the approach or style expected for the piece of writing.

Account for Give reasons for; explain why something happens.

Analyse Examine in very close detail; identify important points and chief features.

Comment on Identify and write about the main issues, giving your reactions based upon what you have read or heard in lectures. Avoid purely personal opinion.

Compare Show how two or more things are similar. Indicate the relevance or consequences of these similarities.

Contrast Set two or more items or arguments in opposition so as to draw out differences. Indicate whether the differences are significant. If appropriate, give reasons why one item or argument may be preferable (see Chapter 9).

Critically evaluate Weigh arguments for and against something, assessing the strength of the evidence on both sides. Use criteria to guide your assessment of which opinions, theories, models or items are preferable.

Define Give the exact meaning of. Where relevant, show that you understand why the definition may be problematic.

Describe Give the main characteristics or features of something, or outline the main events.

Discuss Write about the most important aspects of (probably including criticism); give arguments for and against; consider the implications of.

Distinguish Bring out the differences between two (possibly confusable) items.

Evaluate Assess the worth, importance or usefulness of something, using evidence. There will probably be cases to be made both *for* and *against*.

Examine Put the subject 'under the microscope', looking at it in detail. If appropriate, 'Critically evaluate' it as well.

Explain Make clear why something happens, or why something is the way it is.

Identify Outline the salient points, indicating their significance or implications.

Illustrate Make something clear and explicit, giving examples or evidence.

Interpret Give the meaning and relevance of data or other material presented.

Justify Give evidence which supports an argument or idea; show why decisions or conclusions were made, considering objections that others might make.

Narrate Concentrate on saying *what* happened, telling it as a story.

Outline Give only the main points, showing the main structure.

Relate Show similarities and connections between two or more things.

State Give the main features, in very clear English (almost like a simple list but written in full sentences).

Summarise Draw out the main points only (see 'Outline'), omitting details or examples.

To what extent Consider how far something is true, or contributes to a final outcome. Consider also ways in which the proposition is not true. (The answer is usually somewhere between 'completely' and 'not at all'.)

Trace Follow the order of different stages in an event or process.

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Critical analysis for essays: Reading

Critical analysis when reading

For a student essay, much of your time is spent reading around the subject and working with the material that you read. You would be expected to apply the critical skills that you have covered in earlier parts of this book, such as:

- making good critical decisions about the relevance of reading material, given the specific essay title and the issues relevant to it;
- making good critical decisions in what you choose to read, based on the quality of the materials and the evidence base that these draw upon (see pp. 111–15);
- the breadth of what you read and the variety of perspectives considered.

Demonstrating critical reading skills

You demonstrate your critical abilities to your tutors through the ways that you make use of reading material. They will look at:

- what you choose to refer to – and what you omit;
- how well you query the findings and opinions in what you read, especially in the light of contradictory findings and judgements;
- whether you can recognise the relative merits and flaws of different perspectives.

Reading for balanced analysis

There is a fine balance in writing essays between ensuring that your position is clear, and not appearing to be dogmatic about it. If you have strong personal opinions or beliefs, then it can be especially difficult to maintain a balanced approach. The following guidelines can help.

Balanced reading

- ***Use quality resources*** Justify the position that you believe in on reasons and arguments drawn from good quality sources; if you cannot find these, it will be difficult to present a strong academic case that gains good marks. Cite the sources that you have read that support your position (pp. 167–8).
- ***Balance your choice of reading*** Read good quality resources for all sides of the argument and select the best evidence for all viewpoints – not just for your own position. If you do this, you demonstrate that you are making a reasonable effort to make objective judgements.
- ***Be even-handed*** Bring a similar level of criticality when reading sources that support your own position and those that support positions you reject.
- ***Acknowledge good counter arguments*** Even if you are not convinced by alternative viewpoints, be fair in presenting their strengths. If your reading suggests that there are good arguments against your position, then acknowledge the strengths of these. Cite sources for these just as you would for your own position.

What if the evidence is against you?

If the evidence appears stronger for positions that you do not believe in, your reader would then want to know why you are holding to your position against the available evidence. In such a case, it is possible to state that the evidence base, or academic argument, is currently strongest for a particular point of view without this meaning that you accept this personally or have to change your beliefs. Alternatively, you can indicate where you feel evidence may become available in the future that would support your position.

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Critical analytical essays: Introductions

Parts and components of an essay

An essay consists of three parts, known as:

- the introduction;
- the body of the essay;
- the conclusion.

Into these, a critical, analytical essay integrates the following five components:

1. clarification of the issues;
2. a statement of your own position;
3. analysis of compelling arguments in support of your position;
4. analysis and rebuttal of counter arguments;
5. potential synthesis.

The weighting and sequence of these will vary depending on the issues and the material to be covered, but some guidelines follow.

The introduction to the essay

An essay isn't like a mystery novel where the plot gradually unfolds and the unexpected suddenly occurs. Rather, in a good essay, the scene is set for the reader through a formal introduction. This should be succinct, precise and brief; it can be as short as a single paragraph and shouldn't be longer than a tenth of your essay.

As part of your introduction, without at this point going into detail, you should make the following clear to your potential reader.

The core issues

Refer to the main issues that will be addressed in the essay and, if this can be done succinctly, include an indication of why these are of significance.

Key perspectives

Indicate the major differences of opinion, perspectives or schools of thought on the issues embedded within the essay title. Avoid straying into the more detailed analysis which would come later in the essay.

Define the terms

Define what you mean by words or phrases that are potentially open to different interpretations, or which are contentious, or that could be used in different ways to produce alternative conclusions. Avoid defining obvious terms, covering every word in a title, or giving dictionary definitions.

Your position

Ensure that your position comes across clearly and that this is consistent with what you write in your conclusion. If possible, indicate the critical perspective that you are taking, such as the school of thought that you lean towards. Your own position should either be stated explicitly or else should be easily inferred from the rest of your introduction.

The direction of your argument

If there are twists and caveats intrinsic to your argument, then mention these succinctly, indicating how they support or modify your overall line of reasoning.

Get to the point

- Avoid giving unnecessary background information that an interested reader can find out for themselves.
- Be specific about 'significance'. Give reasons for why an issue is important now – because of new or high-profile research findings, or recent political or social events, or the anticipated future impact in the sciences, arts, the economy, or in a region, or for a large number of people, or in an area of research where solutions have been hard to find, or where previous research findings have been overturned.
- Avoid the urge to make broad generalisations such as about 'science', 'humanity' or 'the world' as a means of getting started.

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Structured argument: The body of the essay

Clarification of the issues

Typically, essays are set on subjects that are multi-faceted and complex – that enable the analysis of different points of view. You are not expected to cover every issue and perspective. The wording of the essay title gives clues to the expected focus for that essay; this narrows down the issues to be addressed. To orientate your reader, clarify the issues either in the introduction or in the opening paragraph of the body of the essay.

Bring out why these issues are significant, controversial or complex. If this seems difficult, it is likely that you have not read sufficiently around the subject. It may help to browse extracts of articles and book reviews for the past few years in key journals for your subject in order to gain a feel for the developing issues.

Your position

Ensure that you provide a clear line of reasoning (pages xiv and 43) such that your own position as stated in your introduction comes across clearly and coherently throughout your essay.

Analysis of arguments

The bulk of the essay consists of examining, critically, two or more viewpoints on the issues identified. In doing so, you will identify compelling reasons for espousing some arguments and for dismissing others.

Some arguments may appear persuasive at first but less so when scrutinised further. These could be the dominant views or most interesting recent research. Opinion on these might be different in a few years' time. It is worth examining them from that perspective and seeing if you can identify any weak points for yourself.

Other viewpoints may have little following or may be so new that the evidence base is still weak. Nonetheless, in the future, these might become dominant perspectives in the discipline. It is worth considering what further research might strengthen these viewpoints or give them more credibility.

Make critical judgements

- Make critical judgements on the relative merits of each perspective or school of thought, stating the strengths and weaknesses of each.
- Be prepared to state if there are gaps in the evidence base that make it difficult to make a definitive judgement on a particular aspect.
- Be clear about *exactly* what the evidence does support; it may be very convincing, or weak, in some conditions, but that might not be the case if the conditions were changed in specific ways. Demonstrate that you have given thought to when, where and why an argument might apply in one case but not another, and to where further research is needed to support the case.

Make critical choices

Select only the most compelling arguments 'for' and 'against'. For an undergraduate essay, there is unlikely to be space within the word limit for you to go into detailed analysis of all differences and points of interest.

Your selectivity provides evidence of the strength of your critical thinking skills; your tutors will be looking to see whether you can recognise the most salient arguments and material to include.



Reflection: Structured argument in essays

In your own essays:

- How clearly does your own position come across? (Do you tend to 'sit on the fence'?)
- Do you engage in a critical way with different perspectives on the subject – evaluating the arguments of those who have written about the subject?

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Essays: Bringing the argument together

Structure your argument

- Use a separate paragraph to analyse each specific point or, if these can be stated briefly, each set of related points.
- Check that paragraphs are sequenced in the best possible order for bringing out your line of reasoning to your reader.
- Use the first or last lines of your paragraphs, and 'signposting' vocabulary, to help indicate the flow of your argument to the reader (see pages 153–8).
- Sometimes, it can be difficult to maintain a sense of clear direction when there are complex issues to analyse. If so, give an interim summary of the arguments so far.

Synthesis

It is likely that, in considering different perspectives or theories, you will find some aspects compelling and other aspects less so. You might feel, for example, that some arguments apply well to particular circumstances, but less so in others. You may feel that all arguments contribute something to the debate even if not in equal weight, but that none are entirely satisfactory in all respects.

Your ultimate position in the essay may be a combination, or synthesis, of the most compelling arguments from several different perspectives that you analysed. If so, avoid generalisations such as 'they are all helpful' or 'none are perfect'. Rather, be specific about what you would take from each perspective. Make a judgement about whether your synthesis also has limitations: you should subject your own synthesis to critical scrutiny.

Conclusions in essays

Essay conclusions shouldn't come as a surprise to the reader. Rather, they should appear to be the inevitable end-point of your line of reasoning. The purpose of the conclusion is to stand back from the more detailed reasoning and evidence that you presented in the body of the essay, and clarify the key messages that have emerged. This may include reasons why it is difficult to form firm conclusions, such as gaps you had identified in the evidence.

Your conclusion draws together:

- your position, which may be a synthesis of other viewpoints;
- the most compelling reasons that support that position;
- the strongest counter arguments and a summary rebuttal of these;
- if relevant, it can refer to strengths or weaknesses in the evidence;
- if relevant, what specifically needs to be researched further in order better to clarify the issues or strengthen the arguments.

Give a sense of an ending

Make sure that your conclusion wraps up your essay neatly. It isn't appropriate to introduce new arguments or evidence here, as any new material should be subjected to critical analysis. If you start to analyse these, then you are, in effect, continuing your argument rather than concluding.

Your conclusion does not have to refer back, point by point to all the issues you have already covered, but it should at least allude to these and serve to reinforce your overall argument. Your conclusion should link back, clearly, to the points you raised in your introduction and, especially, to the essay title. If possible, find a way of doing this in your final sentence.



Reflection: Conclusions

In your own essays:

- Do you aim at making a synthesis of the best arguments on the issue?
- Does your final position on the issue come across strongly?

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Citing and referencing your sources

It should be evident from previous sections that good written critical analysis will involve drawing on reputable sources of evidence.

It is part of the convention of good critical analysis, whether as a student, writer, artist, inventor or more generally, to acknowledge the sources that you use.

Citations

When you draw on someone's work or ideas in your writing, you should acknowledge this straight away at that point in your writing; this is known as a citation. Typically, after making the point, you write the surname(s) and date of publication: (Bloggs, 2012).

References

References are provided at the end of your work as a list of the full details of all those sources you cited.

Bibliographies

Bibliographies are a list of materials that you used as background, but didn't cite in your text.

'Referencing' in critical writing

For all academic writing, you are required to provide details, or 'references', of all the source materials used to produce your work. For students, as much of your work is based on reading, most references will be to academic books and articles. Depending on your subject, you may use other sources in the public domain such as research papers, newspapers, government reports, parliamentary speeches, websites, television or radio programmes, museum or gallery catalogues and brochures.

What is the purpose of referencing?

For the author of the original source

- It is a courtesy to those whose work you have used or drawn upon to give them proper acknowledgement.

For the reader

- It provides transparency about where your ideas and evidence were drawn from.
- It enables your reader to find the source quickly and easily.
- Readers can go to your source to check the accuracy of your use or interpretation.

The advantages to you

- It strengthens your argument if it is well researched and draws on the authority of reputable sources.
- You will be better able to recall where your ideas came from, either if you wish to use those sources in the future, or if the integrity of your work were to be questioned.
- It is a sign of your integrity to acknowledge your debt to others.
- As a student, there is the added advantage of demonstrating to tutors that you have engaged in background reading, as expected.
- For students, it is a convention that you are required to follow or there will be severe penalties.

You don't reference ...

- Common knowledge (names, dates and well-established facts, such as a writer's date of birth or that bees make honey).
- Conversations with friends and students, unless these were formally conducted as part of an agreed research methodology.
- Other students' essays or academic work, as you should not use these for your own work. The exception would be if these had been made available through the university, specifically for reference purposes.

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What do I include in a reference?

It depends on the source

The detail that you need to provide varies depending on whether the source is a book, chapter in a book, journal article, newspaper article, government paper, manuscript, website, radio or TV programme, audio file and so on.

It depends on the style of referencing

The detail that you need to include, and the precise order in which to present this, will vary depending on the referencing style used for your programme of study. Your tutors will advise you which method is used. Typical styles used for academic study are:

- Harvard (widely used);
- Vancouver (especially for science and health subjects);
- Modern Language Association (MLA);
- Modern Humanities Research Association (MHRA);
- OSCOLA (for legal sources).

Typical information to include:

- authors' surnames and initials;
- date of publication;
- full title of the chapter, article or book;
- full title of the book or journal in which a chapter or article is found;
- the edition of a book (if not the first);
- place of publication and the name of the publisher, for books;
- series and volume number, for journal articles;
- location, month and year for conferences;
- Digital Object Identifiers (DOI) for internet pages.

Examples

See pages 144–5 on note-making to support referencing, and examples of references.

How exact do I need to be?

In academic contexts, there are specific conventions for citing sources and referencing. Each type of source, from medieval manuscripts to blogs and tweets, needs to be cited in a very particular way.

Follow referencing rules exactly

You cannot normally pick and choose between referencing systems: whichever system is used on your course should be followed exactly. If in doubt, follow rigidly the examples that are provided by your institution.

'Exactly' for referencing means:

- using the correct referencing style for your programme;
- including a citation within your writing and the full reference at the end, or as a footnote, depending on the style of referencing;
- providing every detail required for the type of source and for the style of referencing;
- providing the information in the exact sequence required;
- using punctuation and abbreviations as stipulated, even down to the use of capitals, brackets, commas, etc.

What if I don't reference my work?

- Weak skills in citing your sources are penalised by poor marks or possibly even a fail. Check your references carefully at least twice.
- If you do not use citations and references correctly, it may appear that you are trying to pass off other people's work or intellectual property (their research or ideas) as your own. This is plagiarism.
- Universities have software and other methods for detecting cheating and plagiarism.
- Plagiarism is treated as an offence and receives severe penalties. You may have to redo part or all of your work for a capped low mark or a zero. You could be removed permanently from your course and college or university.

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Summary

Critical writing draws on many of the skills developed earlier in the book, such as developing an argument, analysing, evaluating and selecting evidence, making judgements, and structuring reasons in a logical way towards a conclusion.

However, spoken arguments can draw on devices such as body language and voice modulations to emphasise points, and the dialogue itself can divide the argument into manageable sections. For critical writing, the writer must take care to use language and structure to organise the argument and to signal different stages within it.

In a written argument, care must be taken to set the scene so that readers know from the outset what conclusion the author wants them to draw. Writers normally present their own position, their conclusions and their own supporting reasons first, so that they orientate the reader to their own perspectives early on. It is important to provide just sufficient for the reader to understand the background. Similarly, at the end of the argument, and at points within it, the writer needs to draw out the conclusions clearly.

In other words, throughout a piece of critical writing, the writer must keep the reader in mind constantly. The aim is not to baffle readers with jargon and clever use of language or to bombard them with so much information that they lose sight of the argument. Instead, the writer must select, group, sequence and structure the best reasons, evidence and details, so that the reader can easily make sense of what is written. Once this has been planned into the writing, signal words can be used to signpost the reader to any changes of direction in the argument and to conclusions.

Critical writing usually follows certain conventions, which were outlined at the start of the chapter. For example, the final drafts of critical writing must be fine-tuned so that critical analysis takes precedence over other aspects such as description and background information. Such conventions signal to the reader that this is a piece of critical writing, which prompts a particular approach to reading.

Students are usually asked to demonstrate their understanding of an issue through essays that use critical and analytical skills whilst using particular writing conventions that support the development of an argument. These were outlined earlier in the chapter.

In the next chapter, you will have the opportunity to look in detail at two critical essays, so that you can see how all these different aspects are combined.

Information about the sources

Marco Polo and unicorns: Eco, U. (1998) *Serendipities: Language and Lunacy* (London: Weidenfeld & Nicolson).

Responses to discoveries of the Americas: Elliott, J.H. (1972) *The Old World and the New, 1492–1650* (Cambridge: Cambridge University Press).

Rocks and minerals: Farndon, J. (1994) *Dictionary of the Earth* (London: Dorling Kindersley).

Productionism: Lang, T. and Heasman, M.A. (2004) *Food Wars: The Global Battle for Mouths, Minds and Markets* (Sterling, VA: Earthscan).

RNA and DNA: Postgate, J. (1994) *The Outer Reaches of Life* (Cambridge: Cambridge University Press).

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Chapter 11
Where's the analysis?
Evaluating critical writing

Learning outcomes

This chapter offers you opportunities to:

- compare two critical essays, to better identify the characteristics of good critical writing
- compare your evaluations of extended pieces of critical writing against a commentary, to check your skills in evaluation
- use a structure for critically evaluating your own writing

Introduction

In this chapter, you have the opportunity to compare two longer pieces of writing on the same subject. These essays are based on the texts found on pages 229–33, which were also used for the reading and note-making activities in Chapter 9. Assume that the authors of the essays have access to all the texts on pages 229–33, and, therefore, are making choices about what to include from those materials, and what to leave out.

On page 172, you will find a checklist to structure your evaluation of Essay 1, followed by the essay and then a commentary. A similar set of materials is provided for Essay 2. The checklists are provided as a tool, and you do not have to use them if you prefer to take a different analytical approach.

As you read each essay, consider how far it meets the requirements for critical thinking that you have covered in the book so far, and what an editor or tutor might provide as feedback if you were to hand this in as your final copy. Note your comments down, either on the checklists or as notes, so that you can compare these with the printed commentaries. The numbers given as superscript in the text (e.g. text[1]) indicate where a note is provided in the commentary.

An adapted checklist has been provided on pages 184–5 to help you evaluate your own critical writing.

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Checklist for evaluating Essay 1

Use this checklist to analyse Essay 1 on the following page. Compare your analysis with the evaluation and commentary on pages 175–7.



Aspect	Yes/No	Comments
1 The writer's own position on the issues is clear.		
2 It is clear what the reasons are for the writer's point of view.		
3 The writer's conclusion is clear and based on the evidence.		
4 Reasons are presented in a logical order, as a line of reasoning.		
5 The argument is well structured and easy to follow.		
6 Reasons are clearly linked to one another and to the conclusion.		
7 All the text is relevant to the assignment (in this case, about whether stealing is always wrong).		
8 The main reasons and key points stand out clearly to the reader.		
9 The writer makes good use of other people's research as supporting evidence to strengthen the argument.		
10 Does the writer make a reasoned evaluation of other people's views, especially those that contradict his or her own point of view?		
11 Does the writer provide references in the text when introducing other people's ideas?		
12 Does the writer provide a list of references at the end of the essay?		
13 Has the writer successfully removed any non-essential descriptive writing?		
14 Does the writing contain any inconsistencies?		
15 Are the writer's beliefs or self-interests unfairly distorting the argument?		

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'Stealing is always wrong.' Discuss with reference to file sharing via the internet

There are many forms of stealing. Although most reasonable people would argue that some forms of theft such as burglary or mugging are always wrong,[1] other areas are less clear cut. In this essay, I shall look at file sharing via the internet as a grey area.

Stealing has probably existed since the beginning of time, and certainly as long ago as the Old Testament, where it was banned by the commandments. All religions regard stealing as wrong, so you would think that there were universally understood principles about what is stealing and what is not. However, this is not the case. This is also true of many other types of ethical issue. Despite this long-standing agreement that stealing is wrong, many people steal. In fact, it is a very common crime, so it is worth considering why this has persisted for so long.[2]

Before the internet became popular, people used to tape music from the radio. Lee (2015) says no one was bothered by this because it was impossible to catch people.[3] Everyone knew it happened but music sales remained high so it clearly had no real impact on artists and labels.[4] Because of this, although home taping was technically illegal, it was only record companies who were worried about profits who could really call it 'stealing'. Nobody knows how much music was copied and it continues today in activities such as songs being downloaded for use as ringtones.

Lee goes on to say that just because it is possible to catch people who file share via the internet doesn't make it any worse than people using images for screensavers or songs for mobile phone ringtones.[5] Carla (2016) agrees with Lee and says that file sharing is a 'useful service to art'. She states that without this service art would be 'extremely bland and middle-of-the-road'. Hibbs (2016)[6] says that most people have now shared files online without paying. Because everyone is doing it, it cannot be a bad thing and cannot be considered wrong.[7]

The real reason file sharing gets classed as stealing is that big media companies do not like to see their profits escaping from them. Spratt (2014) states that media companies are not even that bothered about ordinary people file sharing. They are only worried about companies who make and sell pirate copies of their media. So why do they continue to prosecute small-scale file sharers? This can only be about greed, especially as it is the poorest people who have to file share as they are priced out of legal downloads.[8]

Cuttle (2015)[9] says that people should pay for the products that they consume and that if they cannot pay then they should go without. He sees file sharing as stealing. Kahliney (2013) agrees with this. He says that small companies cannot afford to lose money through illegal file sharing. Even a few instances of this have a bad effect on companies who employ only a few staff and they might have to make people redundant.[10] The type of media these companies produce tends to be quite obscure and unpopular so there is little effect on the majority of media consumers.[11]

Carla (2016) says that new artists are often overlooked by the major music and film companies and are only picked up by small independent companies.[12] These companies are often only able to distribute media on a limited basis. Many have very small staff and resources and cannot get out on the road to sell music to shops across the country, never mind worldwide. Bigger companies can employ sales teams to take the product out to the market, either promoting it in shops, or even arranging tours to schools to promote the music to school children. School children buy music in the largest numbers so a

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band promoted well to children is likely to sell well and become better known to the general public. It is unrealistic to expect that every act can tour schools, as schools limit how many artists can visit in a term as they have other things to fit into the school day, and, furthermore, many bands couldn't afford the cost of going on tour. This is where file sharing provides a service to the less well-known artist.[13] When people share music for free, it actually helps to get it heard by a range of people who would not know about it otherwise.[14]

The public, especially people with little money, should not have to lose out because of the interests of big business. Business is only motivated by profits. It's in the interests of big business to prevent people from file sharing. Their argument is all about money, at the end of the day. They were not so bothered about music being taped from the radio, because the quality of the reproductions was bad. If they really had a moral concern about stealing, they would have objected to home taping as much as they do file sharing.[15]

There are some forms of stealing that are clearly always wrong, such as mugging a person or robbing a house. We have seen in this essay that stealing is a long-standing ethical problem, and that even though there have long been strictures against stealing, the moral position has not prevented people from stealing. This essay has looked at some areas which are much less clear cut. There are arguments for and against why file sharing via the internet might be considered wrong. These depend on what viewpoint is taken – companies worried about profit will always see it as wrong but ordinary music listeners and film viewers think they are providing a helpful service. We also have to think about artists, both what they can earn and also whether it is good to have their work reach a wider audience.[16] Not everyone will agree with the arguments presented by either side. This is an interesting debate and one that will doubtless continue for many years.[17]

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Evaluation of Essay 1

Aspect	Yes/No	Comments
1 The writer's own position on the issues is clear.	No	It is not stated clearly, but can be guessed.
2 It is clear what the reasons are for the writer's point of view.	No	These are not clearly stated.
3 The writer's conclusion is clear and based on the evidence.	No	The essay appears to favour evidence that supports unpaid downloading as acceptable, but this is not formulated into a conclusion. The final paragraph only summarises the arguments.
4 Reasons are presented in a logical order, as a line of reasoning.	No	The reasons appear to be given in a random order, as the author has not stated clearly what their position is or what conclusions the reader should draw.
5 The argument is well structured and easy to follow.	No	The writing hops back and forward between points. It isn't clear what each paragraph contributes to the argument. The lack of a clear authorial position and conclusion makes the argument hard to follow.
6 Reasons are clearly linked to one another and to the conclusion.	No	It isn't clear how one reason relates to the rest. Interim summaries of the argument would help the reader, as would phrases or sentences to link reasons, and to signal changes of topic.
7 All the text is relevant to the assignment (in this case, about whether stealing is always wrong).	No	The material is mostly related to the subject in some way, but some of it is rather tangential. Too much irrelevant material is included.
8 The main reasons and key points stand out clearly to the reader.	No	The author's opinions, such as about the greed of big business, stand out clearly, but the reasoning is confused. Most reasons don't stand out clearly.
9 The writer makes good use of other people's research as supporting evidence to strengthen the argument.	No	The writer has made little use of research evidence and does not make use of the texts that look at ethical issues (Texts 7, 10 and 12).
10 Does the writer make a reasoned evaluation of other people's views, especially those that contradict his or her own point of view?	No	The writer introduces some views that appear to contradict his or her own view. However, these are dismissed too quickly, without considering the implications in any detail.
11 Does the writer provide references in the text when introducing other people's ideas?	Yes	The writer provides references in the text.
12 Does the writer provide a list of references at the end of the essay?	No	A list of references is not provided so the reader cannot follow up the references. Without this, the references in the text are not much use.
13 Has the writer successfully removed any non-essential descriptive writing?	No	The second paragraph, and the paragraph about music in schools, contain unnecessary description.
14 Does the writing contain any inconsistencies?	Yes	The writing describes the music of small bands as if it is unimportant and 'unpopular', but later argues as if it is a good thing to make such 'unpopular' music better known.
15 Are the writer's beliefs or self-interests unfairly distorting the argument?	Yes	The writer's beliefs come across more strongly than the reasoning or argument.

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Commentary for Essay 1

The numbers of the points given below refer to the numbers provided in the text for Essay 1.

1 'Although most reasonable people ...' – the author is making an assumption that the reader will agree with his or her point of view by appealing to them as a 'reasonable' person. There may be validity in this point of view but there is no evidence given of a universal agreement on which areas of stealing are considered wrong. See Chapter 7, p. 98, for more about this kind of flawed reasoning.

2 This paragraph consists mainly of over-generalisations and repeats the main idea expressed in the first paragraph. It would be considered as 'waffle' by an editor or tutor. It is a waste of the words available.

3 The author states Lee's position on home taping and asserts that record companies were happy to overlook it. The author does not refer to any counter arguments on this issue, which weakens the point. Record companies did, in fact, make strenuous efforts to deter home tapers (such as the 1980s campaign 'Home Taping is Killing Music'). The author assumes that the only possible concern about home taping could have been profit and does not mention the possibility of ethical arguments such as the use of the artists' intellectual property. The author either has not considered the issues in sufficient depth, or is attempting to misrepresent the argument (see p. 103).

4 This is an assumption: the author does not provide convincing evidence that artists were not affected by such copying. Sales might have been even higher if copying had not taken place. Artists might have received a small proportion of the profits, so any reduction in sales may have affected them disproportionately.

5 The author makes uncritical use of Lee's and Carla's texts in this paragraph. Although the 'Lee' source is relatively credible, the 'Carla' text is sourced from a website for supporters of free downloading and therefore respondents have a vested interest in their own arguments, which makes them less credible (see p. 113). There is an unquestioning acceptance that the music industry would be 'bland' without downloading, without critical consideration of this assumption. For example, it is worth considering how music has developed and changed across the centuries, and develops today within many cultures, without use of the internet.

6 The credibility of the 'Hibbs' source is questionable and yet the author re-states these views as if they were 'facts', without any analysis or discussion of what is being said.

7 It is flawed reasoning to argue that because 'everyone' does something, it is then acceptable. See page 104.

8 The author's argument becomes very polemical at this point. The main thrust of the author's argument is that 'greedy' media companies are desperate to protect profits. It would have been useful to provide supporting evidence to back up this argument. The author would need to do some research to see if there are links between a decline in media company profits and an increase in file sharing via the internet. Similarly, the author would have needed to find some supporting evidence to convince the reader that the main reason people file share is because they cannot afford to pay. Alternative points of view are presented in the texts that the author has chosen not to use. The author jumps to conclusions and appears to select facts to support his or her own interests. The position may be justifiable, but it has not been supported by the evidence presented.

9 There seems to be a sudden shift in the line of reasoning here, as the author lists arguments to support the view that internet downloading is wrongful theft. A linking paragraph is needed here to summarise the author's previous arguments and signal the intent to focus on a new topic. See page 153 on words that signpost the direction of the argument.

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10 A linking word or phrase, such as 'however' or 'on the other hand', is needed here.

11 The arguments by Cuttle and Kahliney presented here against illegal file sharing appear quite plausible. However, the author dismisses these opposing arguments too quickly, without analysing the evidence. The line of reasoning is flawed, as the conclusion made at the end of the paragraph focuses on how many media consumers are affected, which is irrelevant to whether or not file sharing is stealing. Even if it were true that the music or film was obscure, file sharing might still be considered as stealing. Without further exploration of the author's thinking it is difficult for the reader to see how this interim conclusion has been reached.

12 The argument has switched back to supporting free downloading as a valid activity. Again, the author does not summarise the previous argument to help the reader follow what has been said so far, and does not signal that the topic is going to change.

13 A disproportionate amount of attention is given to talking about school tours. Most of this paragraph is too wordy and irrelevant, and the main argument becomes lost.

14 This paragraph makes unquestioning use of material from Text 1. It may be a good argument that downloading for free helps music made by lesser known groups to reach a wider audience, but no evidence is provided to support this. Moreover, this argument is not consistent with the view raised by the author earlier, that such music is obscure and unpopular. This point also ignores other complexities raised in the texts about the small record companies needing sales in order to survive, and about the legal rights of artists and businesses.

15 An interesting point, but the argument has hopped back to points already raised earlier.

16 Although, in the final paragraph, the author summarises two positions on unpaid downloading, this paragraph does not state the author's own position or draw a logical conclusion.

17 The essay's final sentence is very weak and contributes nothing to the argument.

Overall, the author has shown an ability to describe and summarise texts, but does not demonstrate good reasoning skills.

In this essay, the line of reasoning is not clear and the author's position is not reflected in the conclusion. Much of the material is irrelevant or based on sources that are not very credible.

The final sentence draws attention to the weakness of the overall argument. There is little critical analysis of the evidence. The author has said the case is a 'grey area' but has not supported this point of view by identifying what factors make something a 'grey area'. The essay ends on a weak point because it uses such a vague generalisation.

Some whole paragraphs, such as the forth and sixth paragraphs, do little more than paraphrase one source after another. These are summaries rather than critical analysis. See p. 49.

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Checklist for evaluating Essay 2

Use this checklist to analyse Essay 2 on the following page. Compare your analysis with the evaluation and commentary on pages 181–3.

Aspect	Yes/No	Comments
1 The writer's own position on the issues is clear.		
2 It is clear what the reasons are for the writer's point of view.		
3 The writer's conclusion is clear and based on the evidence.		
4 Reasons are presented in a logical order, as a line of reasoning.		
5 The argument is well structured and easy to follow.		
6 Reasons are clearly linked to one another and to the conclusion.		
7 All the text is relevant to the assignment (in this case, about whether stealing is always wrong).		
8 The main reasons and key points stand out clearly to the reader.		
9 The writer makes good use of other people's research as supporting evidence to strengthen the argument.		
10 Does the writer make a reasoned evaluation of other people's views, especially those that contradict his or her own point of view?		
11 Does the writer provide references in the text when introducing other people's ideas?		
12 Does the writer provide a list of references at the end of the essay?		
13 Has the writer successfully removed any non-essential descriptive writing?		
14 Does the writing contain any inconsistencies?		
15 Are the writer's beliefs or self-interests unfairly distorting the argument?		

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'Stealing is always wrong.' Discuss with reference to file sharing via the internet

There are many different forms of stealing, from the theft of property, muggings and burglaries, to the theft of ideas through plagiarism. Although there are legal sanctions against many forms of stealing, the issue of moral and social sanctions has always been more complex. For example, Robin Hood, who stole from the rich to give to the poor, is held up to be a great hero. Piaskin (1986) suggests that ethical issues are not simply questions of right and wrong but should be regarded as 'dilemmas'. In this essay I shall use the example of file sharing via the internet to highlight these complexities but, contrary to the view held by Piaskin, to argue that in this case stealing is always wrong.[1]

In recent years, there have been a number of high profile cases against people who have shared files for free on the internet. Prior to the development of the internet, music was commonly shared via home taping. Lee (2015) argues that although home taping was technically illegal, no one pursued this as perpetrators cannot be caught.[2] Because it is possible to catch internet file sharers, Lee argues that they are being unfairly punished. Whilst there may be a practical basis to this argument – it is easier to catch downloaders than it was home tapers – this does not mean that one behaviour should be considered acceptable and the other should not. This kind of argument is a rationalisation, used to make unacceptable actions appear acceptable.

Indeed, this point is made by Cuttle (2015). Cuttle, a legal expert, states that 'piracy of videos, music and software is stealing' and makes it clear that all such copying is illegal.[3] Given that there is a legal argument against both home taping and internet downloading, it appears reasonable to assume that both should be considered as wrong.[4] However, it is important to explore the moral arguments in order to evaluate whether such behaviours should also be considered 'wrong' from an ethical perspective.[5]

Research by Mixim, Moss and Plummer (1934), as well as later studies by Mixim et al., suggest that most people do maintain an ethical sense of right and wrong, even in areas where stealing appears to be more socially acceptable. Their findings suggested that people's ethical sense wanes when payment methods are difficult but they do not forget what is ethically right. Ebo, Markham and Malik (2014) examined the effect on internet downloading of easier payment schemes. During the study there was a dramatic decrease in illegal downloads with the majority of users choosing to make use of the easy payment scheme. This indicates that the majority of people in the study acknowledged that to download music for free, in effect stealing it, was wrong.[6]

A different ethical perspective is suggested by those authors who support unpaid downloading, especially those who use ethical and artistic arguments to counter economic arguments. A number of authors, such as 'Carla' (2016), an internet file sharer, assert that the main argument against downloading comes from media companies who are primarily concerned with their own profits.[7] Economic arguments are treated by such writers as if they are intrinsically weaker than artistic ones. 'Carla' develops this argument to suggest that true artists are driven by a desire to have their music heard by others and welcome the 'service' provided by file sharers. Hibbs (2016), a member of the public, also argues that file sharing is a kindness between friends. These kinds of arguments can sound convincing as they make downloading appear to be altruistic, and altruism appears to have the ethical advantage over the rush for profits. On the other hand, it could be argued that this is altruism at someone else's expense. The economics of free downloading do not help less well-known artists, so not paying for downloading of their work is unethical.[8]

Furthermore,[9] those who defend downloading often act as if they know the 'real' wishes and interests of artists. Carla, for example, refers to 'true artists' without defining what a 'true artist' is, or

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providing evidence to show what such a 'true artist' would want. Authors such as Carla and Hibbs do not provide evidence to show that artists regard free downloading as being more in their interests than the actions taken by businesses. As music sales are usually of direct financial benefit to artists, many artists may also disagree with free downloading.[10]

Moreover, Cuttle (2015)[11] asserts that arguments such as Carla's and Hibbs's are invalid in free market terms.[12] Publishers have a right to charge the highest price that they are able to obtain, and consumers can choose whether or not to purchase. In that case, business is not in the wrong to charge whatever price the market will sustain. However, there are other economic, and indeed artistic,[13] arguments against Carla's and Hibbs's positions.[14] Such authors assume that objections to downloading come mainly from large corporations that can be dismissed as 'greedy'. Kahliney (2013) argues that small, independent companies and recording artists are most likely to suffer from the effects of downloading as their overall reliance on sales is greater. Given that sales for independent artists tend to be low anyway, falling sales could mean the collapse of small labels. Whilst artists could still have their music heard for free via downloads, their position is unlikely to remain financially viable for long. Ironically, this increases the likelihood of a music industry populated by the types of 'bland' or 'middle of the road' acts that Carla complains would exist without internet downloading: they will be the only artists that can guarantee reasonable sales.[15]

In conclusion, I have demonstrated in this essay that there are arguments to support the view that all stealing can be viewed as 'wrong'. This holds true even in relation to complex areas such as internet downloading and file sharing, where social behaviours may appear to support the view that downloading without paying is acceptable.[16] Indeed, in the case of unpaid downloading, there are legal, ethical, economic and artistic arguments to support the view that stealing from the industry is wrong. There are counter arguments, such as that downloading offers a service to music and small artists, but there is little evidence to support such views or to suggest that they represent the views of the majority. On the contrary, when given accessible, affordable payment options, most people chose not to steal, thereby acknowledging that free downloading is wrong. Although moral positions can easily be influenced by practical circumstances, such as how easy it is to pay, research suggests people maintain an ethical sense that stealing is always wrong.

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Evaluation of Essay 2

Aspect	Yes/No	Comments
1 The writer's own position on the issues is clear.	Yes	This is clearly stated in the opening paragraph and again in the conclusion, and helps to orientate the reader.
2 It is clear what the reasons are for the writer's point of view.	Yes	The author presents legal, ethical, economic and artistic reasons.
3 The writer's conclusion is clear and based on the evidence.	Yes	The conclusion is based on the reasons given, and especially on research into people's behaviour when given a chance to pay rather than steal.
4 Reasons are presented in a logical order, as a line of reasoning.	Yes	The reasons are clearly grouped, starting with the legal reasons and progressing to ethical and other considerations.
5 The argument is well structured and easy to follow.	Yes	The argument groups similar points and details together. It is clear when the argument is changing to consider a different point.
6 Reasons are clearly linked to one another and to the conclusion.	Yes	Good use is made of interim summaries to sum up arguments so far. Linking sentences and signal words are used to develop the line of reasoning.
7 All the text is relevant to the assignment (in this case, about whether stealing is always wrong).	Yes	The text is comprised almost entirely of reasons, summaries, linking phrases, evaluations and judgements.
8 The main reasons and key points stand out clearly to the reader.	Yes	The reasons are stated very clearly and are not lost amongst unnecessary opinion or description.
9 The writer makes good use of other people's research as supporting evidence to strengthen the argument.	Yes	The writer makes effective use of experts, such as the legal expert, to support their own position.
10 Does the writer make a reasoned evaluation of other people's views, especially those that contradict his or her own point of view?	Yes	The writer includes reference to ideas that contradict the main line of reasoning, indicates where opponents may have a point, and provides reasons to show why their points of view are not acceptable overall.
11 Does the writer provide references in the text when introducing other people's ideas?	Yes	There are references to all sources used.
12 Does the writer provide a list of references at the end of the essay?	Yes	A complete list of references is provided and set out correctly.
13 Has the writer successfully removed any non-essential descriptive writing?	Yes	There is very little description, except in the introduction, on Robin Hood, to set the scene.
14 Does the writing contain any inconsistencies?	No	
15 Are the writer's beliefs or self-interests unfairly distorting the argument?	No	The writer may have strong beliefs but the argument is firmly based on reasoning.

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Commentary on Essay 2

The numbers of the points given below refer to the numbers provided in the text for Essay 2.

1 The author sets out their position clearly in this opening paragraph. The author acknowledges that there are complexities to the issue, but nonetheless, the text clearly states the author's position. We know from the outset that they will be taking up the position that stealing is always wrong.

2 The author begins to create the argument by taking a piece of evidence that appears to contradict his or her position and analysing its argument. In refuting this evidence, the author is establishing their own argument and building credibility for this by weighing it up against a counter argument.

3 The author starts to tease out the different layers within the argument. The argument is clearly based on a legal approach to the issue. Good use is made of a quotation from a legal expert to support the author's position.

4 In this sentence, the author makes an effective interim summary of the argument so far. Good use is made of tentative language, 'it seems reasonable to assume', in order to indicate an awareness that the argument has not yet been won.

5 The final sentence of the paragraph is helpful in signalling to the reader that the argument will now consider a different perspective on the issue, the moral issue. Good use is also made of the signal word 'however', to indicate a change of topic.

6 The author makes good use of research in the field to suggest that most people's behaviour, when they are given a chance to pay or steal, supports the view that downloading for free is recognised as wrong.

7 The author places Carla's position in the context of her being an internet downloader. The author does not explicitly state that Carla's beliefs are necessarily the result of self-interest, but by placing her comments in context the author helps the reader make sense of why she might hold these views.

8 The author strengthens the overall argument by showing why counter arguments can appear convincing, but undermines these counter arguments effectively by questioning who is paying the cost of altruism.

9 The use of a linking word, 'furthermore', indicates that the argument is being continued in a similar vein, but that a new angle is being introduced to strengthen the point being made.

10 The author points out weaknesses and flaws in the counter arguments, using good critical analysis. The author makes a detailed critical analysis of some aspects of the arguments, such as the use of 'true artist' as emotive vocabulary, and points out gaps in opposing arguments. As the counter argument makes suggestions about what artists would think, the author puts forward reasons why artists might have alternative views.

11 By beginning the paragraph with the word 'moreover', the author signals to the reader that a further point will be made to support the current line of reasoning.

12 The views of an expert are again used to support the argument, along with an allusion to a theoretical position, that of the 'free market economy'.

13 The author states that opposing arguments can be dismissed in absolute terms by considering the suppliers' right to charge whatever price they wish. However, this reason may not be persuasive for some audiences, so the author rightly builds further on this argument by considering other angles.

14 Throughout the writing, the author has helped to clarify the nature of the argument by categorising the reasons. Previously, the author stated he or she would refer to legal and moral reasons, and here the text signals that there are also economic and artistic reasons that support their position.

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15 The author effectively undermines the counter argument that free downloading prevents a 'bland' music world, by showing how it could lead to an increase in 'middle-of-the-road' music.

16 This paragraph draws the conclusion that stealing is always wrong. This conclusion has been well supported by the line of reasoning throughout the essay, so should not come as a surprise to the reader. The author summarises their position in the conclusion, clearly asserting this position and recapitulating the key points of the argument. The reader may not agree with this position but will be clear about what the author believes and why.

Overall, this is a much stronger piece of critical writing than Essay 1. The author's position is clear, and the writing is consistent in providing reasons to support this. Good use is made of expert sources to support the author's position, so that it comes across as more than personal opinion. The author makes a careful consideration of opposing arguments, making it clear why these opposing arguments might be attractive, but drawing attention to gaps and flaws in opponents' arguments.

Throughout, the author is careful to cite sources and, helpfully, gives an indication of whether sources are authoritative or not. This is done by alluding briefly to 'studies' and 'research', whether someone is a legal expert, or a member of the general public. The list of references is complete and set out correctly and consistently.

The argument could have been even stronger if it had been more questioning of some of the underlying theoretical arguments for the author's position. The writing takes the position that the law and the free market economy are right, without providing any challenge to this point of view.

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Evaluating your writing for critical thinking

You can copy this self-evaluation tool to use for future reports and assignments.

Self-evaluation	Yes/No	Action
1 I am clear on my position on this subject and the reasons for my point of view.		Write your position down as a statement in one or two sentences. If you cannot do so, this suggests that your position isn't yet clear in your own mind. If possible, also check whether your point of view is clear to a friend or colleague who knows little about the subject.
2 My conclusion and/or recommendations are clear, based on the evidence, and written in tentative language where appropriate.		Write your conclusions first. Read these aloud; check that they make sense. Imagine someone tells you that your conclusion is wrong. What reasons would you give to defend it? Have you included all these reasons in your writing? For language: see p. 159.
3 The material included is the most relevant to the subject.		Double-check that your line of reasoning meets the task requirements, such as meeting the project brief or answering the questions set for an essay. Does it match the statement you wrote about your position?
4 All sections of the assignment or report are relevant to the exact specifications of the task.		Read through each section or paragraph in turn, checking how the information contributes to your line of reasoning, leading to your conclusion or recommendations. Check that each meets the project brief, or is necessary to answer the set question.
5 I have analysed the structure of my argument. Reasons are presented in the best order and lead clearly towards the conclusion.		If not, write the reasons out in brief and consider how each is linked to the conclusion. Check whether the argument 'hops' from one point to another. Cluster similar reasons together and indicate how each contributes to the main argument or conclusion.
6 The argument stands out clearly from other information. I have selected the best examples.		Check you have not presented so much detail that the main argument is lost. An analysis of a few examples or details is better than a superficial approach to lots of material. Select carefully to meet the task requirement.
7 My reasons are clearly linked to one another and to the conclusion(s).		Check that each paragraph opens with a clear link to what has gone before or signals a change in the direction of your argument using 'signal words' such as those suggested in Chapter 10.
8 My main reasons and key points stand out clearly to the reader.		Take a marker pen and highlight the sentence that sums up the main point or reason covered in each paragraph. If you find this difficult, it is likely that your reader will find it hard to identify your points. If large sections of a paragraph are highlighted, then it is probable that you haven't summarised its main point sufficiently.

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Self-evaluation	Yes/No	Action
9 My facts are accurate.		Don't rely on opinion or memory. Check that your sources are reputable and up to date. Investigate whether anything published more recently gives different information. Check that you have reported the facts accurately, and without distortion.
10 I have included reference to relevant theories.		Find out the schools of thought or theories related to this subject. Make a critical evaluation of these to identify where they support or conflict with your argument.
11 I make use of other people's research as supporting evidence to strengthen my argument.		Check what has been written or produced on this subject by other people. Include references to relevant items that best support your point of view.
12 I have cited the source of information for evidence and theories to which I refer.		Write out the details of the references in brief within the text, and in full at the end of the writing.
13 I include a reasoned evaluation of views that do not support my own argument.		Find out what has been written that contradicts your point of view, and consider any other potential objections that could be raised. Evaluate these as part of your line of reasoning. Make it clear why your reasons are more convincing than opposing points of view. Identify any flaws, gaps or inconsistencies in the counter arguments.
14 My writing is mainly analytical and contains only brief, essential descriptive writing.		Check whether all sections of descriptive writing and background information are essential to understanding your reasoning or are part of the conventions of the type of report you are writing. Keep descriptions very brief, look for ways of summarising them and link them clearly to your main argument. Beware of wordy introductions.
15 I have checked my argument for inconsistencies.		Check whether any of the reasons or evidence you have used could be interpreted as contradicting what you have written elsewhere in the piece of writing.
16 I have given clear indications of levels of probability or uncertainty.		Check that your writing indicates your judgement of how likely it is that the conclusion is accurate and irrefutable. If there is a chance that research findings could be interpreted differently by someone else, use appropriate language to indicate a level of uncertainty or ambiguity. See p. 159.
17 My current beliefs are not unfairly distorting my argument.		If any section of your assignment covers a subject where you have strong beliefs or interests, be especially careful that you have checked the evidence supports your reasoning. It is important that your arguments come across in a calm and reasoned way that will convince your reader. Check several times, and be careful not to include emotive language or poorly substantiated opinions.
18 I have covered all the required aspects of the assignment.		Check the assignment's details carefully. Tick (✓) aspects already completed so it is clear what else you must do.

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Summary

This chapter provided the opportunity to evaluate two pieces of critical writing on a similar subject and to compare these with pre-written evaluations and commentaries. One aim of this activity was to build upon your skills in critical evaluation by applying them to extended pieces of text. However, the key aim was to help you develop the skills to evaluate your own critical writing.

The commentaries provide you with a critical evaluation of two essays, drawing out their weaknesses and strengths. This is the kind of approach that an editor or tutor will take when you submit your own writing. When you produce critical writing for assessment purposes or for publication, you should make an equally rigorous evaluation of your own work before submitting it.

Evaluation, in this case, means making a critique of your work as a single, completed piece of critical writing, checking how all the different components contribute to the strength of the written argument. Before getting to this stage, you should have evaluated, already, the different component parts such as the quality of your evidence, the validity of your selections (what you have chosen to include and what to leave out), whether your reasons support your conclusion, and the validity of your conclusion.

There isn't one correct way to evaluate your overall piece of work. You may find it easier to make rough notes in the form of critical commentary on your text. Alternatively, you may find it easier to use one or more structured checklists, looking for particular aspects in your writing. You may prefer to combine both methods, moving back and forward between them depending on what works best for the way you write.

The important point is that, having made a good critical analysis of your source materials, you apply an equally critical approach to your own writing to ensure that you have presented your argument in a structured, logical and convincing way.

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Chapter 12
Critical reflection

Learning outcomes

This chapter offers you opportunities to:

- understand what is meant by critical reflection and appreciate its challenges and benefits for academic study and professional practice
- understand how to undertake critical reflection and decide on the most appropriate approaches for your purposes
- formulate your own model of critical reflection
- understand the difference between phase 1 and phase 2 reflection
- know how to relate theory to practice
- recognise good and bad critical reflection
- understand how to present critical reflection to others, especially for academic assessment

Introduction

It is easy to become caught up in everyday routine such that we lose sight of the reasons for thinking, feeling, believing and acting as we do. Diverse aspects of our experiences and emotional responses – and our interpretations of these – can become entangled in ways that may not be apparent to us. This can distort our perspective and block understanding, which isn't helpful to us in the longer term.

Critical reflection is used increasingly in professional and academic contexts as a means of focusing our attention back onto our own experience. Reflection in this context refers to specific kinds of mental discipline. It involves clarifying our thinking, deepening understanding and reinforcing learning in ways that, ideally, lead to transformation and change.

Although 'reflection' sounds as though it should be easy, in practice, there is a great deal of challenge in working with the raw material of our experience, and especially in doing so with integrity, in ways that support learning, and doing all this within a framework of academic conventions.

This chapter helps you to understand what is required when undertaking critical reflection for academic and professional purposes. It provides structures that can help you to negotiate the various phases and processes of critical reflection, from initial selection of a topic through to presentation of your reflection for assessment.

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What is critical reflection?

More than just 'thinking'

In everyday language, the term 'reflection' is used loosely to refer to:

- being engaged in thought;
- vague musings or day dreaming;
- going back over an event in our minds.

Critical reflection for academic and professional purposes is different from this. It is structured, focused and conscious, with the end purpose of developing our understanding. If you undertake critical reflection as a requirement of your programme or profession, there are likely to be expectations about the form this takes and how to present it.

Critical reflection: Characteristics

There isn't a single kind of critical reflection, but typical characteristics include:

1. **Selection**: select an aspect of experience, learning or professional practice for analysis.
2. **Changing perspective**: analyse experience from different angles and different levels of detail.
3. **Returning to experience**: once, periodically, or frequently, as best fits the issue.
4. **Analysis of own role**: look at reasons for, and consequences of, your own actions, rather than those of others.
5. **Drawing on received wisdom**: make use of theory, research, professional knowledge.
6. **Deepening your understanding**: look actively for meaning, recognising what is significant, and learning from this.
7. **Using insights to effect change**: use your new understanding to do things differently in the future – ideally to the benefit of others as well as yourself.

These are examined in more detail below.

'Actions' in this context refers to all salient behaviours, speech, feelings and thoughts, including such matters as body language, assumptions that influence behaviour, and failure to take action.

1 Selection

Effective critical reflection is an intensive activity that takes time, mental energy and often, too, emotional energy. Realistically, you won't be able to bring the same level of critical analysis to all areas of your experience. Select one or two areas where that kind of investment of your resources will bring the best returns for you. See pages 192 and 203.

2 Changing perspective

In our everyday routine, we tend to observe our actions from a given perspective and level of detail, referring mainly to our own opinions. It would be difficult to function on a day-to-day basis if we didn't do so. For critical reflection, we look to develop understanding partly by altering our focus, using different levels of detail from those employed on an everyday basis – as if zooming in and out with a camera lens to see surroundings with fresh eyes.

Analysing in detail

This means putting our experiences under a magnifying lens so as to identify component parts and contributing factors. For example, by clarifying an exact sequence of events, we might identify potential causes and effects overlooked at the time. Alternatively, we might recall things that we said or did that did not seem significant at the time but which take on new meaning when we reconsider the context, our motivations or anxieties. We can use these observations to elicit patterns and themes.

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Considering the bigger picture

This means standing back to look at our experience from broader perspectives, as if through a wide-angle lens:

- actively searching out themes or patterns within our reflections;
- testing our personal experience against published theories, research and professional knowledge;
- considering the influence of broader political, social or ideological contexts.

3 Returning to experience

The appropriate timing and frequency will depend on what you want to gain from your critical reflection. You could select:

A significant, one-off, incident to analyse in detail, to help understand better what happened and your own role and reactions.

A recurrent situation to reflect on after each occurrence, so that you build your observations, your sense of potential cause and effect, and of your own role.

An issue or theme, and return to this at frequent and regular intervals so that you can identify patterns in your actions and emotional responses over time.

A particular project or challenge, and record your observations and reflections over its duration. Draw conclusions about your own role and actions, the reasons for these, their consequences for the project, yourself or others, and for future action.

4 Analysis of your own role

Critical reflection requires you to look at your own experience – it is not intended to be an occasion to blame others or focus on their actions. Depending on the context, this means looking at such aspects as:

- the impact or consequences of what you did or didn't do – and why;
- the reasons for your actions, moving beyond superficial and immediate reasons to look at latent fears and motivations, displaced emotions, assumptions you had made, and whether these helped or hindered the situation;
- changes in the way you acted or responded over time.

5 Drawing on received wisdom

For critical reflection in academic contexts, you are expected to relate your own experience to theory. Ways of doing this are addressed on page 194.

6 Deepening your understanding

Critical reflection is more than detailed observation or a narrative of events. It involves active working with the raw material of your experience so as to make sense of situations and events, their dynamics, your own role within them and the influence of the broader context. This means sifting through your reflections, selecting key insights and identifying why these are of significance to your performance in study or work, the way you work with others, or your life generally.

7 Using insights to effect change

A key aim of critical reflection is to transform the way you see the world, or an aspect of it, such that you think and act differently. Change of some kind is an implicit aim: this may be a small change but one that will make a difference to you or others. It often means owning responsibility for one's own role in contributing to events, which can be challenging but also liberating. At its best, reflection can have a transformative quality, with a profound effect upon your being and sense of self, others and the world around you.

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Why engage in critical reflection?

Challenges of critical reflection

Although we may be aware that we would benefit from putting time aside for reflection, it is not easy for most of us to do so. It takes commitment and discipline because of the following factors.

- *Distractions*: There are usually more compelling things to distract us.
- *Confusion*: We may confuse everyday, unfocused 'thoughtfulness' with the more purposeful and structured critical reflection required.
- *Feelings*: If we select difficult material, this can bring its own challenges such as uncomfortable feelings, unexpected emotional responses, or unwanted conclusions about our own role – it is natural to want to evade these.
- *Skills*: It takes skill to balance a critique of experience with the requirements of academic writing, especially relating to the personal and the theoretical.

It can help to know that these challenges are not unusual. Managing such challenges forms part of the purpose of undertaking critical reflection – and contributes to the sense of achievement.

The benefits of critical reflection

A helpful mental discipline

Critical reflection develops a mental discipline that brings diverse and often unexpected benefits – potentially the transformative results referred to above. Academic or professional requirements for critical reflection are helpful in that they provide an external incentive, and often a framework, that we may otherwise lack for undertaking such in-depth analysis.

The more we create a routine for critical reflection, the more we gain in terms of self-management, critical observation, emotional maturity, and in actively using experience to the benefit of ourselves and others.

Space and time for development

When we commit ourselves to critical reflection, we need to create appropriate conditions, such as making the right mental and emotional space and focusing on ourselves in ways that we wouldn't do otherwise. It helps us organise our time and enables different types of thinking.

More effective study and learning

You can use your own learning as the focus of critical reflection. Stepping back from study, you can benefit from reflecting on your attitudes to learning, your strategies, the effectiveness of these for different subject matter, and what assists or prevents you from achieving well.

You can better understand your own performance through looking critically at:

- your educational history and its impact on achievement and self-belief;
- how your current attitudes and study strategies evolved, and whether these best fit your current context;
- whether your self-perceptions assist or hinder your performance;
- whether, unconsciously, you sabotage your own performance;
- if you wish to pursue this in more depth, a wide range of materials are available in *Skills for Success – Personal Development and Employability* (Cottrell 2015).

To support professional practice

Critical reflection enriches professional practice. By developing your understanding of what you are doing and why, and the consequences of your actions, you are better placed to manage new and unexpected situations in a professional manner that also supports your own needs. This is explored in more detail on pp. 205–6.

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Decide your approach and purpose

Because of the challenges identified above, it can be difficult to:

- get started on a piece of critical reflection;
- keep going with it once started;
- give shape to your reflective thinking so that it becomes meaningful;
- come back to it over time to see what it is really telling you;
- work with your reflective material productively and creatively so that you can learn from the experience and use it.

Some people generate their best reflective thinking by working in impromptu ways, using the moment, and letting the creative flow direct their thoughts. If this is true of you, then go with what works best for you. In that case, you may not need to use a formal approach until you come to shape your final reflective summary for an external audience.

However, for many people, especially those for whom critical reflection is a relatively new activity or one for which they need encouragement to keep going, it is easier to use a more structured approach. This gives a starting point and direction and helps you to produce stronger, more focused critical reflection.

In general, if you spend time early on elaborating what you wish to gain from your critical reflection and the approach you will take to best achieve that outcome, then you are likely to reap benefits in terms of the quality of your reflection and effective use of your time. This chapter outlines some key considerations for formulating your approach.



Reflection: What is your style?

How could you produce better critical reflection by:

- Working in a freer, creative way?
- Using a structured approach?

Deciding on an approach

Key considerations are:

1. purpose;
2. type of desired outcome;
3. focus;
4. model of reflection to employ;
5. your methodology;
6. audience;
7. relating experience and theory.

These are examined in detail below.

1 Identify your purpose

Start by deciding what it is that you want to achieve from undertaking this reflection – in terms of knowledge, understanding or performance. For example, for particular aspects of your study, life or work, you might want to understand:

- why something went well, and whether the lessons learnt could apply to other situations;
- why a particular situation arose or unfolded the way it did;
- why things don't seem to work out as you hoped or expected;
- why your contributions to seminars or at work aren't as effective as you would wish;
- why you think or react the way you do in particular circumstances.

Once you are clear about your purpose, it is easier to decide on other aspects of your approach, such as the expected outcomes, an appropriate focus, your methods and the model to use.

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2 Decide on type of outcome

Before you start your reflections, consider how you will make use of them. This will help you to decide the form that you want your conclusions to take, such as:

- A list of lessons learnt?
- A critical, analytical piece of writing based on your reflections?
- Recommendations for personal action?
- Guidelines for future action?
- Instructions for future action?
- A combination of these?

If you are undertaking reflection as a requirement of your programme of study or professional practice, you may be given guidance about how to write up these outcomes.



Reflection: Outcomes

- What kinds of outcomes are expected for the reflection you are expected to undertake for your programme or profession?

3 Select a focus

Select a meaningful focus for your reflection. Choose a particular incident or type of incident, or a recurring issue, concern, set of relationships, or similar that you consider relevant to achieving your purpose above.

You need to select a focus that is relatively challenging, so that there is sufficient material to enable you to develop your understanding. However, the focus should be sufficiently narrow so that you can examine it from multiple perspectives yet without being too superficial.

4 Choose a model of reflection

There are various models of reflection that you can draw upon to provide a framework for your reflection – though this isn't always required or necessary. Particular frameworks may be preferred within your discipline or professional area. If not, you may find it helpful to draw upon one or more of the models outlined on pp. 201–4.



Reflection: Using a model for reflection

- Is a particular model of reflection recommended for your course or professional area?
- Do you feel that using a model to structure your reflections would stimulate helpful reflections, or do you feel using a model would stifle your creativity?

5 Method

There isn't a single method of critical reflection. You need to consider:

- whether there is a required methodology for your discipline;
- if you have a choice, the method that will help you engage most effectively.

The following considerations may help you to select what will work best for you.

How best to record your reflections?

For your purposes, would it be better to record your initial reflections on paper, in a notebook, blog or electronic record? Would you prefer to record them digitally?

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With whom?

In relation to your identified purpose, are you likely to develop your deepest insights if you do this entirely on your own, or if you use a colleague, friend or peer group to open up your thinking?

Frequency and regularity?

For the purpose and outcomes you have identified, would it be better to undertake critical reflection on a weekly basis, or more or less frequently? Would you gain most by planning specific times each day, week or month, or making notes as and when there are new developments for you to consider?

First steps/prompts?

Consider how you will get started on your reflection. Would you work best by:

- Jotting down your thoughts as they come, and then going back over these for further consideration?
- Using the same set of questions each time, in order to structure your thinking sessions?
- Using a theory as a starting point?
- Discussing an event with a colleague, and then jotting down your thoughts?
- Applying a model of reflection in a systematic way on each occasion?
- Using a pro-forma to guide your reflection?

Working the material

Your initial reflections will generally provide raw material, or phase 1 reflections, that will require further work. Consider when and how you will work with your initial reflections so as to arrive at deeper understandings through phase 2 reflection (pp. 198 and 200).

End-point

Consider how you will draw your thoughts together towards conclusions. What format will this take? For example, you may be required to submit a blog that contains your developing ideas, and a piece of reflective writing that draws together key insights and conclusions.



Reflection: Methodology

- Which aspects of your methodology do you think you need to develop the most?
- How will you go about this?

6 Audience

If you are undertaking critical reflection for your own purposes, then you can decide on the format it takes. However, if you are required to share it with a tutor, peers, work supervisor, assessors, or others, consider how you will adapt your written account in order to take your future audience into consideration.

This may include:

- using different content and styles of writing for phases 1 and 2 (see pp. 198–200);
- including only information and reflections that you are comfortable for others to read;
- providing a well-structured and referenced critical summary;
- ensuring confidentiality and anonymity for anyone referred to in reflections – and permission to use workplace material.



Reflection: Confidentiality

- What kinds of information would you not feel comfortable sharing with your peers, tutors or others?
- What kinds of confidentiality issues are there in general for the kinds of reflection you might be asked to undertake for your course?

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7 Relating theory and practice

For critical reflection on experience for academic work, you are expected to relate experience to received wisdom or 'theory'. You can approach this in different ways, some of which are listed below.

Check the research basis first

When selecting an area of experience for critical consideration, choose an aspect for which you know there is published research. Ideally, there should be research findings that support different theoretical perspectives. If not, it may create difficulties when you come to write up your reflections for assessment. It is much harder to undertake reflection and then start looking for relevant theory and research.

Update yourself on the issues

Investigate the research findings and accepted professional practice for the area you have selected for reflection. Read reputable journals or recently published books on the subject, and find out what key issues are being addressed in that area. What are the 'hot topics'?

Ensure that you are aware of:

- established and new theories related to 'hot topics' or to other areas that you want to investigate through your critical reflection;
- research that supports these;
- any criticisms of that research;
- the direction of debate on these issues within your subject discipline or profession.

In this respect, your background reading will be similar to that undertaken for other kinds of academic work.

Do your experiences support the theory?

Consider when and how your own experiences match or do not match what you would expect from the key theories you have selected to draw upon. If there isn't a good match, consider the reasons for this, such as differences that there might be in the context or data. What is true of one situation may not be the case in your own.

Do theories support your findings?

It is possible that you will develop insights or draw conclusions on matters that were not covered in the initial research items that you used. If so, investigate whether there are theories or research findings that support or contradict what you have found. It adds weight to your own position if you can relate your conclusions to a broader research base. Research into the broader social, political and cultural issues may also help.

Learn from the experts

Use reputable sources of research and their conclusions in order to throw light onto areas of practice or study that you find difficult. Consider what these sources offer you in identifying ways of doing things differently. For example, theories of how memory works may help you in preparing for exams.

Your analysis and your consideration of salient theories should help you to understand better:

- what has been happening up to now and why;
- why you, personally, act as you do and the impact of your actions;
- how wider issues impact upon the everyday and the individual;
- how you can act differently so as to bring about different outcomes.

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Decide your approach: Summary

Aspect	Decide ...
Purpose	What is it that you want to understand or do better as a result of this reflection?
Type of outcome	When you look back over your reflections to draw out conclusions, what form do you want or need these to take? For example: • A list of lessons learnt? • Recommendations for personal action? • Guidelines for action? • Instructions on applying what you have learnt?
Focus	What will provide a meaningful focus for your reflection (a particular incident, recurring issue, concern, set of relationships, etc.) that you consider relevant to achieving your purpose above? What makes it worth the time and emotional effort to focus on this?
Model of reflection	• Are you expected to apply a particular model of reflection? If not, would adopting or adapting an existing model help you to structure your reflection? • Would it help to design your own model?
Method	How will you go about your reflection – if you have a choice? • *Recording*: On paper, in a notebook, blog or electronic record? • *With whom*: Individually? With a colleague or peer? In a group? • *Frequency*: Weekly? Daily? As needed? A one-off reflection? • *When*: As a daily log or blog? At a particular time of day? • *First steps/prompts*: How will you get started on your reflection (see phase 1 reflections on pp. 198–9)? Would it help to use free association? Using a series of questions or a pro-forma to provide structure? Using a theory as a starting point? Comparing and contrasting with another example? • *Evolution*: How will you develop your initial reflections (see phase 2 reflection, p. 208)? • *End-point*: How will you draw your thoughts together towards a conclusion?
Theory	• Which theories or research are you going to draw on to bring insight and more depth to your reflection? • How will you go about relating experience and theory? • Are there broader social, political, cultural, ideological, economic or technological issues that you could research that would provide helpful additional perspectives?
Audience	If your written reflection will be seen by peers, tutors or others, how will you adapt it to achieve the following? • Make the key points stand out? • Ensure confidentiality? • Ensure you are reasonably comfortable with them seeing any personal information or reading about personal feelings?

An outline copy of this summary is provided below. You can copy this and complete it for your work if you would find this helpful.

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Resource: Outline approach to reflection

Use the table on page 195 to guide your own approach to the reflection you need to undertake. Write your decisions about your approach, using the right-hand boxes, below



Aspect	Decision
Purpose (p. 191)	*What I want to understand better (and why this is significant):*
Type of outcome wanted? (p.192)	*Conclusions, recommendations, critical observations, personal guidelines, etc.?*
Focus (p. 192)	*Area of experience to focus on:* *How would this focus be useful in achieving my purpose?*
Model (p. 192 and pp. 201–5)	*I will/won't be drawing on a formal model of reflection because …* *The model I will use is:*

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Resource: Outline approach to reflection (continued)

Aspect	Decision
Method (p. 193	*With whom:* *Frequency:* *When:* *First steps/prompts:* *Evolution (working with your reflective material).* *End-point:*
Theory (p. 194)	*Which theoretical perspectives and approaches I am going to draw upon:*
Audience (p. 193)	*To help the reader, I will organise the writing up of these reflections by …* *Confidentiality issues I need to address and how I will do so:*

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Reflection phases 1 and 2

The process of reflection can be considered in terms of two distinct phases;

- Phase 1: Generating ideas and raw content;
- Phase 2: Analysis and synthesis.

Phase 1: Generating ideas and raw content

First-phase reflection refers to jotting down reflections when the details, thoughts and emotions are still fresh, recording these as a stream of initial thoughts. These might be captured in a notebook, file, blog, digital recording or whatever best suits your way of working.

Characteristics of phase 1

- Immediate: written whilst still fresh in the mind.
- Chronological: written 'as it happens' so it may read as an unfolding story.
- Emotional: it provides a chance to explore your feelings – and more emotion may arise as you identify what was happening for you emotionally and why.
- Detailed: likely to contain information and details that seem relevant at the time but which may become less significant as time goes by.
- Fragmented: different things will appear to be significant at different times.

Phase 2: Analysis and synthesis

It is this aspect of reflection that brings depth to your thinking and learning. It is most likely to be phase 2 reflections that are assessed or gain the most marks. It should be clear how phase 2 reflections relate to your phase 1 reflections. In effect, you consider both the issue and yourself in relation to the issue.

Characteristics of phase 2

- Holistic: you are looking back over an experience, event or project *as a whole.*
- Distance: you stand back to gain perspective. For example, you look at the feelings that were evoked without becoming caught up in them to the same extent.
- Analytical: you analyse your reflections critically.
- Synthesis: you bring together disparate and scattered thoughts and analyses into more rounded judgements and conclusions.
- Extrapolation: you are engaged in a process of teasing out meaning, insights and significance.

At phase 2, you work with your reflections as a whole, looking for significant trends, patterns and conclusions. You may benefit both from sitting quietly with these for a while to see what further thoughts emerge, and also from working with them actively, asking questions of yourself, and drawing out the implications of what you have learnt.

Understand yourself in relation to the issue

At both phases, but especially for phase 2, you need to consider your own role and actions, the impact of these at the time or later, and the consequences of these for yourself or others. For example:

- How well did you handle a situation? How good were your decision-making and other skills?
- How did your views and your understanding develop through the process of reflection?
- What difference will your new understanding make to how you approach similar issues in future?
- What further training, research, reflection or support would help you to develop personally and professionally?

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Examples of phase 1 reflection

For a project for their degree in medicine, Sophie and Charlotte created a resource to help parents and carers communicate with young children about cancer. They each completed a weekly reflective log. The following examples from their logs illustrate the flavour and range of phase 1 reflection.

Their feelings as they undertook new tasks and stretched existing skills

I am slightly worried about my computer abilities and unsure how to create professional looking graphics which could be used in the resource …

Evaluations of their current experience

My experience of dealing with bereavement in children is minimal and it is an area that I feel I will have to research in order to help me develop a greater understanding.

Their feelings about dealing with such a sensitive and difficult issue

Although I was excited about the prospect of filling this apparent need and helping parents, I was also anxious about tackling a major topic such as cancer.

Meetings with the tutor

We arrived at the first session with ideas which we had created individually, with a view to discussing these and narrowing them down to create a specific project brief. I felt apprehensive prior to the meeting as both my partner and I were coming up with individual ideas and had not agreed a direction for the project. After … I felt less anxious about the project.

The research they undertook

… we found many study and journal articles, mainly from the BMJ*, which focused directly on this issue. This encouraged me to proceed … as initially I was concerned that there would not be much literature for us to base our research on or to indicate that this would be a worthy project. My experience of dealing with bereavement in children is minimal and it is an area that I will have to research in order to help me develop a greater understanding.*

Their responses to their reading

I was shocked to read that many parents put off telling their children about cancer. As a result, children find out themselves through overhearing conversations and finding leaflets around the house.

Lessons they learnt as they went along

I was taken aback when I realised that producing visually appealing cards was not the main aim of the project. This was slightly disheartening as I was looking forward to [that] … I then began to understand that the thought process, rationale and development of ideas were more important than creating storyboard cards. With hindsight, it perhaps would have been more useful if [we] had decided on a specific idea … prior to the meeting.

How their ideas developed week by week

This week we also considered how the resource may not only help children but also individuals themselves … After looking at the fairytale card game, I realised we had missed many important aspects of the story. We then began to reorganise our ideas more logically … [and] discovered new ideas …

Recognising contributions from the team

My partner and I definitely complemented each other in other ways. She had a wider vocabulary … I was better at …

Broader social and cultural issues

… religion, language, ethnic origin and family situation. Upon thinking about and discussing this issue, I have realised that … these differences in culture are likely to result in different methods of coping with bad news and different family situations and support networks. All these need to be thought about and addressed in our resource …

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Examples of phase 2 reflection

Sophie and Charlotte drew together their learning in a reflective summary and project report which drew upon, and developed, selections of material recorded in their weekly logs. Some examples are given below. This phase 2 reflection synthesised their reflections, drawing together:

- how their thinking developed, and the effects of this on their project;
- how they made use of research;
- the broader lessons they learnt;
- changes to the way they viewed their future professional roles as doctors.

How and why their assumptions and goals changed over time

Reflecting back to the beginning of the project, I had fixated in my mind that I wanted to create a resource that looked professional and useable … this initial aim actually hindered me from thinking more deeply about the concept behind the cards … I slowly realised that … the cards did not have to look aesthetically perfect – the more important element was what the card represented.

Research that informed their thinking

Barnes et al. (2000) stated that most of the mothers included in their trial of breast cancer … were not offered any advice on how to break the news to their children. Many stated that they would have welcomed this opportunity. … This is also supported by Kroll et al. which states that research is needed to aid physicians and parents as to what age-appropriate information to give children and at what stage of the disease.

How they learnt from past personal experiences

I have also been forced to revisit some issues which I learnt about during my mother's illness. … The fact that this project has reminded me of those issues … has helped me greatly in thinking through some of the topics which need to be addressed in our resource in order for me to help others.

The deeper lessons extrapolated

I think one of the most important aspects that I take away from this project is the effect of cancer not merely on the individual but on their whole family and people close to them. I began this project with a scientific knowledge of cancer and its effects, but I take away … an insight into how cancer changes people's lives.

Before this project, I had never considered … that different family set-ups can affect the way in which we cope and rely on people. Your culture definitely has a huge impact upon the way you view and deal with disease and adversity. I think that as a doctor, it is vital to bear this in mind and adapt your advice accordingly.

Personal awareness and change

From completing this work for the resource pack, it has made me very aware of a lot of cultural influences which direct me to choosing a certain type of image. I have never before thought about the fact that a picture I might choose to represent a certain issue could convey a completely different meaning to someone from a different culture than me. I think it is important that I have learnt about this as it will make me think twice about using certain images or words in the future.

Suggested changes for the profession and the reasons for doing so

I also think … we have identified an important fact, that clinicians should look into increased involvement with their patient's family … it is greatly appreciated by patients and their families. This will have a knock on effect … as they may feel more comfortable coming to you with subsequent problems … [with] medical benefit as it can allow for the problem to be addressed and treated earlier, both minimising the effect on the patient and reducing cost and time for the NHS.

Whether a patient is at home or in hospital it is a doctor's duty to ensure that they have enough information and resources about their cancer. [Otherwise] people are more likely to put off telling their children about their cancer …

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Models of reflection

What is a model of reflection?

Models provide frameworks that can assist and structure the process of reflection. From a practical perspective, models of reflection vary in complexity, depending on how many steps, or stages, they use to break down the process of reflecting on experience, and how many prompts or cues they provide as potential memory joggers for each stage.

The underlying concept

Reflective models share three basic assumptions. These are that we can:

1. think back over our experiences;
2. understand them at a deeper level; and
3. use that understanding to do things differently in the future – that is, to effect change through learning.

Staged models

Stages present a nominal order for considering different aspects of an experience. Each stage serves as a useful reference point, prompting a different approach to working with an experience. In practice, you are likely to move fluidly between these stages and to include material not covered explicitly by them.

Three-stage model

There are various three-stage models, such as Borton (1970) and Driscoll (1994), that, in essence, simplify the reflective process to:

1 What?
2 So what?
3 Now what?

These map onto the basic assumptions described above broadly as follows.

1 *What?* Go back over the experience and identify what happened.
2 *So what?* Describing your experience is not enough – you need to work out why this is significant. What do you now understand better?
3 *Now what?* Now that you have a better understanding of your experience, what will you do with the insights gained? How will you be different? Or do things differently?

Multi-stage models

It is a complex process to work usefully with experience so as to learn from it. There are many potential components, from identifying what is relevant, exploring personal feelings and understanding other people's perspectives, through to the insights gained from research findings, or a consideration of sociological, political or other dimensions.

Any of the above components, and others, could be extracted for particular emphasis. This could be framed either as a distinct 'stage' within the model or as a distinct set of prompts. For example, Gibbs (1988) provides a model that includes a separate stage on feelings, whereas other models such as Boud et al. (1985) do not, but build consideration of feelings throughout the reflective process.

Prompt-based models

Johns and Freshwater (1998) have developed a model that is popular for professional practice in nursing. This provides prompt questions that relate, broadly, to five different ways of knowing:

- Empirical: the tested knowledge base, such as scientific research.
- Personal: knowing how feelings and motivations influence your actions.
- Aesthetic: knowing how to make meaningful, creative, unconscious responses in the moment.
- Ethical: knowing what it is right to do.
- Reflexive: knowing how to apply learning gained from experience.

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Deciding on your model for reflection

Before launching into your reflection, consider the following.

Are you required to use a model?

Do you have to draw on a particular a model of reflection for your programme? If so, ensure that you understand its rationale so that you can apply this to your own reflection.

Do you want to use an existing model?

If you have a choice about whether you apply a reflective model, consider whether you would find it helpful to draw on one that already exists, to help give structure to your approach and thinking. If so, research the various alternatives available. The references on page 210 provide a good starting place.

Would you prefer to design your own model?

Alternatively, you can design your own model. The advantages of designing your own model are not only that this should better fit your own purposes and circumstances, but that you also gain practice in planning reflective activity. You could also design your own model through active, critical engagement with existing models. This would deepen your understanding of what critical reflection is about.

Designing your own reflective model

Your own model can be as simple or as complex as you find helpful for motivating you and generating deep and useful reflection. You may wish to adapt your model depending on how well it works for you.

Concept
You would need to take on board the underlying concept outlined on page 201.

Characteristics
You would need to ensure your model addressed the characteristics of reflection outlined on page 188.

Adapting a staged model
You can adapt an existing model to suit you. Consider whether you work better with a small number of overarching stages or with additional stages that draw your attention to areas that are significant to you or your subject discipline or profession. You could adapt either the three-stage model (page 201) or the Core Model (Cottrell, 2010) outlined on page 203. Alternatively, investigate other models and adapt one of those.

Prompts
Consider whether you would benefit from a list of questions to serve as prompts for each stage of your model. Listing these can help you to decide what it is that you really want to address through your reflection. If you wish to use the Core Model for reflection (page 203), you can build from the prompts provided there. Alternatively, a fuller list of prompts is available in *Skills for Success* (Cottrell, 2015).

Name the stages and the model
Naming the stages of your model helps to make it your own. On a practical level, it also helps to clarify whether the stages are distinct and likely to work well together. It will be easier for you to recall the stages whenever you undertake your reflection.

Once you have named your stages, the distinctive character of your model should be clearer to you. Give your model a name that reflects this.

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